

HOL-2770-02-ANS-L

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VMware vDefend Firewall w/ Advanced Threat Prevention (HOL-2770-02-ANS-L)

VMware vDefend Firewall w/ Advanced Threat Prevention (HOL-2770-02-ANS-L)

This vDefend Security lab will enable you to explore the latest security features in vDefend Security, including Advanced Threat Prevention (ATP). In this lab, you will gain hands-on experience with the many security features vDefend Security has to offer. You will be introduced to the Security Services Platform, gain an in-depth understanding of vDefend Intrusion Detection & Prevention (IDPS), Virtual Patching, Network Traffic Analysis, and Network Detection & Response (NDR).

Lab Guidance

Welcome! This lab is available for you to repeat as many times as you want. Use the Table of Contents in the upper right-hand corner of the Lab Manual to jump ahead to any module.

Module	Title	Length	Level
1	vDefend Security Services Platform 5.2 (SSP) Overview	15 min	Beginner
2	vDefend Intrusion Detection/ Prevention System (IDS/IPS)	25 min	Intermediate
3	vDefend Network Traffic Analysis	10 min	Beginner
4	vDefend Network Detection & Response	25 min	Intermediate

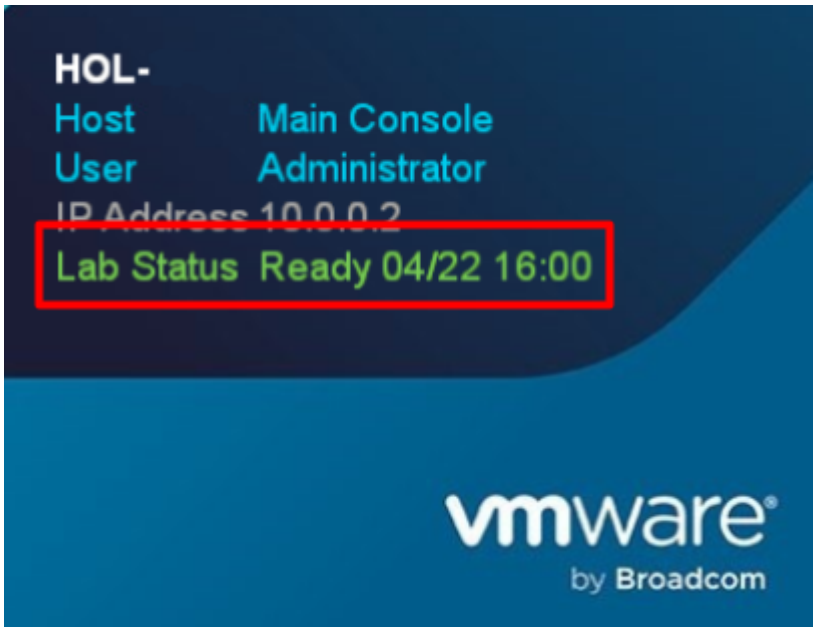
Lab Authors:

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First time using Hands-on Labs?

If this is your first time taking a lab you can review the VMware Learning Platform interface **before proceeding.**

Getting Started



The lab console will indicate when your lab has finished all the startup routines and is ready for you to start. If you see anything other than "Ready", please wait for the status to update. If after 5 minutes your lab has not changed to "Ready", please ask for assistance**.**

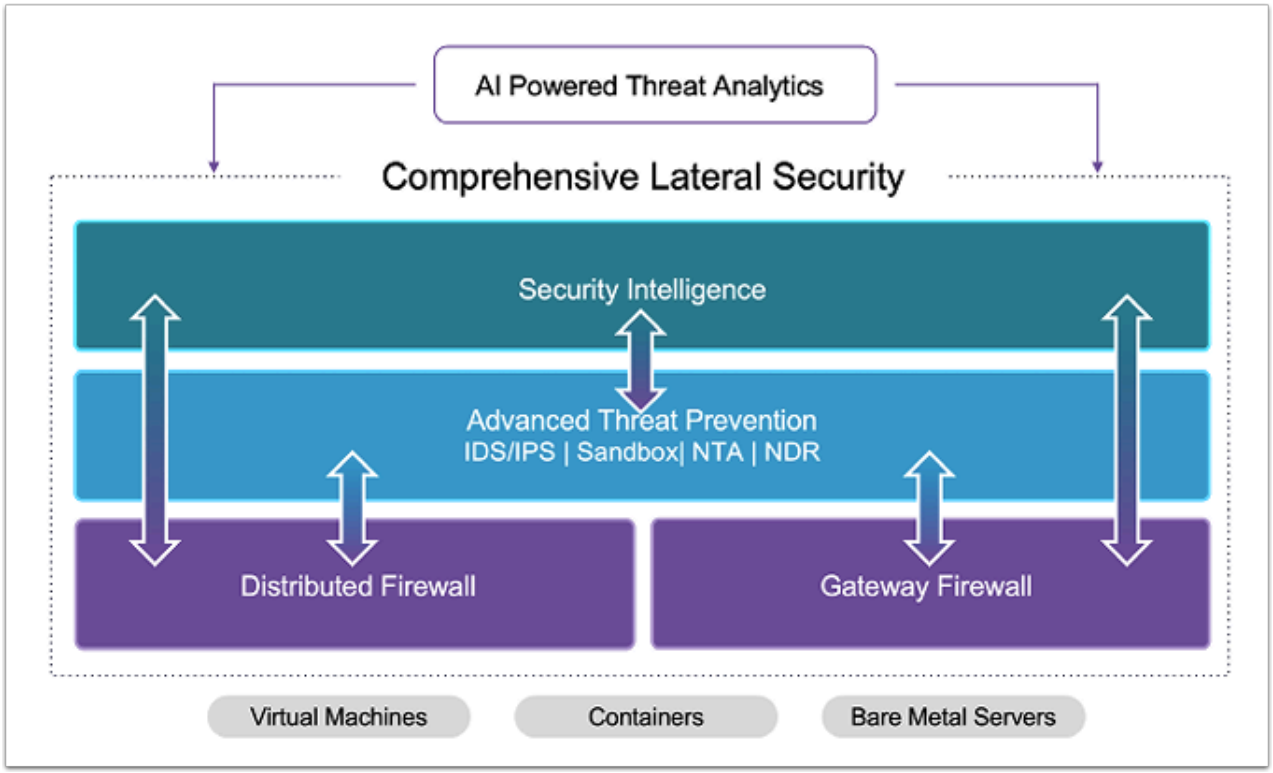
Module 1 – vDefend & Security Services Platform Architecture Introduction (10 Min)

vDefend Overview

VMware vDefend is a software-defined, layer 2-7 firewall and security suite integrated directly into the VMware Cloud Foundation (VCF) hypervisor.

VMware vDefend Advanced Threat Prevention(ATP) combines advanced detection technologies IDS/IPS, MPS, and NTA with aggregation and correlation engines from Network Detection and Response (NDR).

Advanced Networking and Security



VMware vDefend Firewall

VMware vDefend Firewall is a software-defined Layer 2-7 firewall purpose-built to secure virtualized workloads in VMware Cloud Foundation. It provides stateful firewalling capabilities that can be used by organizations to protect against the lateral movement of threats. VMware vDefend Firewall includes a comprehensive suite of security features such as:

- **Distributed Firewall (DFW)**: A software-defined Layer 2-7 firewall integrated into the hypervisor that delivers scale-out micro-segmentation. A Distributed Firewall can be deployed at each vSphere workload.

- **Gateway Firewall (GFW):** A software-only L2-7 firewall designed and deployed for use cases such as zone-based controls for application workloads or policy controls for traffic entering and/or exiting VCF workload domains. A Gateway Firewall can be deployed on a vSphere host, either as a Virtual Machine (VM) or as an ISO image on a physical server.
- **Security Intelligence:** A distributed analytics platform that provides network traffic visibility across virtualized environments – every host and every workload – and automated rule recommendations for micro-segmentation.

VMware vDefend Advanced Threat Prevention

VMware vDefend Advanced Threat Prevention (ATP) provides threat prevention capabilities that protect organizations against advanced threats. ATP combines multiple detection technologies—Intrusion Detection/Prevention System (IDS/IPS), Network Sandboxing, and Network Traffic Analysis (NTA)—with aggregation, correlation, and context engines from Network Detection and Response (NDR).

- **Distributed Intrusion Detection and Prevention System (IDPS):** It inspects network traffic for malicious activities and security vulnerabilities and simplifies compliance audits.
- **Network Sandboxing:** It provides advanced malware analysis of artifacts by deconstructing malicious behavior engineered into a file or URL.
- **Network Threat Analytics (NTA):** It detects anomalous activity and malicious behavior by identifying protocol anomalies (unusual protocol activity), traffic anomalies (unusual traffic activity), and host anomalies (unusual workload behavior).
- **Network Detection and Response (NDR):** It enables the security team to visualize attack chains by condensing massive amounts of network data into a handful of “intrusion campaigns.” It achieves this visualization by aggregating and correlating security events such as detected intrusions, suspicious objects, and anomalous network flows.

vDefend Security Services Platform (SSP) Overview

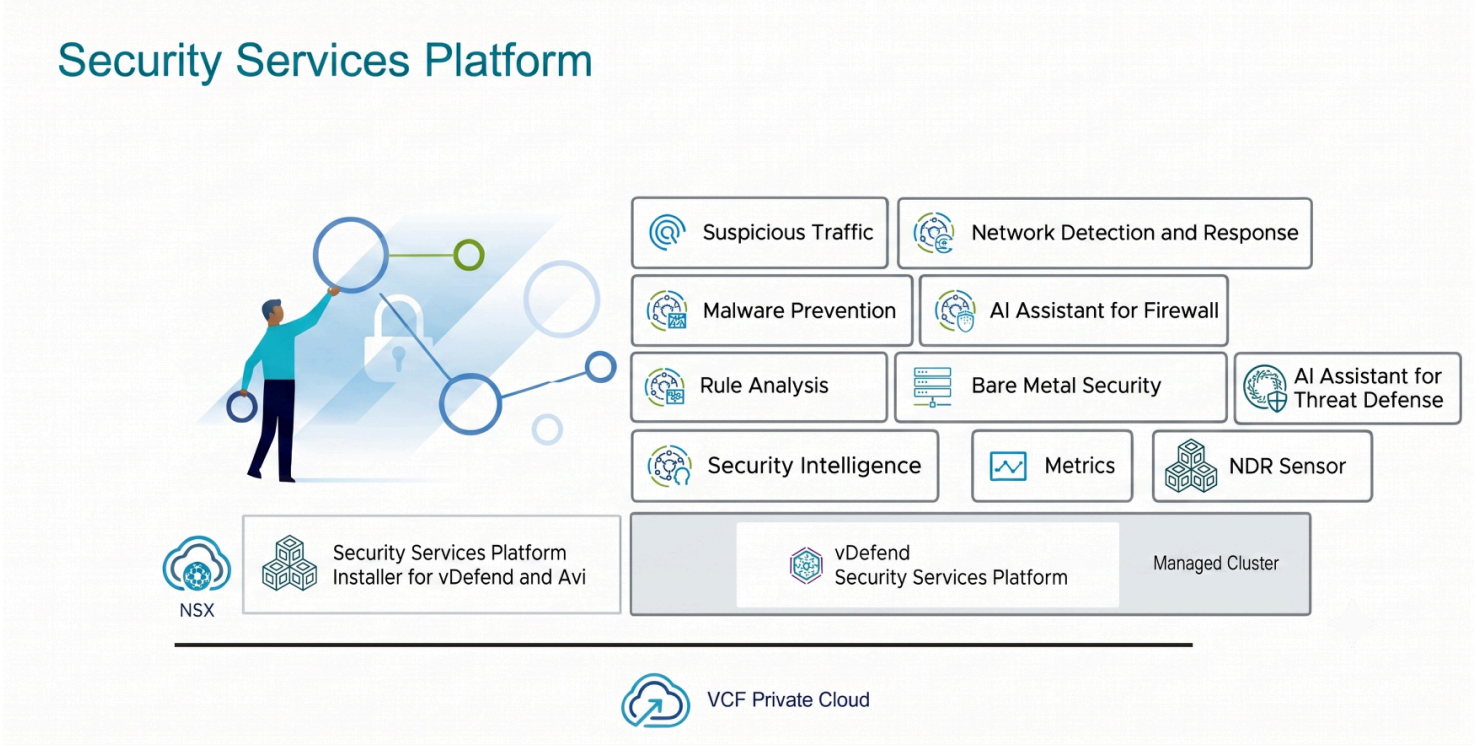
The Security Services Platform (SSP) for VMware® vDefend™ accelerates and simplifies the journey to Zero Trust security within the VCF environment. It is a high-performance and scalable platform that runs vDefend security services, including Security Intelligence, Network Detection and Response, Malware Prevention, and more.

Key Features

The Security Services Platform addresses key challenges that are posed by evolving cyber threats. The platform hosts innovations in VMware® vDefend™ not only react to threats but fundamentally change how you build, operate, and secure your enterprise private cloud.

From SSP, you can manage the following security features:

- **Security Intelligence:** Automatically visualizes network traffic flows and generates recommendations for micro-segmentation policies.
- **Rule Analysis(RA):** Identifies overlapping, redundant, or unused firewall rules to optimize policy enforcement and performance.
- **Bare Metal Security(BMS):** Extends vDefend security policies and visibility to non-virtualized physical workloads.
- **Network Traffic Analysis (NTA):** Monitors and analyzes network flow data to detect suspicious lateral movement.
- **Malware Prevention Service (MPS):** Uses sandboxing and behavioral analysis to inspect files and block malicious payloads.
- **Network Detection and Response (NDR):** Correlates multi-stage network threat signals to provide unified detection, investigation, and response.
- **AI Assistant For VMware vDefend:** Provides generative AI-driven troubleshooting, natural language policy queries, and threat analysis.
- **Metrics:** Displays real-time operational data and system health analytics across all active vDefend security services.


Security Services Platform
Suspicious Traffic
Network Detection and Response
Malware Prevention
AI Assistant for Firewall
Rule Analysis
Bare Metal Security
AI Assistant for Threat Defense
Security Intelligence
Metrics
NDR Sensor
NSX
Security Services Platform Installer for vDefend and Avi
vDefend Security Services Platform
Managed Cluster
VCF Private Cloud

vDefend SSP Architecture

vDefend Security Services Platform Concepts

SSP Installer - SSP Installer is a VM which deploys and manages theSecurity Services PlatformInstance andAvi OperationsInstance. TheSSP Installerhosts packages needed for theSecurity Services Platform instance lifecycle management, which includes deployment and upgrade. The installer features a browser-based user interface for managing the environment.

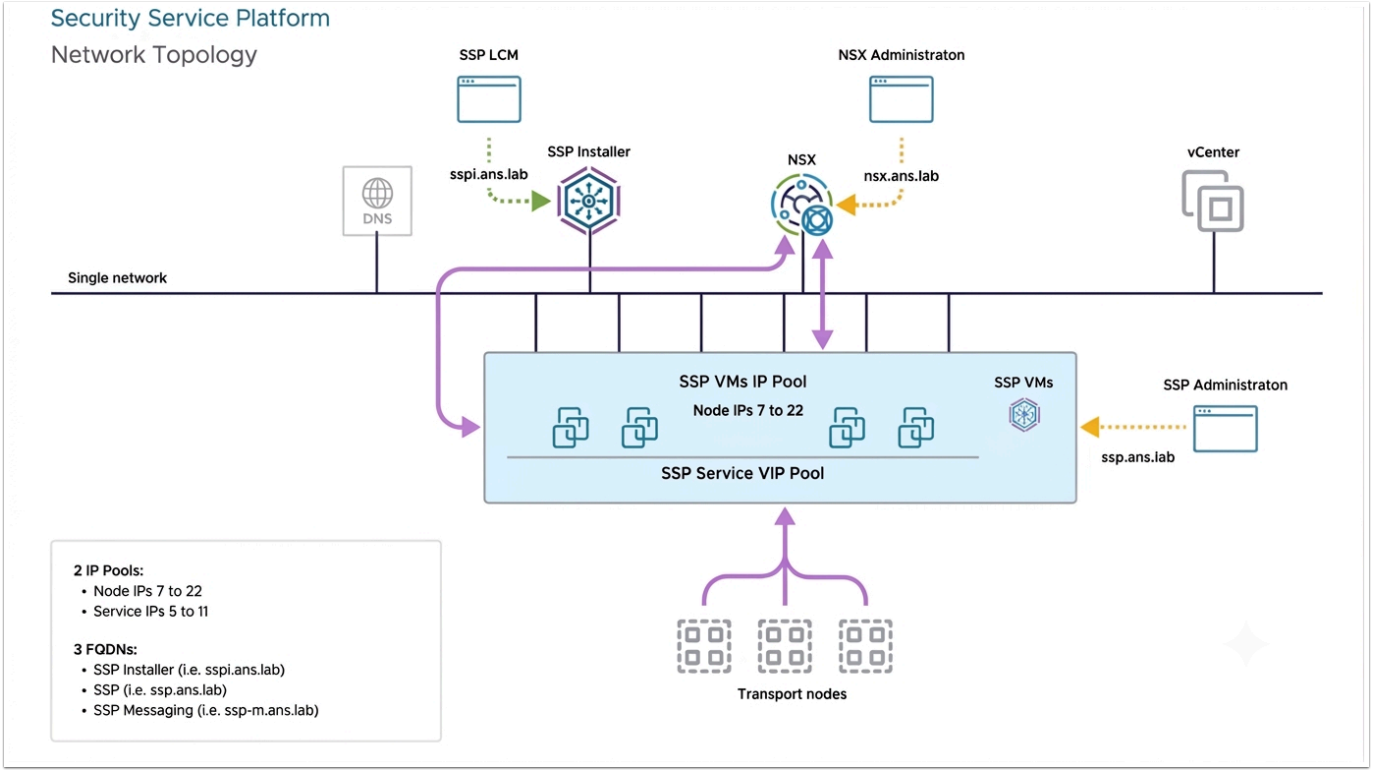
SSP Instance - Security Services Platform is a managed cluster that hosts various core security features likeSecurity Intelligence,Network Detection and Response,Malware Prevention Service, and so on. Multiple instances of the platform can be deployed and managed by the SSP Installer.

Security Feature - A security feature is a network security tool or technology, such as Security Intelligence, Network Detection and Response, Network Traffic Analysis, and so on.

Avi Operations - Avi Operations instance integrates with the VMware Avi Load Balancer Controller and enables AI Assistant capabilities for Avi, facilitating enhanced operational capabilities using generative AI.

Network Topologies

Security Services Platform can work in a single-network environment (for example, a single Virtual Private Cloud (VPC), Local Area Network (LAN), dedicated VLAN, or air-gapped environment) or in a routed-network environment.

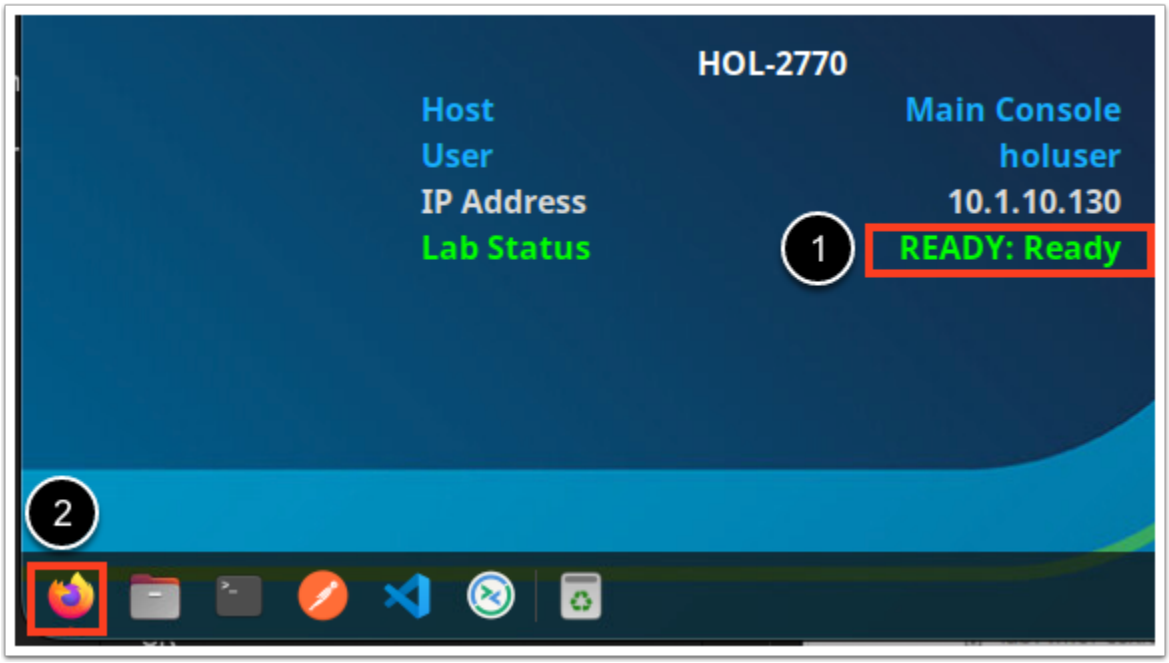


- In a single-network environment, all components reside on the same network. There is direct connectivity across all components, such as vCenter,NSX Manager, transport nodes,SSP Installer, and the SSP instance. The IP addresses of all the components are on the same network.
- Security Services Platform can also be deployed where SSP Installer is on a management network and the SSP instance is on a different network (for example, the network where the transport nodes are on). Both networks are routable.

vDefend SSP Installer Hands On

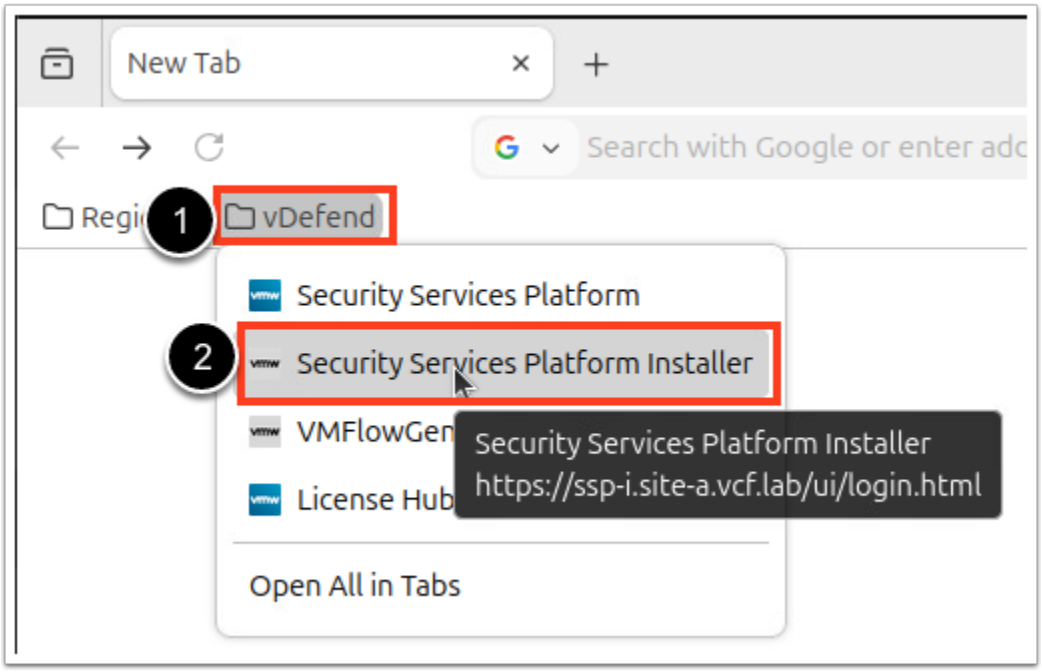
Let's get some hands-on exposure and experience with the vDefend Security Services Platform Installer.

Accessing Firefox



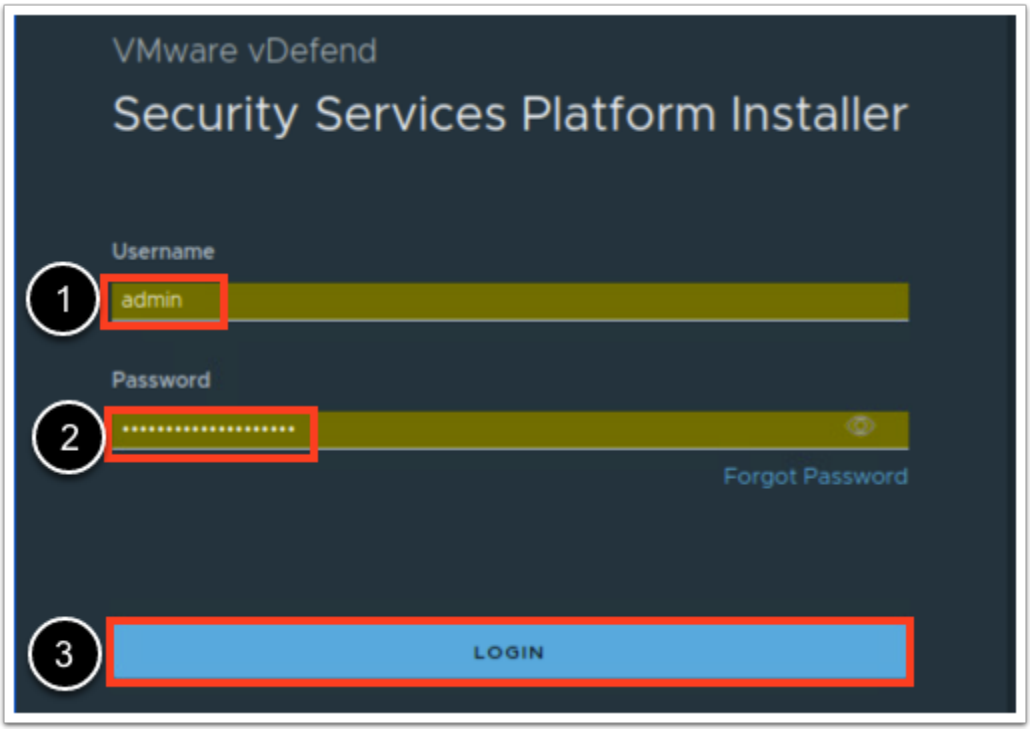
1. Confirm the Lab Status shows **Ready**
2. Click on the Firefox icon to launch the browser

Accessing Security Services Platform Installer



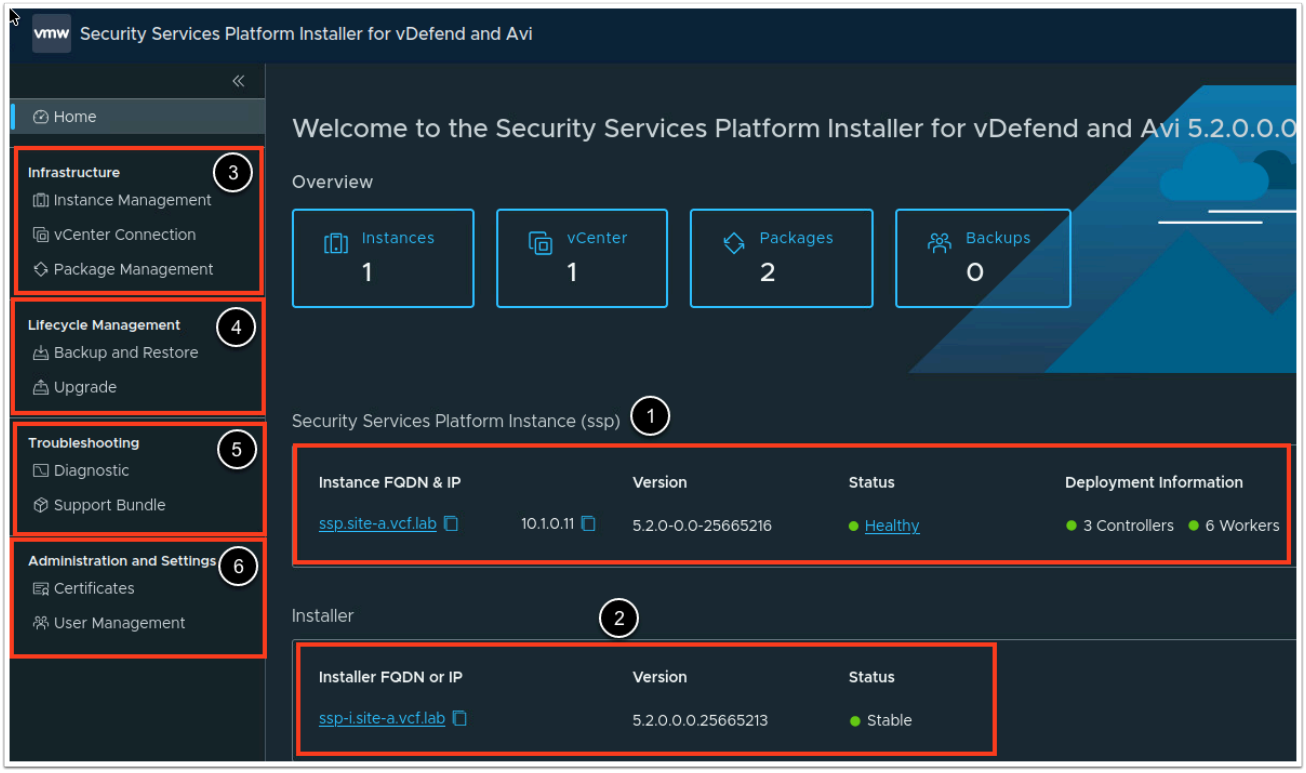
1. Click the **vDefend** Folder on the browser
2. Click on the **Security Services Platform Installer** bookmark.

Accessing Security Services Platform, Log In



1. Username **admin** should be pre-populated
2. Password **VMware123!VMware123!** should be pre-populated
3. Click **LOGIN**

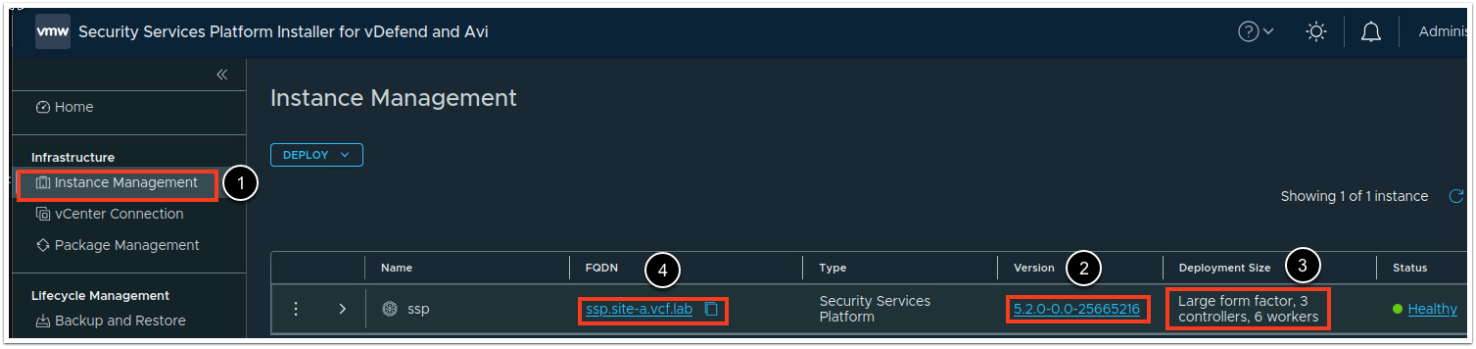
vDefend Security Services Platform Installer UI Overview



In this lab we have pre-deployed the Security Services Platform (SSP) Installer and created a SSP instance.

1. **Security Services Platform Instance (ssp):** Displays the primary operational status, network details, version, and Deployment Information (controllers & workers) of the platform instance.
2. **Installer:** Shows the network address, running software version, and current status of the platform installer utility.
3. **Infrastructure:** Configures core integrations and connections, including cluster instances, vCenter environments, and software packages.
4. **Lifecycle Management:** Handles routine platform maintenance tasks, including system backups, restores, and version upgrades.
5. **Troubleshooting:** Provides system diagnostics (Health chart & Components Grid) and support bundle generation capability.
6. **Administration and Settings:** Manages platform security parameters, including SSL/TLS certificates and user access controls.

vDefend Security Services Platform - Instance Management

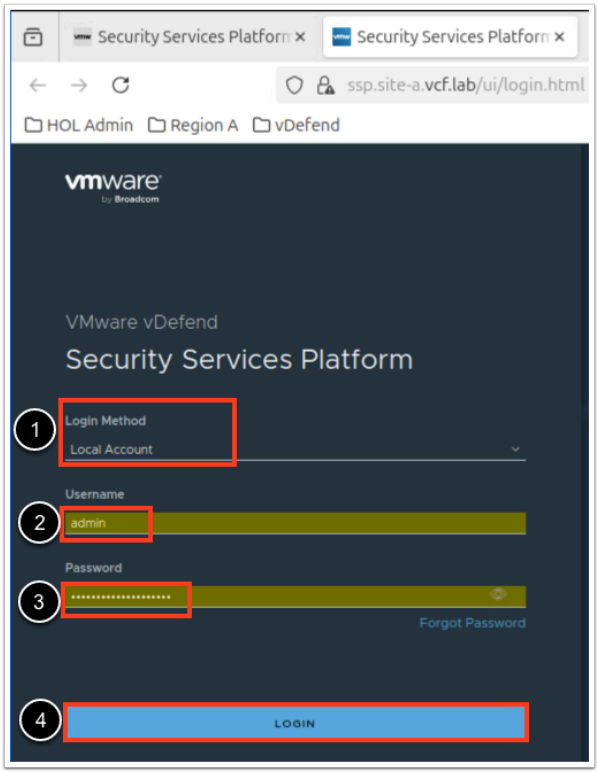


1. Click **Instance Management**
2. **Version:** This indicates the version of the SSP instance deployed.
3. **Deployment Size:** Details the form factor and Deployment Information (number of controllers & workers)
4. **FQDN:** Click on the FQDN **ssp.site-a.vcf.lab** to access the pre-deployed SSP instance.

vDefend SSP Instance Hands-On

Let's get some hands-on exposure and experience with vDefend Security Services Platform features.

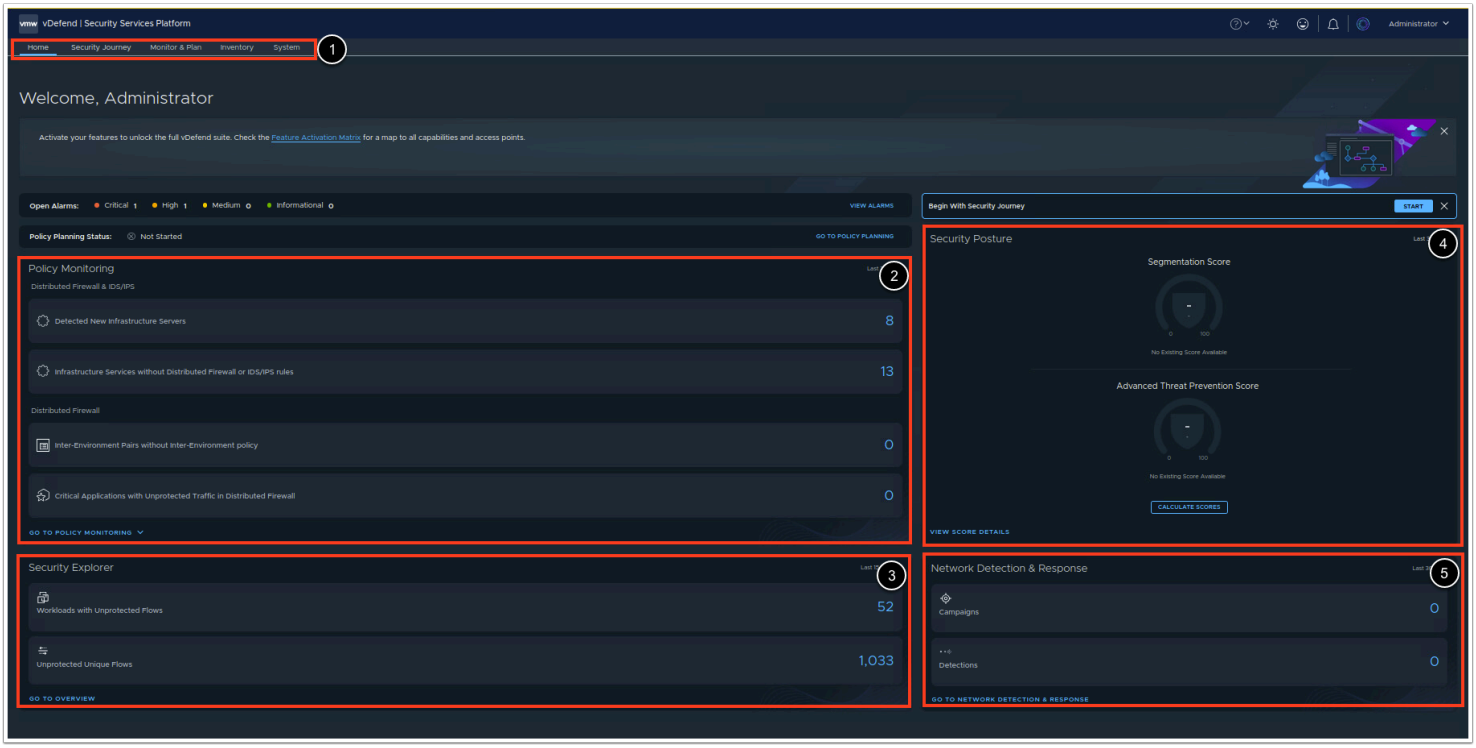
Accessing the vDefend Security Platform Instance



If you skipped the SSP Installer overview, you will need to open the Instance view from the vDefend folder. Else you should be take directly to the SSP instance without having to login and proceed to the next step.

1. **Login Method:** This should already be set to **Local Account**
2. **Username:** should be pre-populated with **admin**
3. **Password:** should be pre-populated with **VMware123!VMware123!**
4. **Click LOGIN**

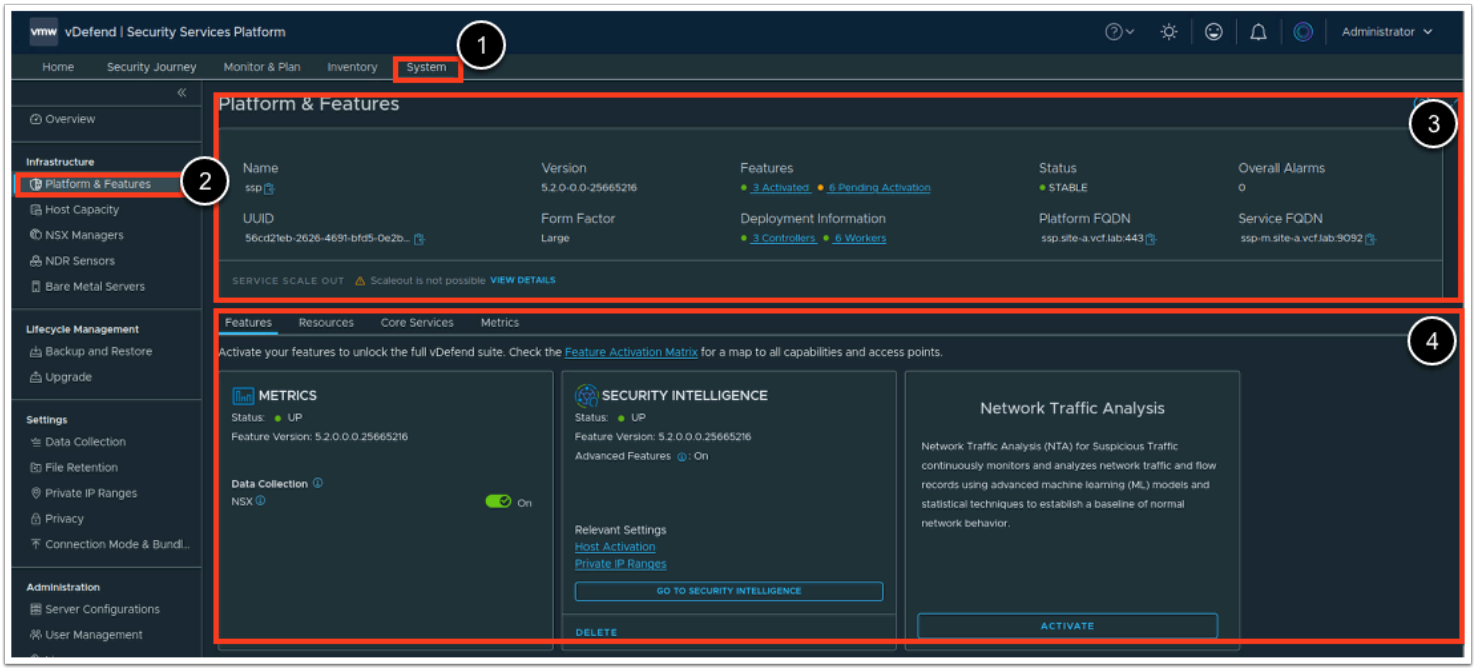
vDefend Security Services Platform Instance - Home Page



The Security Services Platform Instance landing page provides an overview of various features

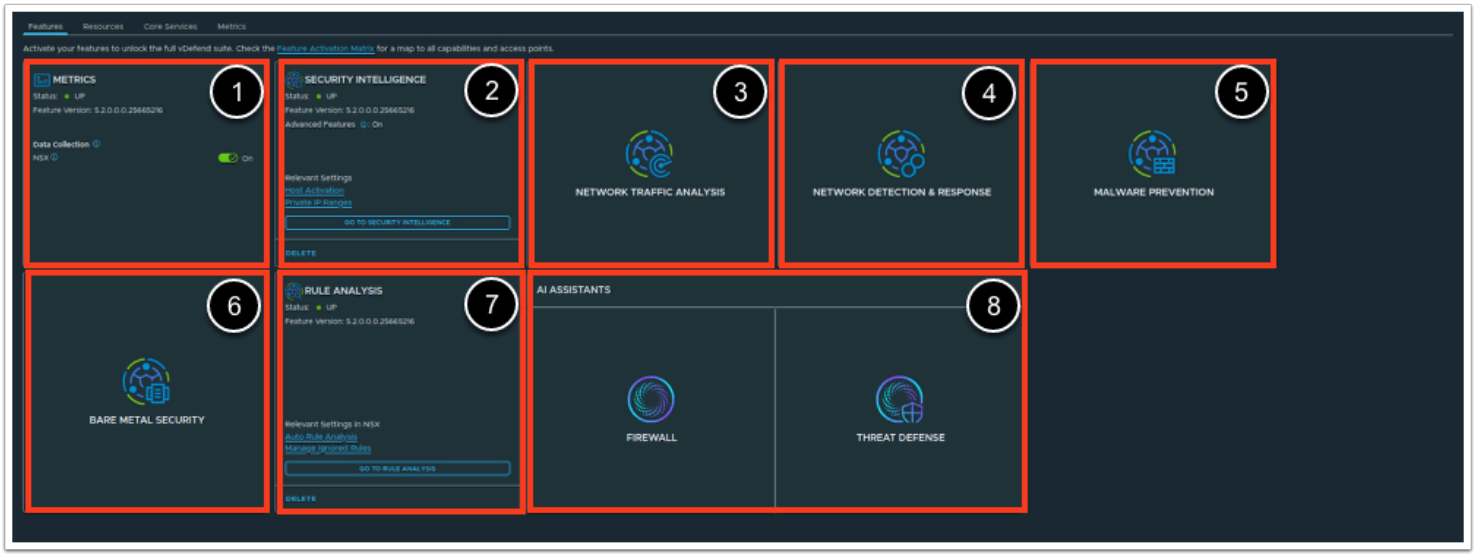
1. The **Menu bar** allows you to switch between various platform functionalities.
2. **Policy Monitoring** highlights critical applications with unprotected flows in Distributed Firewall and IDS/IPS.
3. **Security Explore** provides a summary of the detected Flows.
4. **Security Posture** allows to calculate segmentation score & Advanced Threat Prevention Score.
5. **Network Detection & Response** highlights campaigns & Detections.

Security Services Platform Instance - Platform and Features



1. Click **System**
2. Click **Platform & Features**
3. **Platform & Features:** Here we find the Platform instance details and summary - similar to what we saw in the Installer.
4. **Features:** Here is where we activate the features which will run on the Security Services Platform Instance. You will notice that Metrics and Security Intelligence have already been activated.

Security Services Platform Instance - Features



1. **Metrics** will be enabled by default when deploying the SSP instance.
2. **Security Intelligence** was pre-activated and from here you can access the feature dashboard or configure settings.
3. **Network Traffic Analysis** has not been activated yet, this will be performed later in the lab.
4. **Network Detection & Response** has not been activated yet, this will be performed later in the lab.
5. **Malware Prevention** has not been activated yet, this will be performed later in the lab.
6. **Bare Metal Security** has not been activated yet, this will be performed later in the lab.
7. **Rule Analysis** was pre-activated and from here you can access the feature dashboard or configure settings.
8. **AI Assistants** has not been activated yet.

Security Services Platform Instance - NSX Manager

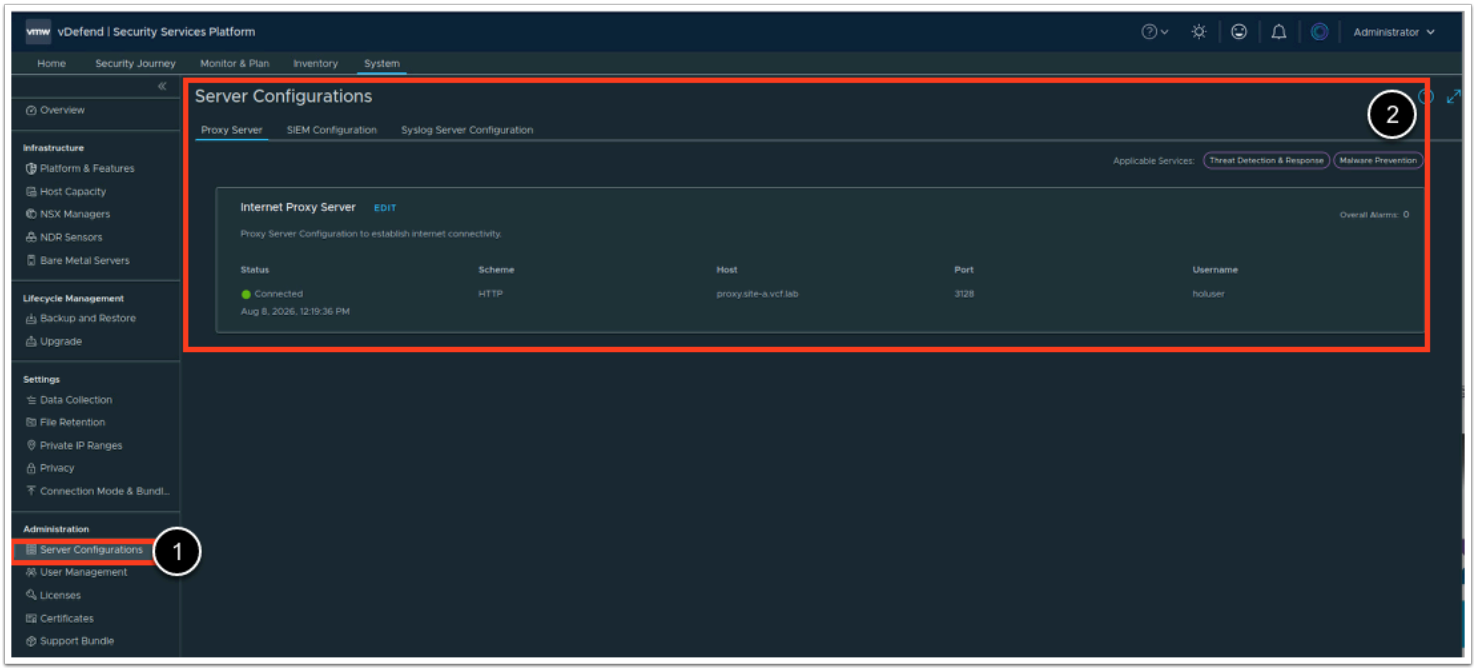


In the current version of SSP 5.2 , **Each SSP Instance** maintains a strict **1:1 relationship** with a single NSX Manager cluster.

In the current version of SSP 5.2 , **Single SSP Installer appliance** can manage **maximum 5 SSP instances**. You do not need to deploy a new installer VM every time you add another environment.

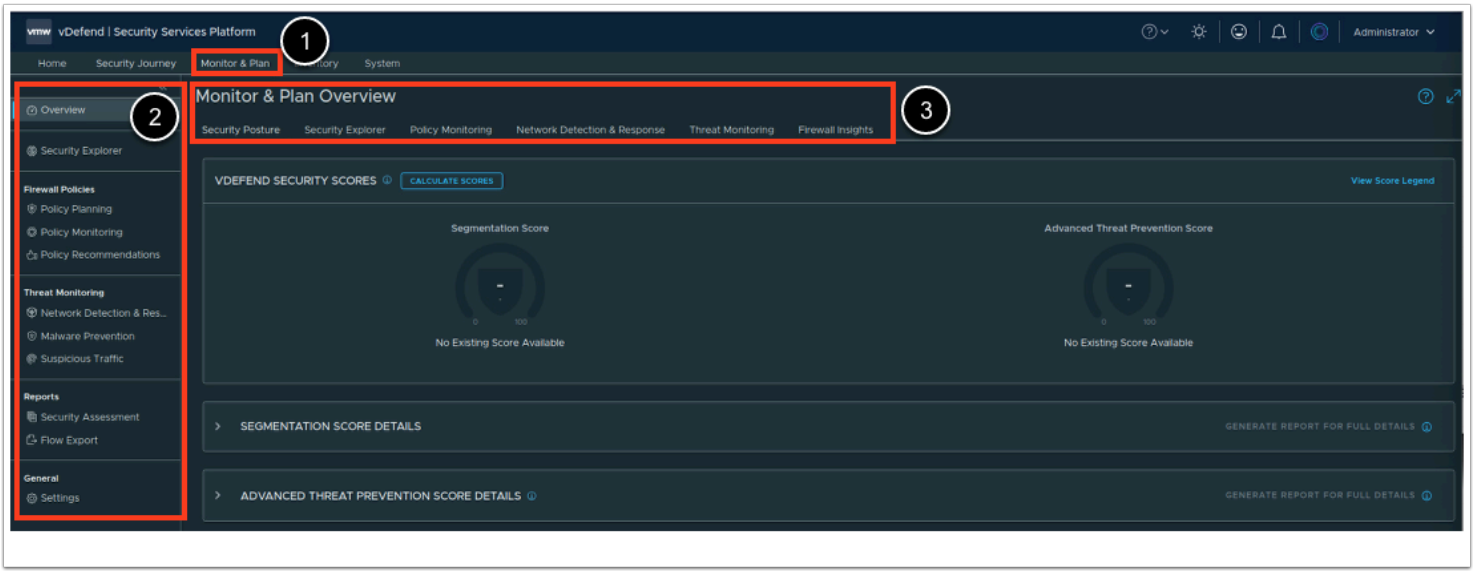
1. Click **NSX Managers**.
2. Here we see the NSX Manager that has been onboarded to this SSP instance.
3. Here we can validate the connectivity and status with the NSX Manager.

Security Services Platform Instance - Server Configurations



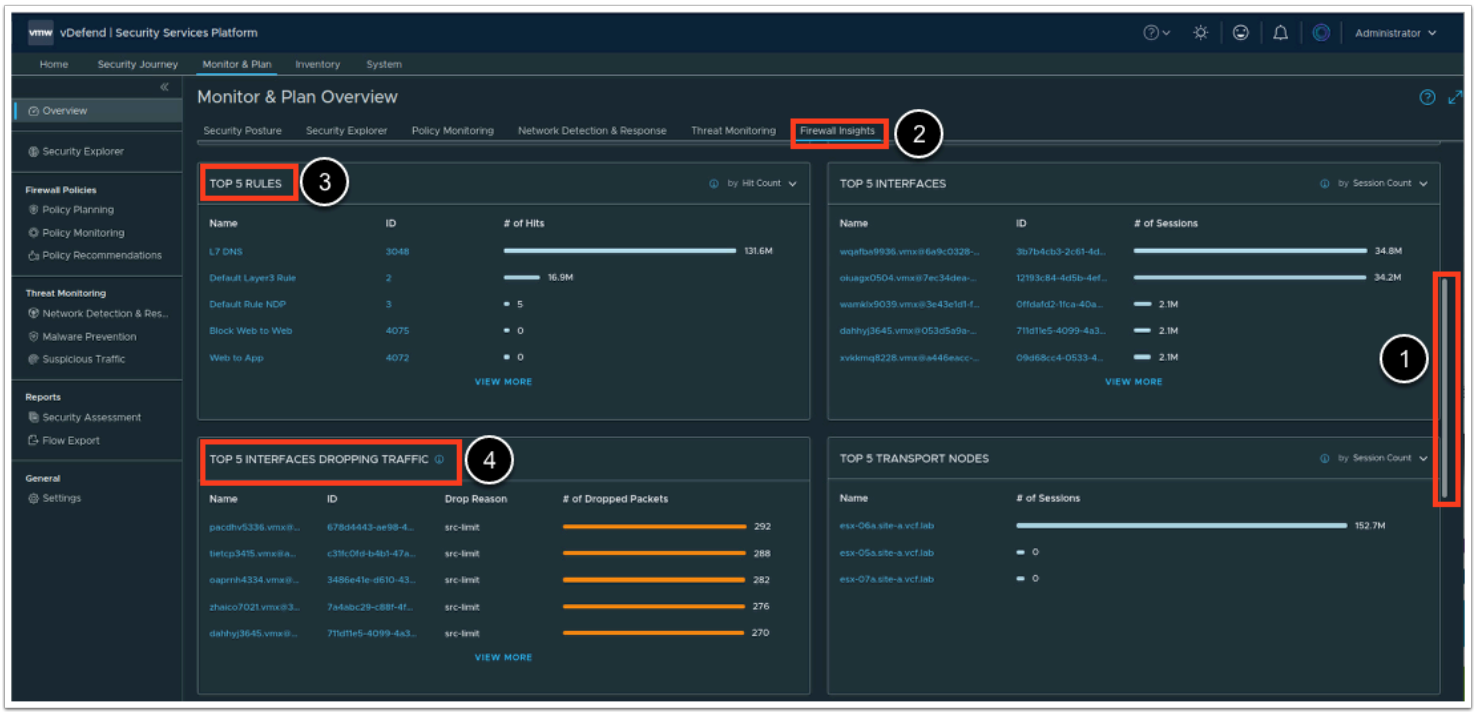
1. Click **Server Configurations.**
2. The SSP Platform supports configuring a Proxy Server, SIEM Server and Syslog Server Configurations.

Security Services Platform Instance - Monitor & Plan



1. Click **Monitor & Plan.**
2. Here are options to review features and capabilities related to Security Intelligence.
3. Here are monitoring tabs for various features.

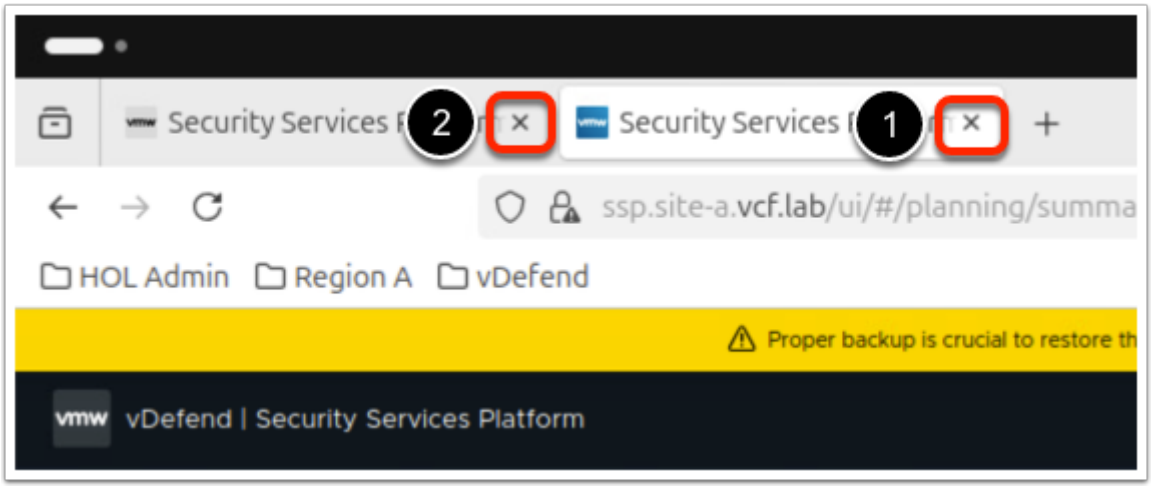
Security Services Platform Instance - Monitor & Plan, Firewall Insights



1. Click **Firewall Insights**
2. **Click and hold** the scroll bar on the right and page down to see all the feature tiles.
3. **Top 5 Rules**; Here you see the Top 5 Firewall rules by hit count.
4. **Top 5 Interfaces**; Here you see the Top 5 VM interfaces by session count.

This module provides a brief overview of these Security Services Platfrom instance dashboard, you may want click around and get familiar with other capabilities or features or continue with the next steps

Closing the browser tabs



1. Click the **x** to close the SSP Instance page.
2. Click the **x** to close the SSP Installer page.

vDefend SSP Summary

In this module, you explored the foundational concepts of the **vDefend Security Services Platform (SSP)** and gained hands-on experience navigating its deployment

Proceed to **Module 2** to learn about vDefend Intrusion Detection/ Prevention System (IDS/IPS)

Module 2 - vDefend Intrusion Detection/ Prevention System (IDS/IPS) (25min) (Intermediate)

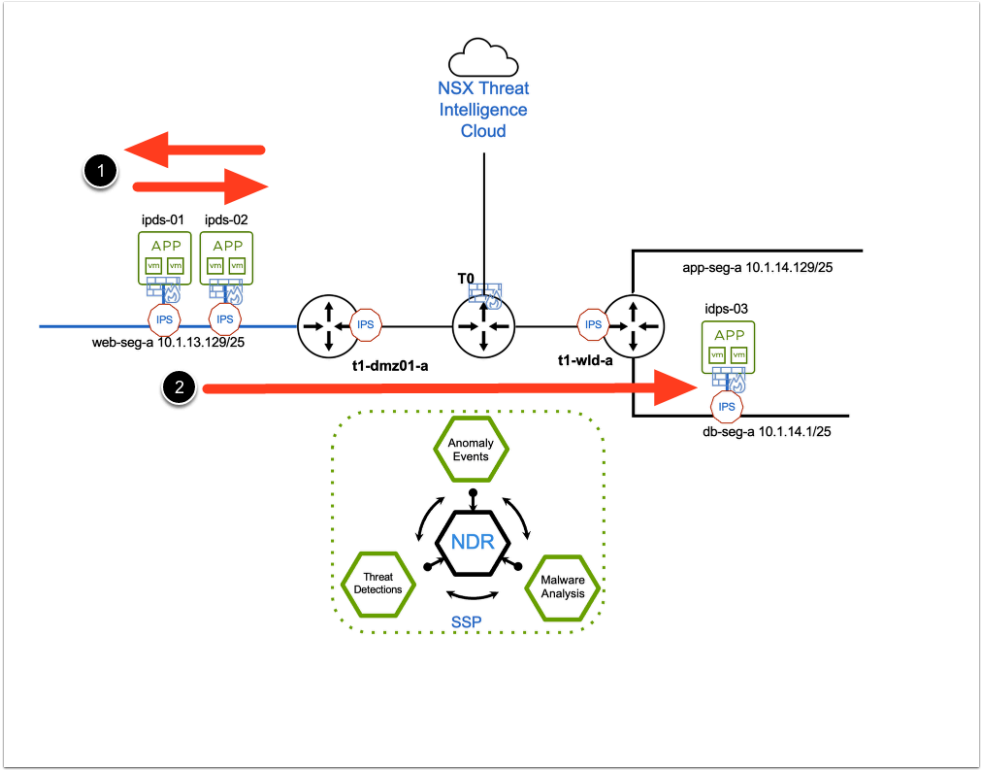
Module 2 - vDefend Intrusion Detection/ Prevention System (IDS/IPS) (25min) (Intermediate)

In this module, we explore both Distributed and Gateway IDS/IPS. You will enable IDS/IPS, configure inspection policies, and generate events, which you will monitor and review.

VMware's Distributed IDS/IPS offers a fundamentally new architecture that breaks this traditional trade-off between breadth of security coverage and operational complexity. It embraces an all-software distributed approach, moving traffic inspection out to every workload and eliminating the need to hairpin traffic to discrete appliances (i.e., moving traffic to and back from a centralized IDS/IPS capability). The operational simplicity of deploying and managing IDS/IPS functionality at each workload ensures comprehensive coverage without any blind spots.

Module 2 Topology

Let's review the topology used in this lab.



This network topology diagram illustrates two primary traffic flows analyzed by the VMware vDefend IDS/IPS security engines:

1. You will generate **lateral traffic** between **idps-01** and **idps-02,** highlighting distributed IDS/IPS on the same segment.
2. You will generate **Cross-Segment traffic** between the web-seg-a segment and db-seg-a triggering the Distributed and Gateway IDS/IPS engines.

The Tools used in this lab

We will be using 3 tools in this lab to visualize traffic flows and the changes to those flows when we apply vDefend IDS/IPS rules to the environment - VCF Operations for Logs, VMFlowGen and the NSX Manager.

VCF Operations for Logs

VCF Operations for Logs acts as a central repository for logs from different sources within the VMware Cloud Foundation environment, including vSphere, NSX, and other components. It supports content packs—plugins that provide pre-built dashboards, extracted fields, and alerts for specific products or log types such as NSX.

VMFlowGen

VMFlowGen is an internally developed tool used to demonstrate Security Intelligence. It deploys light VMs and generates traffic flows between them. In this lab, VMFlowGen builds the VM and traffic flow topology to simulate a real-world environment as closely as possible.

NSX Manager

NSX Manager serves as the centralized management and control plane interface for VMware NSX security services. In this lab, it acts as the primary console used to enable host and gateway security services, configure vDefend Distributed and Gateway IDS/IPS rules, manage threat signature profiles, and monitor real-time intrusion events.

Review the VMs and Flows in VMFlowGen

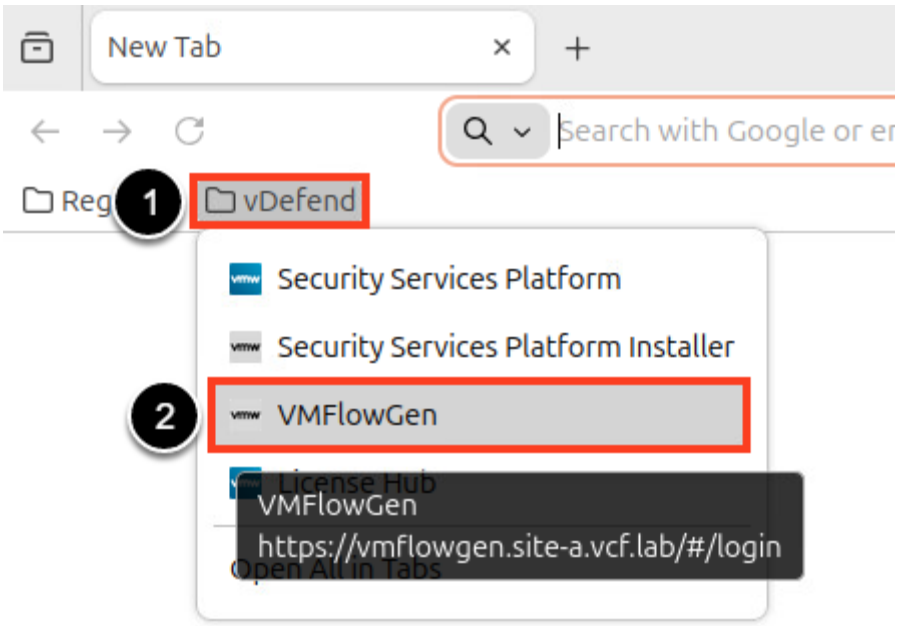
We will start by accessing FireFox.

Launch Firefox



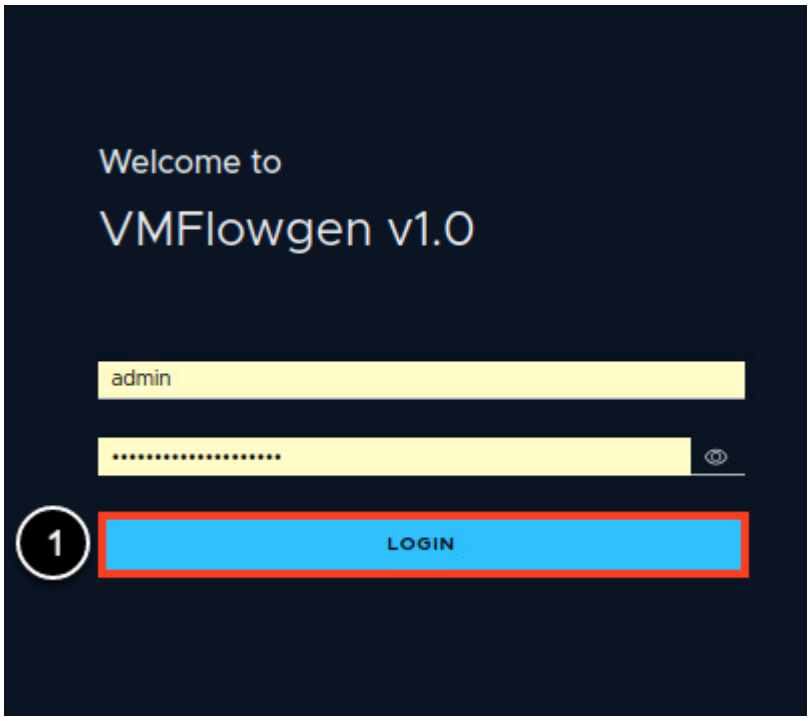
1. Launch Firefox by clicking on the Firefox icon in the lower left-hand corner of the console.

Open VMFlowGen



1. Select **vDefend** from the Bookmark Bar
2. Select **VMFlowGen** from the drop-down me**nu**

Log into VMFlowgen



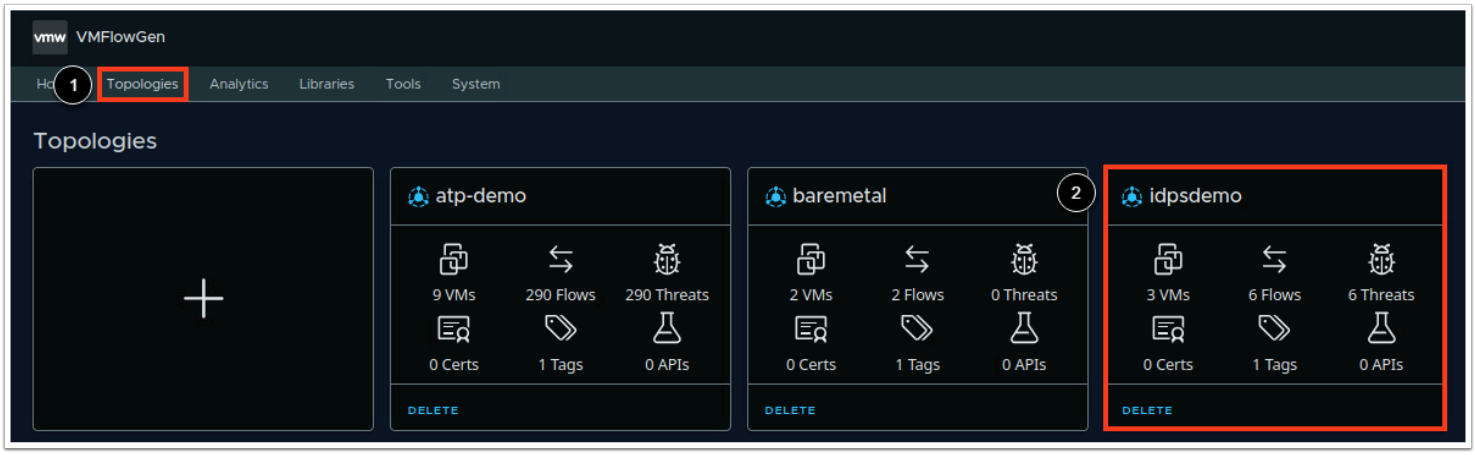
The credentials for admin **should already be cached in the browser window.**

1. Using the pre-populated user account details, select **LOGIN**

   Username: *admin*

   Password: *VMware123!VMware123!*

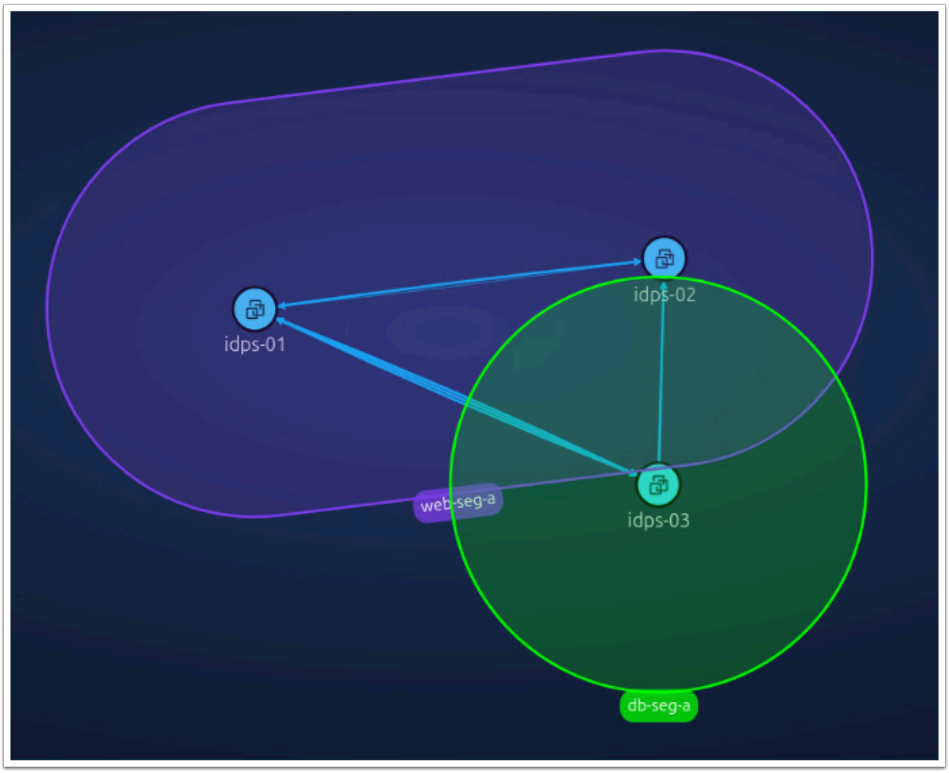
Open the idpsdemo Topology in VMFlowGen



1. Click **Topologies**
2. Click on the **idpsdemo** topology

> The application is currently not deployed as indicated by the blue color of the symbol in the top left of the application card

View Topology in VMFlowGen

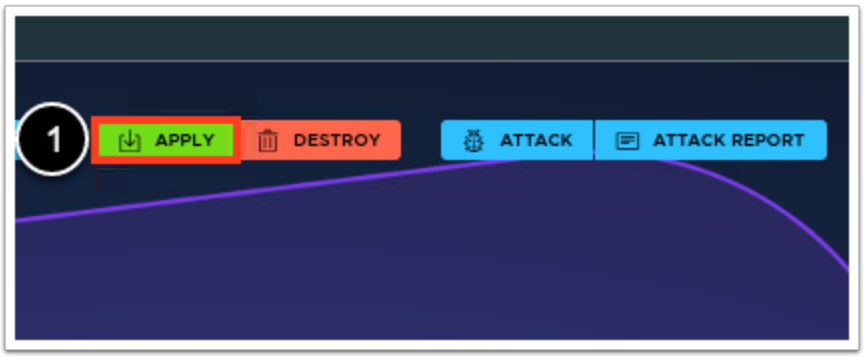


Flow Summary

1. idps-01 and idps-02 are connected to web-seg-a and will be used to demonstrate distributed idps
    1. idps-01 will connect to idps-02 on TCP 80 with a payload simulating AD Account Enumeration
    2. idps-02 will connect to idps-01 on TCP 53 with a payload simulating data exfiltration using dnscat2
2. idps-01 and idp-02 are connected to segments which are attached to t1-dmz-a and t1-wld-a
    1. idps-01 will connect to idps-03 on TCP 3389 with a payload simulating AnyDesk Remote Desktop
    2. idps-03 will connect to idps-01 on TCP 445 with a payload simulating Cobalt Strike
    3. idps-01 will connect to idps-03 on TCP 80 with a payload simulating CnC Activity
    4. idps-03 will connect to idps-02 on TCP 80 with a payload simulating .NET Framework Vulnerability (CVE-2024-29059)

When the topology has not been deployed the VMs will be a blue color.

Apply Topology in VMFlowGen



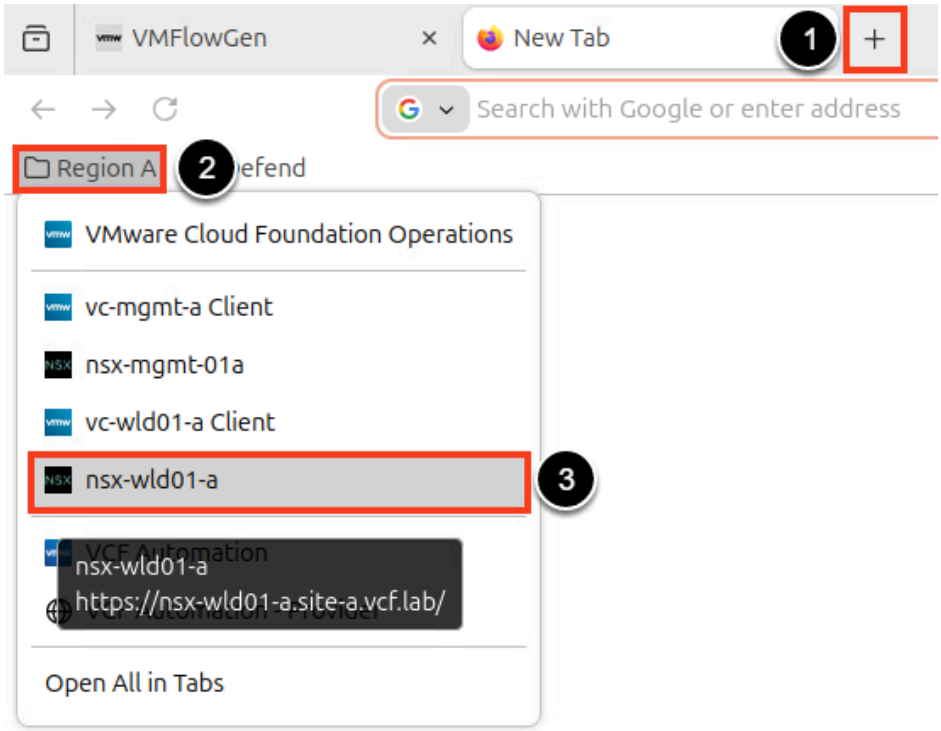
1. Click **APPLY**

> The application will take between 1 to 3 minutes to fully deploy. During this time you will notice a progress bar on the top left of the canvas. If the application fails to deploy, click APPLY again.

Access the NSX Manager

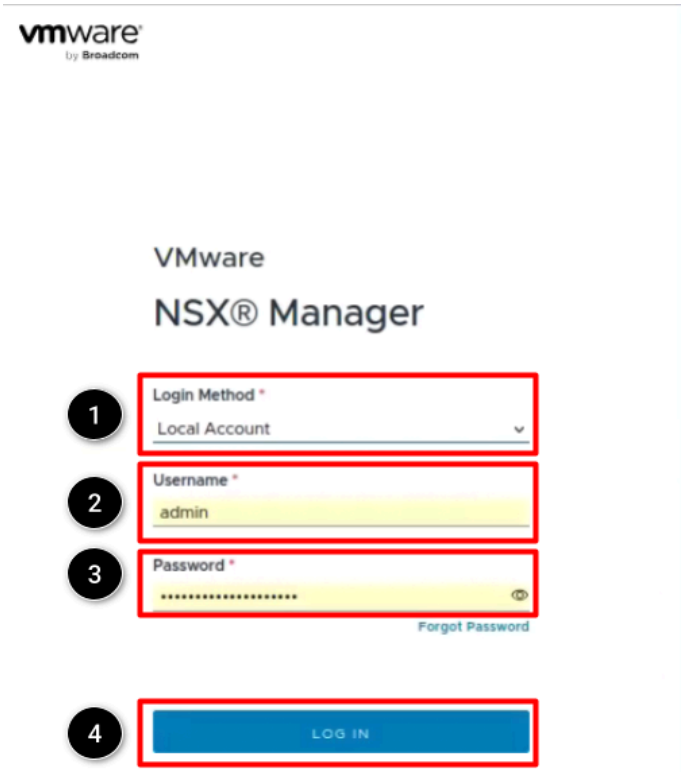
Next you need to access the NSX Manager console.

Open the NSX Manager Log on page



1. Click **+** for a new Tab
2. Click on the **Region A** folder
3. Click **nsx-wld01-a** to launch the NSX Manager login

Log In to NSX Manager



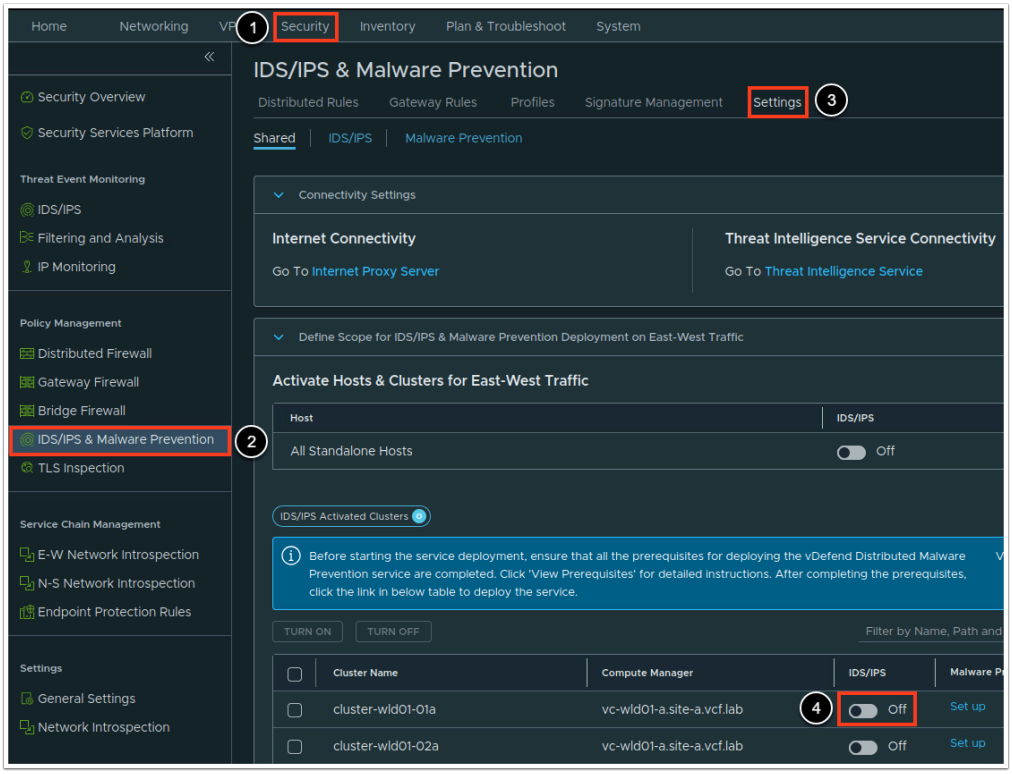
The credentials for admin **should already be cached in the browser window.**

1. Make sure the Login Method is set to **Local Account**
2. Username: admin
3. Password: **VMware123!VMware123!**
4. Select **LOG IN**

Activate Hosts & Clusters for East-West Traffic

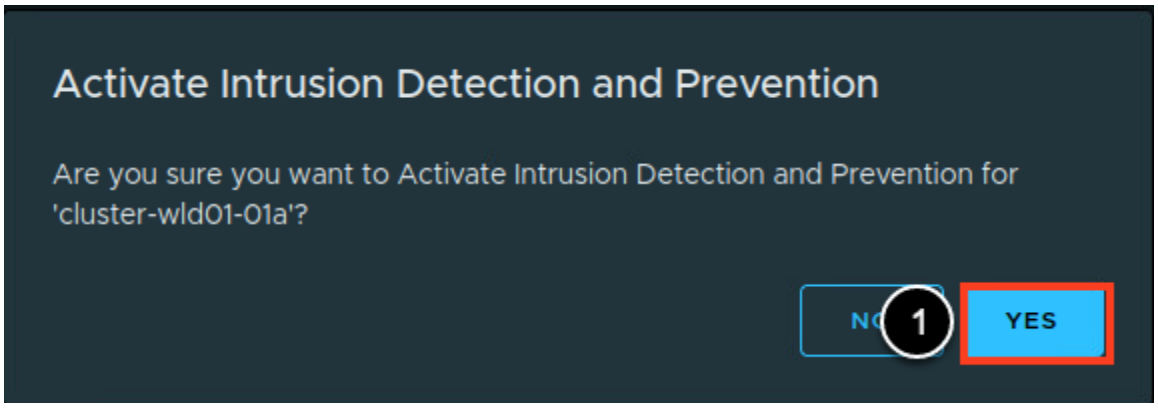
You now need to enable the distributed IDPS engine on the ESXi hosts. Note that for this lab, this action only applies to the **cluster-wld01-01a** cluster.

Activate cluster-wld01-01a



1. Click **Security**
2. Click **IDS/IPS & Malware Prevention**
3. Click **Settings**
4. Click the toggle to **On** under IDS/IPS for **cluster-wld01-01a** to enable the IDPS engine

Confirm the IDPS Activation

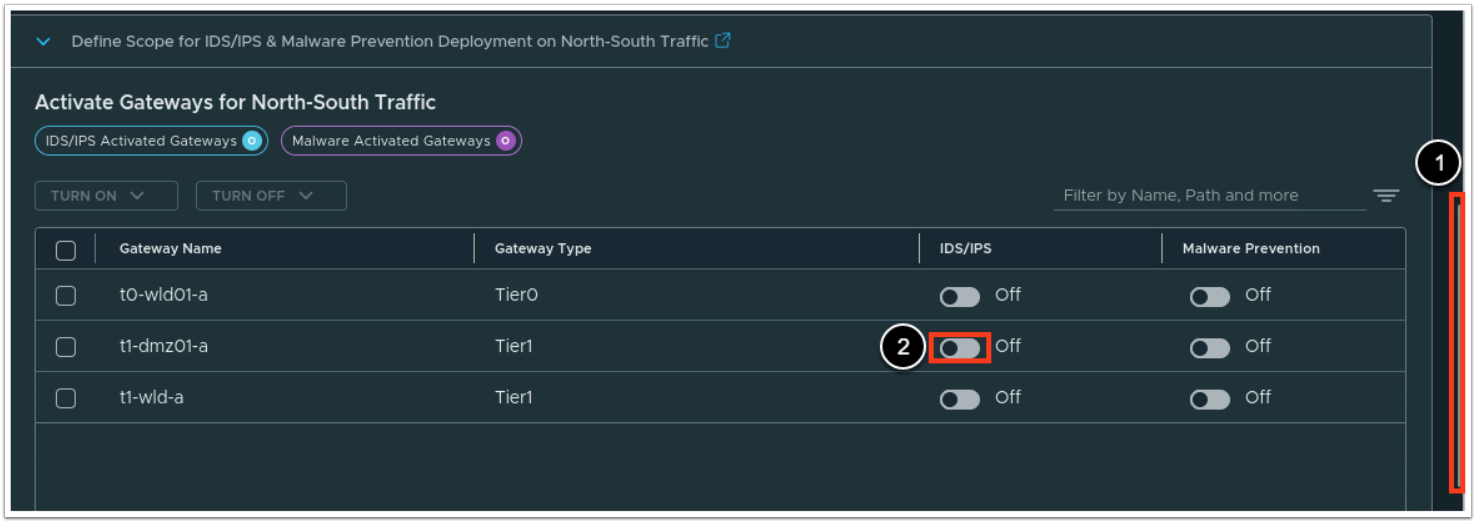


1. Click **YES**

Activate Gateway IDS/IPS for North-South Traffic

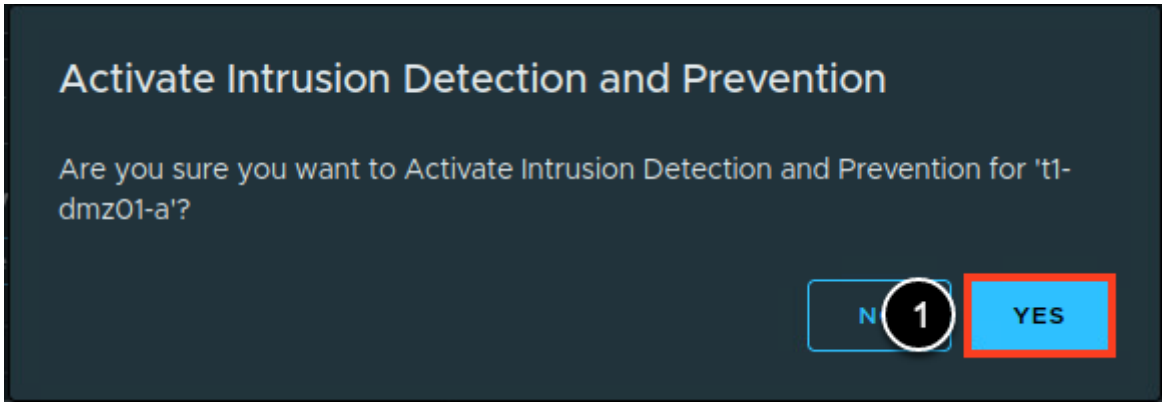
Next you need to activate Gateway IDS/IPS on the target Tier-1 gateways

Enable Gateway IDPS on t1-dmz01-a



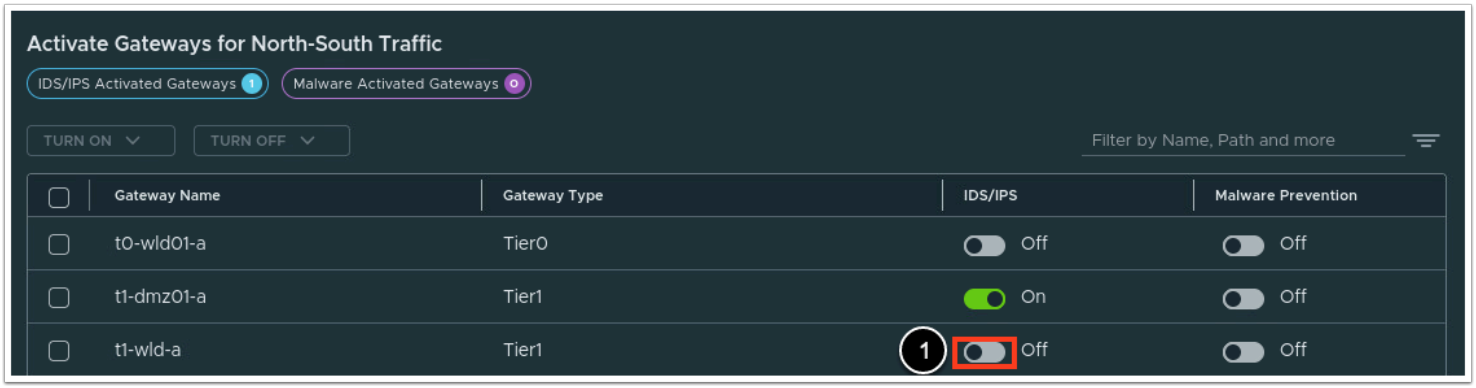
1. **Click and hold** the slide bar and page **down**
2. Click the toggle to **On** for **t1-dmz01-a**

Confirm the IDPS Activation



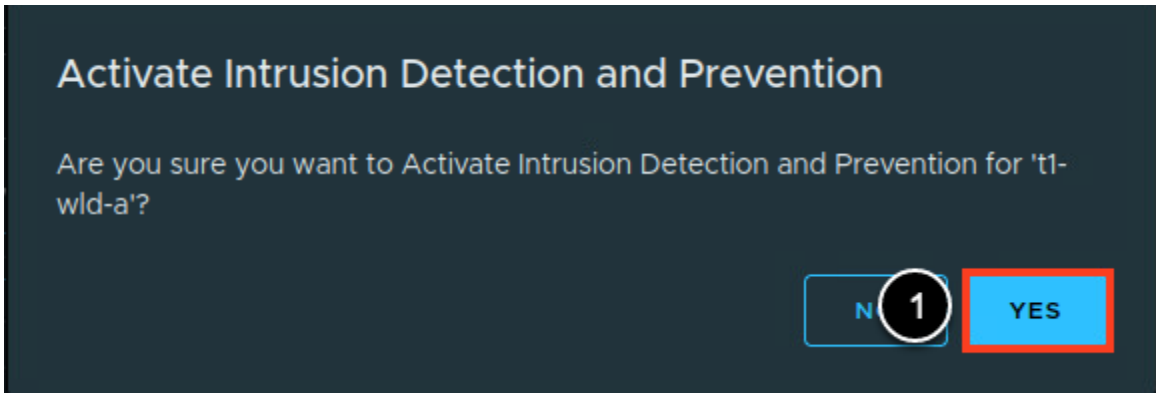
1. Click **YES**

Enable Gateway IDPS on t1-wld-a



1. Click the toggle to **On** for **t1-wld-a**

Confirm the IDPS Activation

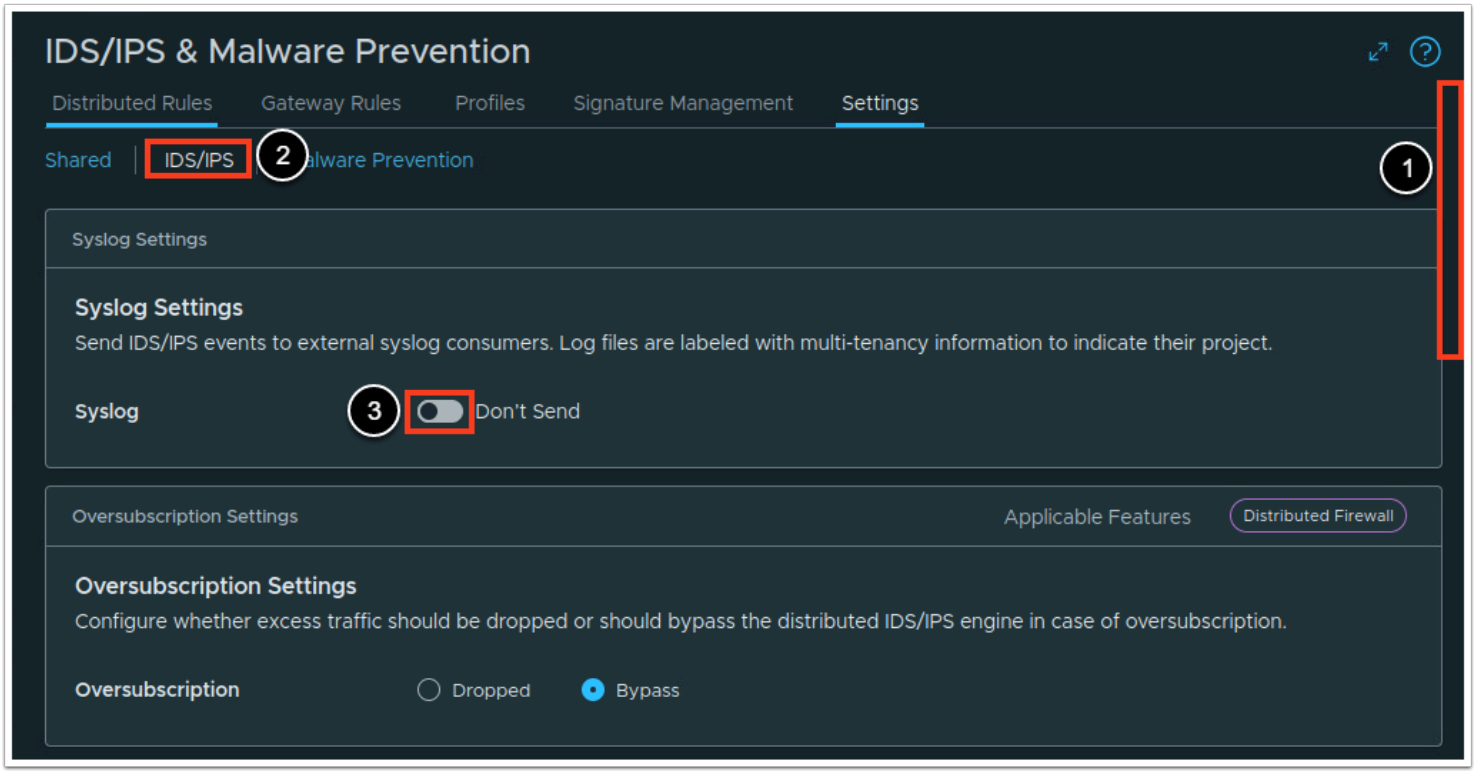


1. Click YES

Enable Syslog

To send IDS/IPS events to external syslog consumers, you enable this globally in the NSX Manager as well as per rule.

Confirm IDPS/IPS enablement

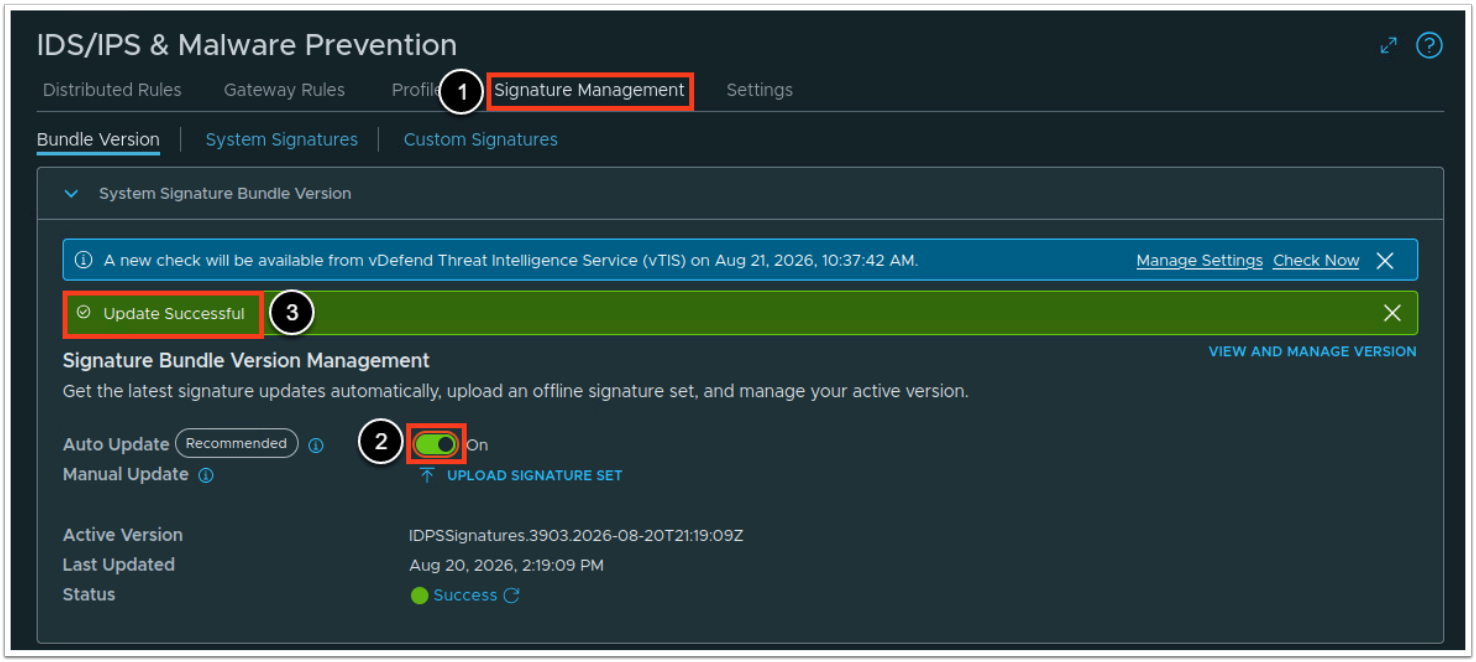


1. **Click and hold** the slide bar and page back up
2. Click **IDS/IPS**
3. If not enabled, click the toggle to enable Send **IDS/IPS events to external syslog consumers**

Signature Management

The NSX Manager supports automatic signature updates or manual signature bundle uploads. In this lab you will enable the Auto Update.

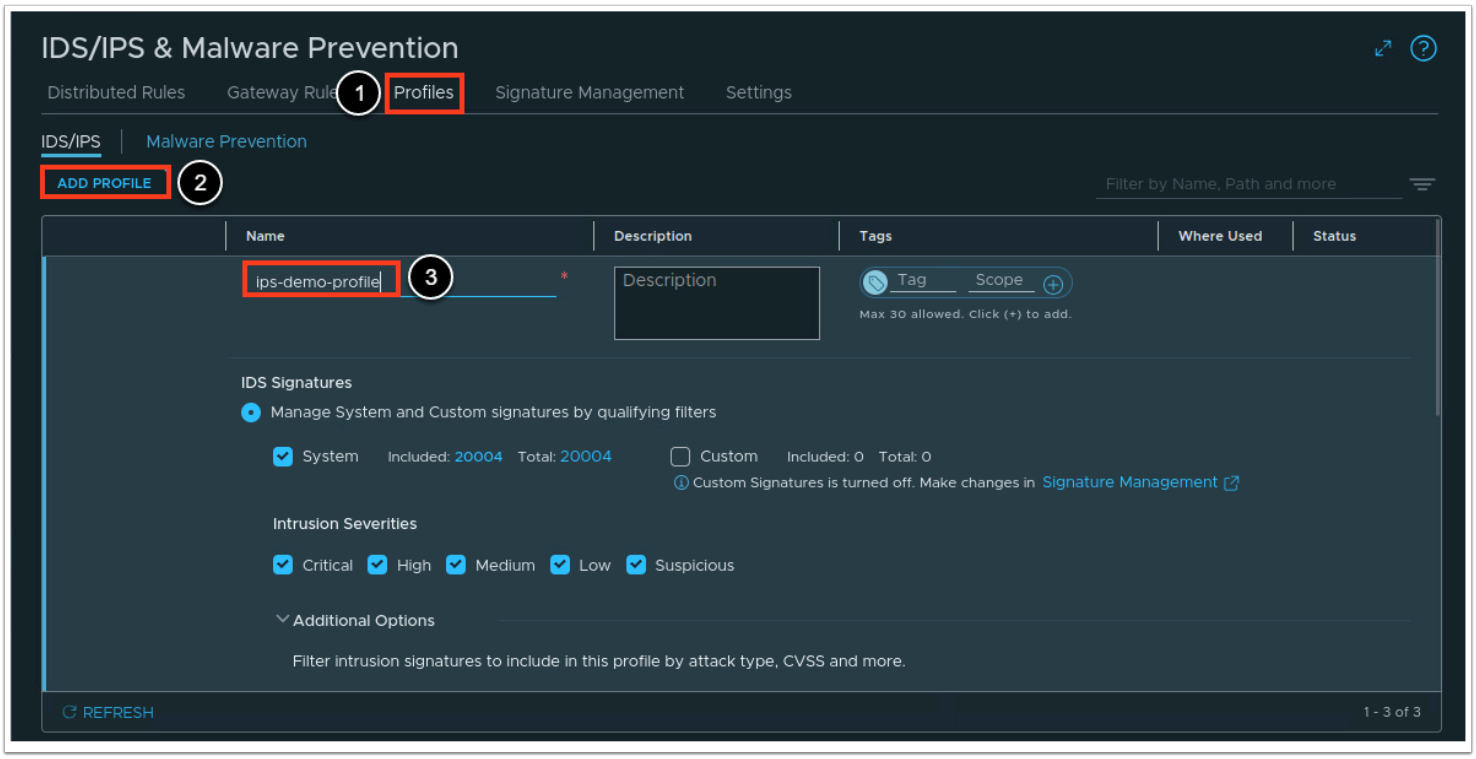
Enable IDS/IPS Signature Database Update



1. Click **Signature Management**
2. Click the toggle to enable **Auto Update**
3. Confirm IDS Signatures have updated

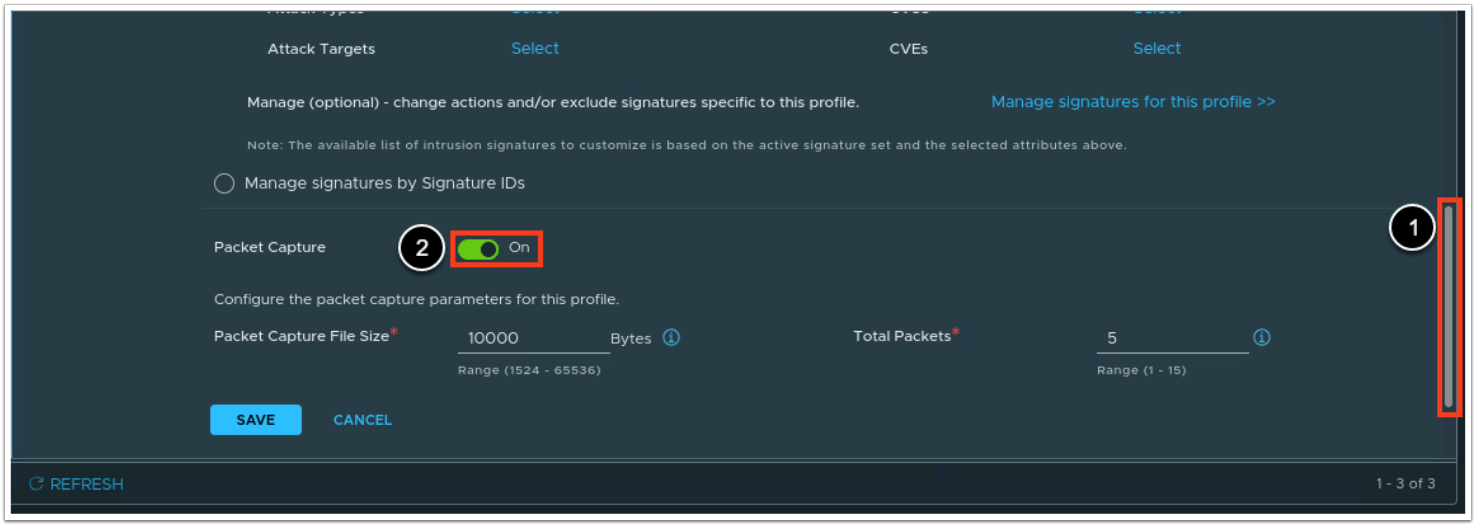
> After you enable Auto Update it will take a few seconds while it "Updates IDS Signatures" - This bar will be blue, once IDS Signatures has completed it will temporary show a green bar that reads "Update Successful"

Create a Custom Signature Profile



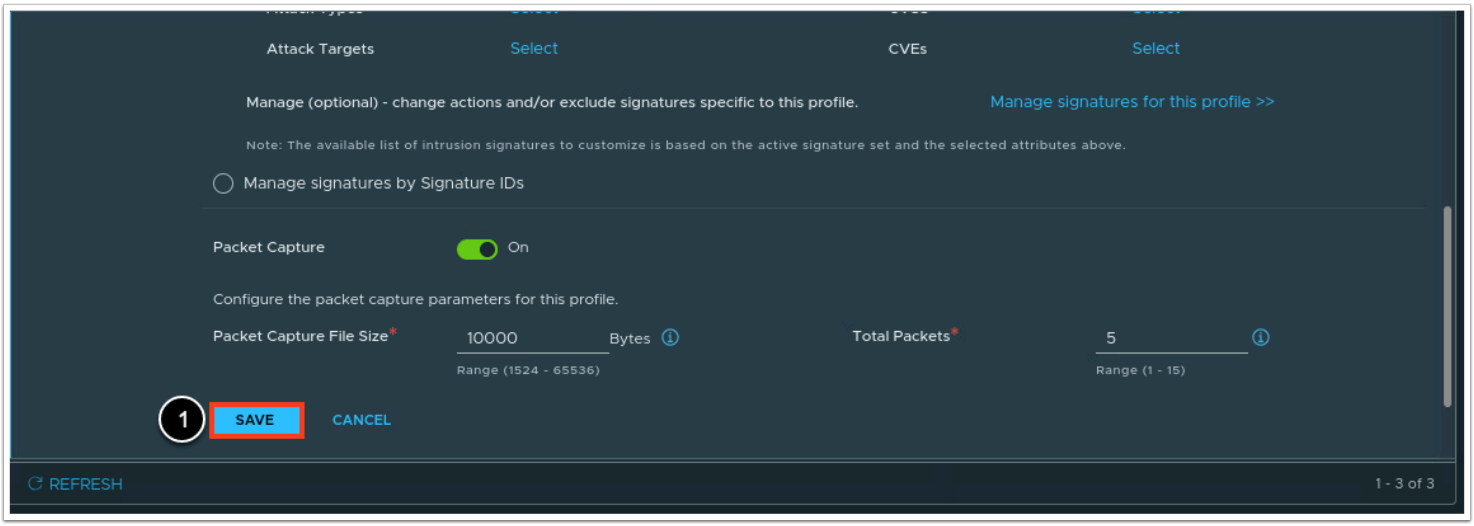
1. Click **Profiles**
2. Click **ADD PROFILE**
3. Name the profile **ips-demo-profile**

Enable Packet Capture



1. Scroll Down on the right side
2. **Enable** Packet Capture

Save The Custom Signature Profile

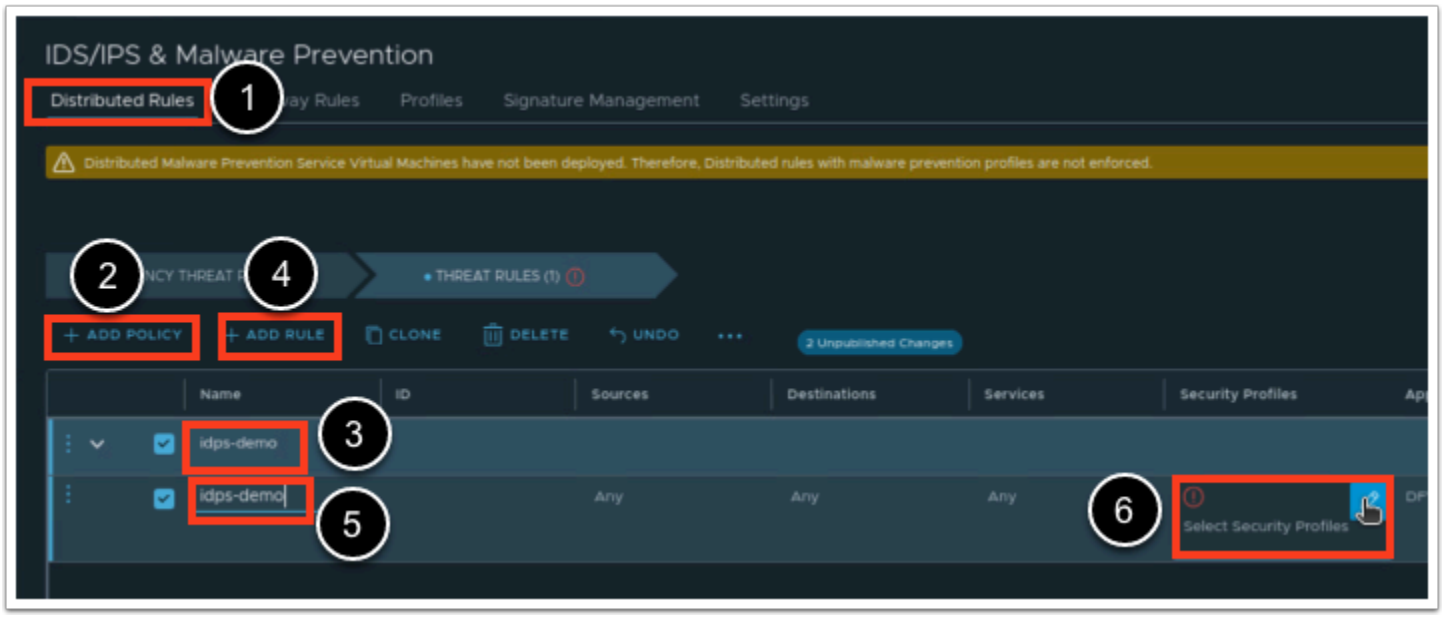


1. Click **SAVE**

Creating Rules for IDS/IPS

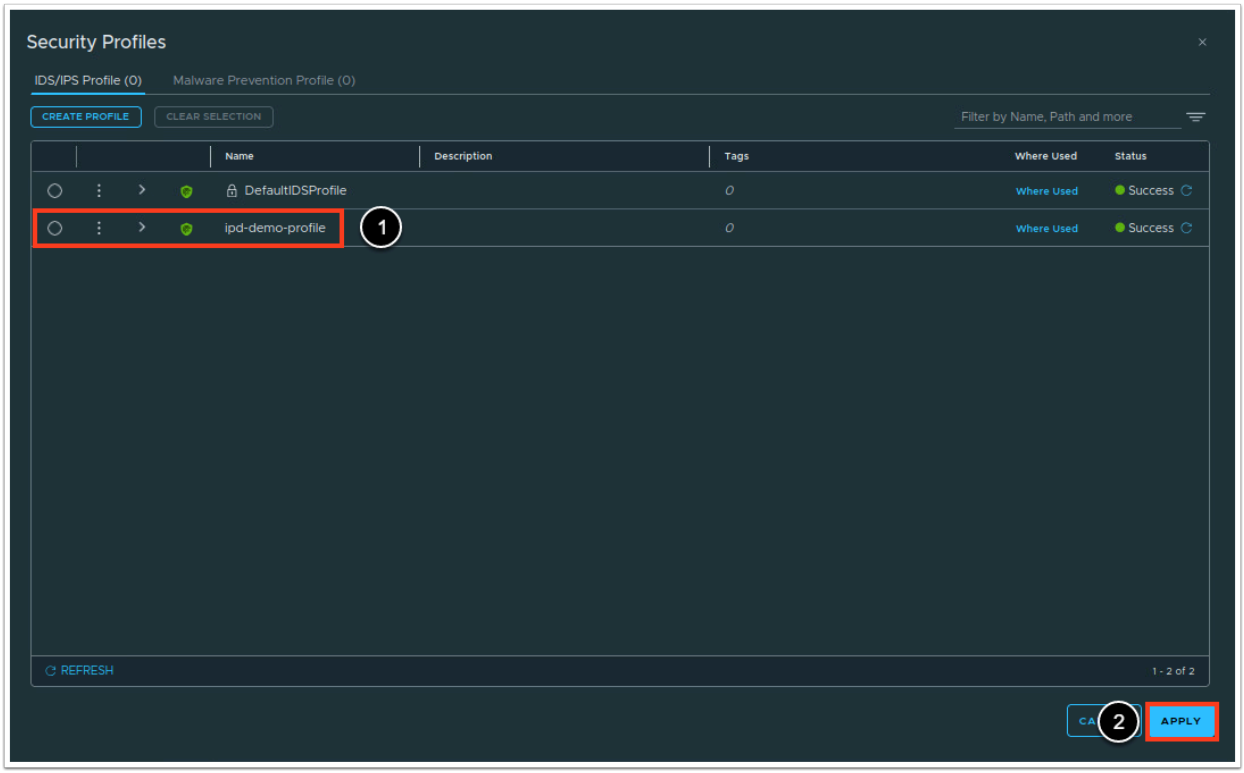
The NSX Manager UI provides a common rule table to add rules for Intrusion Detection/Prevention on a vDefend Distributed Firewall.

Creating Distributed IDS/IPS Policy



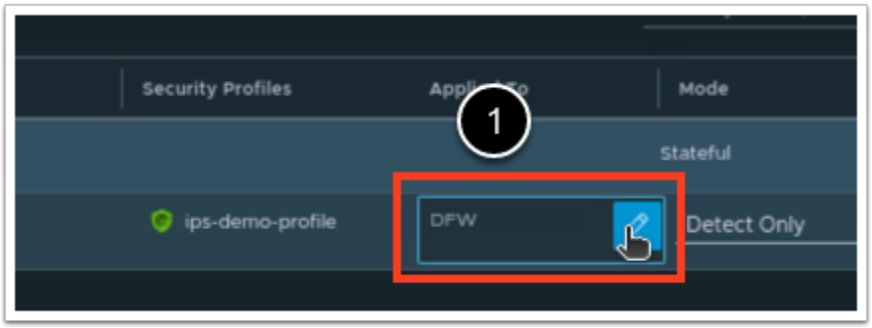
1. Click **Distributed Rules**
2. Click **ADD POLICY**
3. Rename the policy to **idps-demo**
4. Click **ADD RULE**
5. Rename the rule to **idps-demo**
6. Click on the pencil to select Security Profiles - here you will be selecting the custom profile created from the previous step **ips-demo-profile**

Selecting Custom Signature Profile



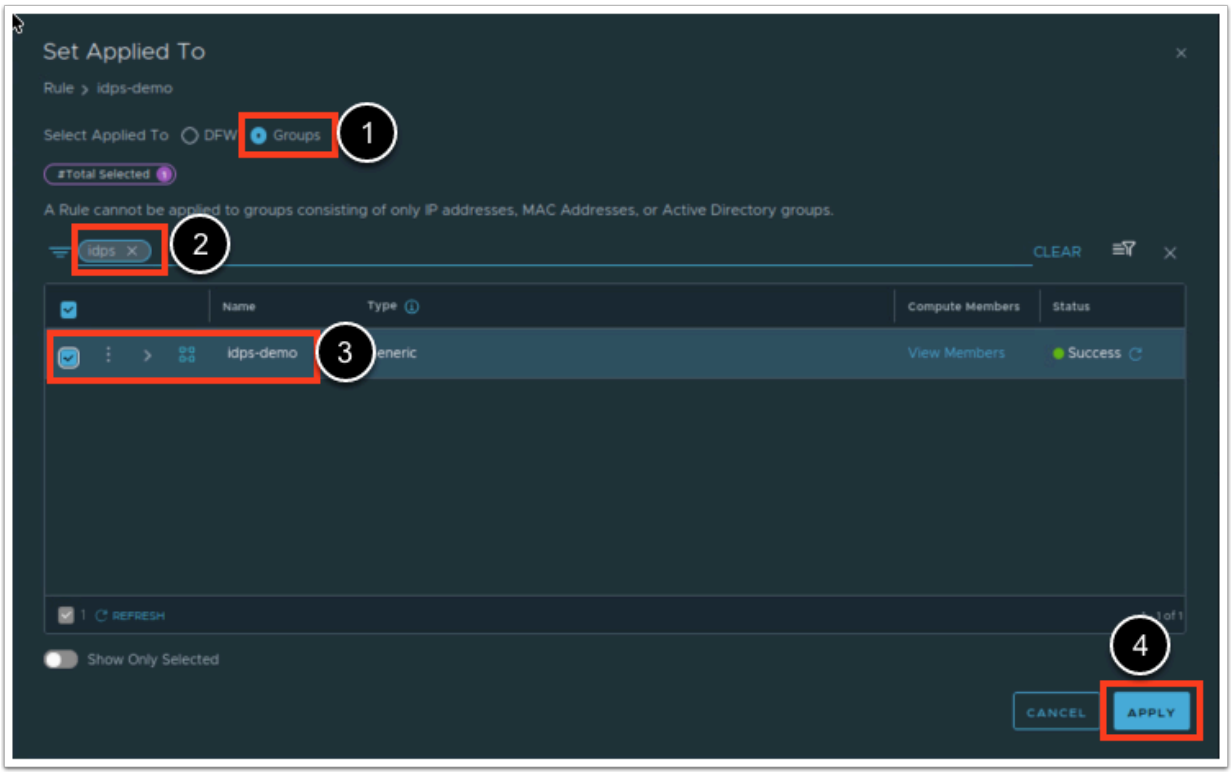
1. Click on the radio button selecting ips-demo-profile
2. Click **APPLY**

Apply the Distributed Policy to the Demo VM's



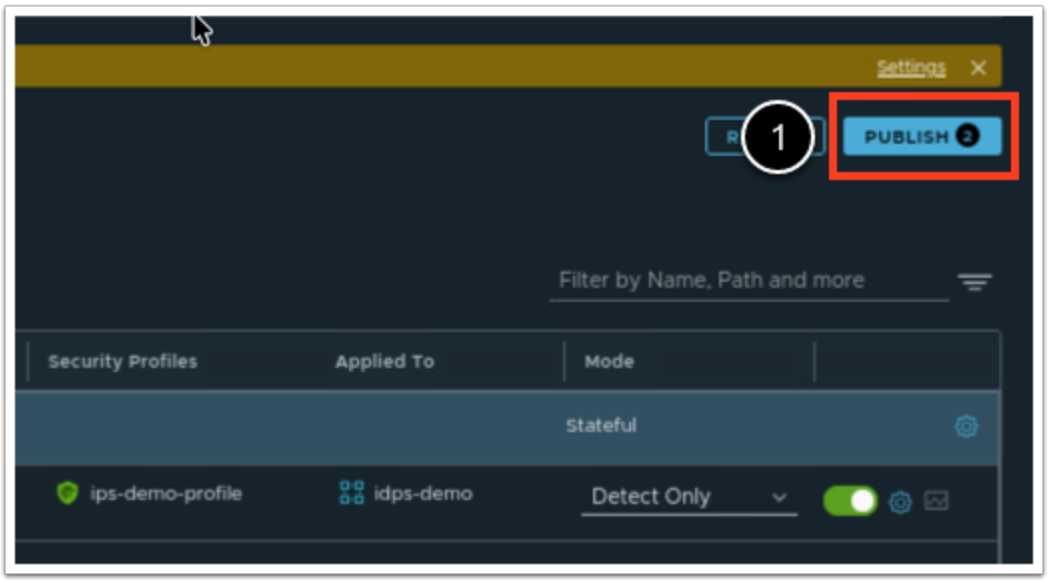
1. Click on the pencil in the Applied To field and edit the default DFW **setting**

Selecting the idps-demo group



1. Select **Groups**
2. In the **Apply Filter, type idps** and **enter**
3. Select the group **idps-demo [This group includes the three idps demo VM's, which can be viewed by clicking on View Members]**
4. Click **Apply**

Publish the Distributed IDPS Policy



1. Click **PUBLISH**

Create Gateway IDS/IPS Policy



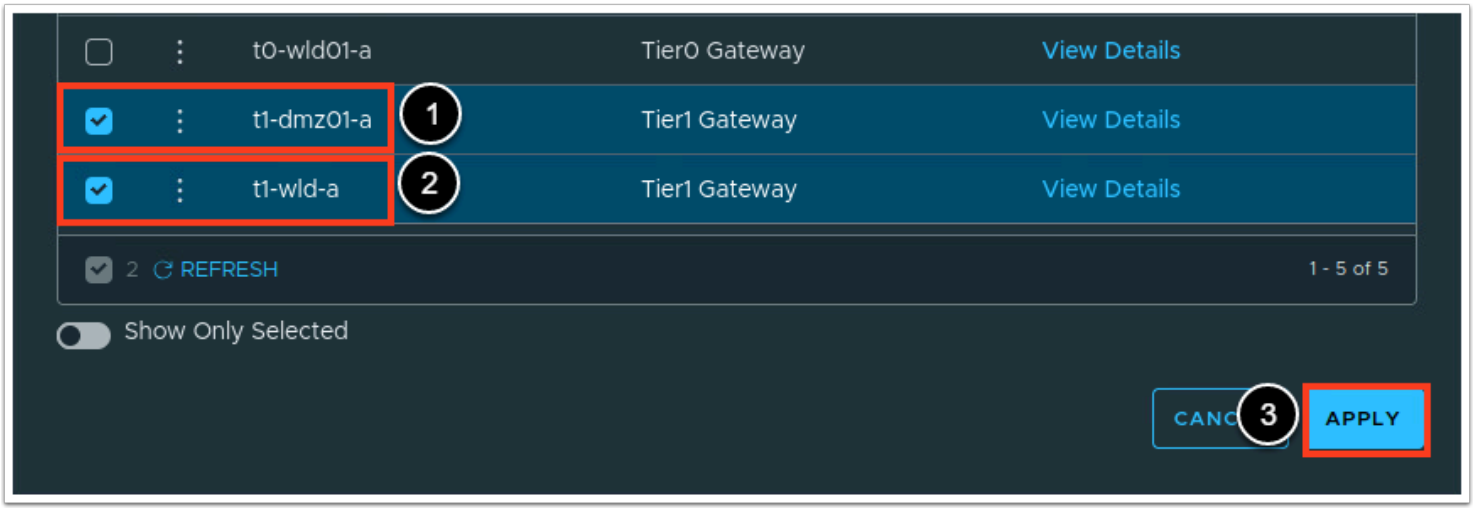
1. Click **Gateway Rules**
2. Click **All Shared Rules**
3. Click **ADD POLICY**
4. Rename the policy to **idps-demo** by clicking in the name section and typing idps-demo

Create Gateway IDS/IPS Rule



1. Click **ADD RULE**
2. Rename the new rule to **idps-gw**
3. **Source** = idps-demo group
4. **Destination** = idps-demo group
5. Select the **ips-demo-profile Security Profile**
6. Click under **Applied To** to select the T1 gateways

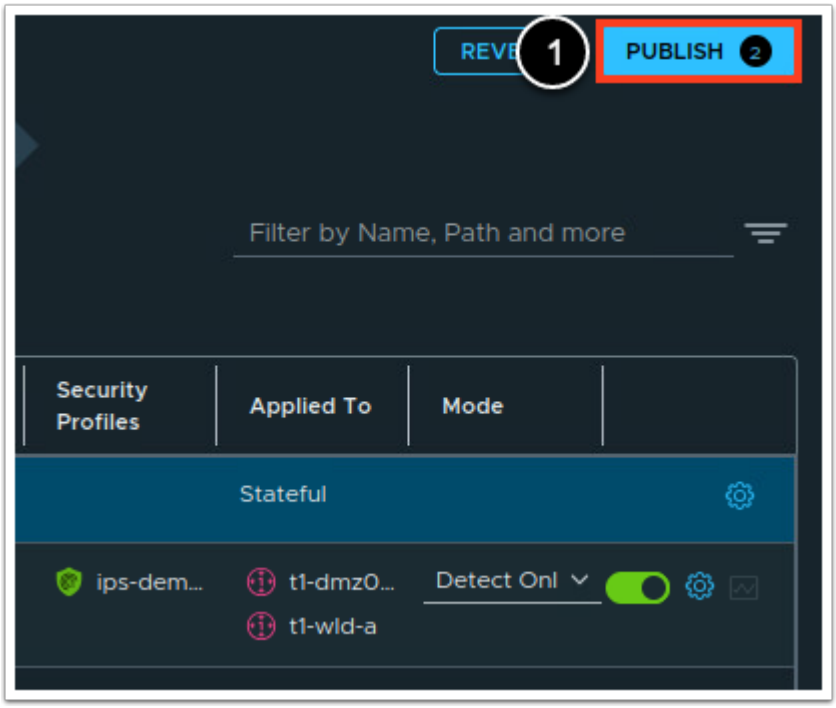
Apply IDS/IPS Rule To T1 Gateways



In the Applied to Field, click on the pencil

1. Select **t1-dmz01-a**
2. Select **t1-wld-a**
3. Click **APPLY**

Publish the Gateway Policy and Rules

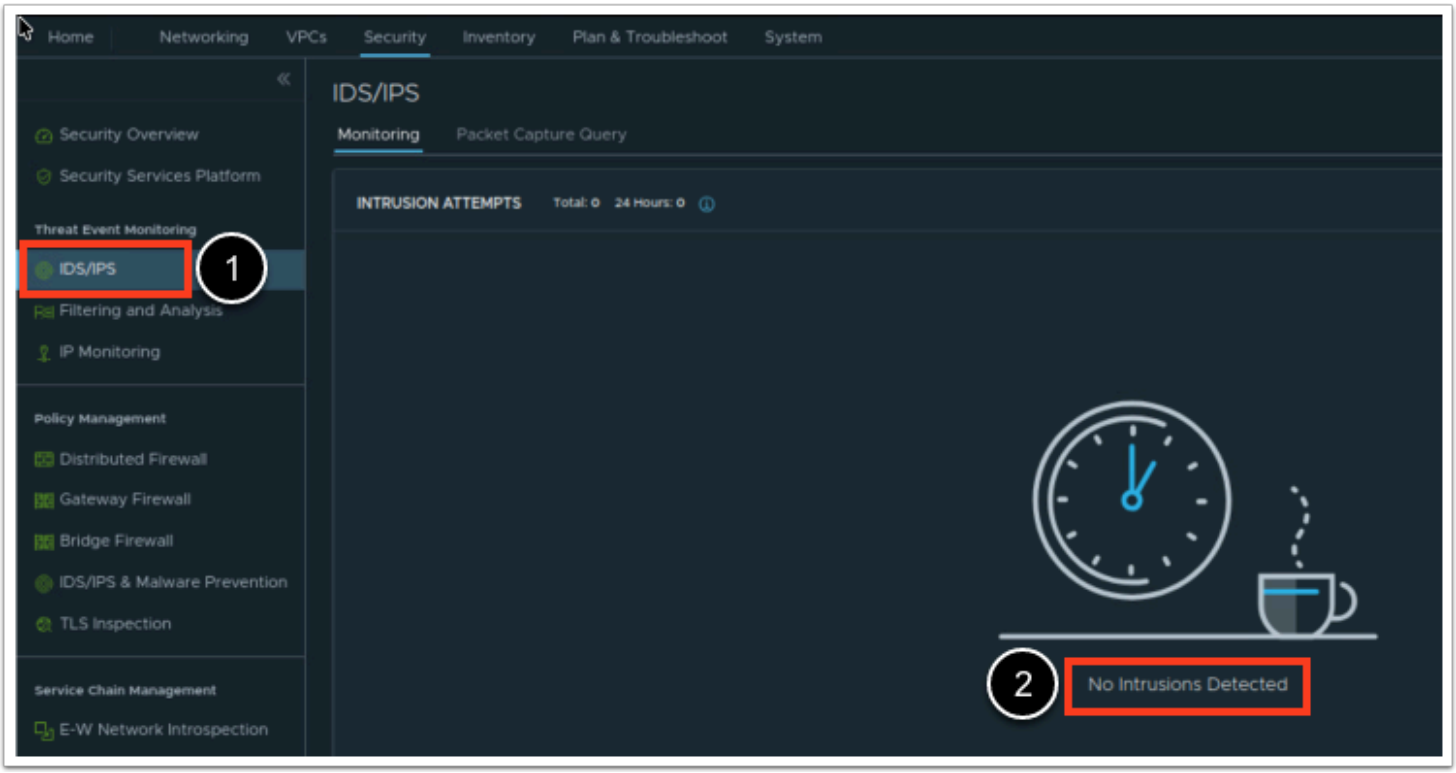


1. Click **PUBLISH**

Threat Event Monitoring - IDS/IPS

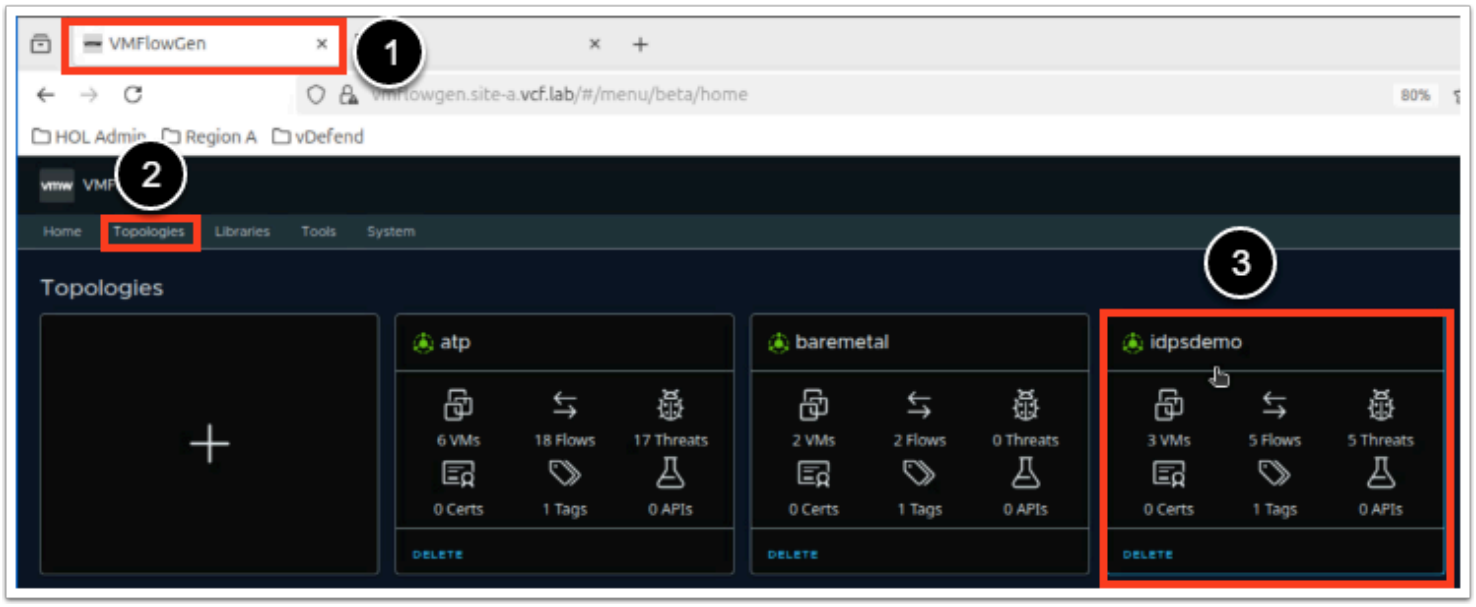
Time to start generating and monitoring IDS/IPS detection events.

Review IDS/IPS Dashboard in the NSX Manager



1. Click IDS/IPS **under Threat Event Monitoring**
2. Validate **No Intrusions Detected**

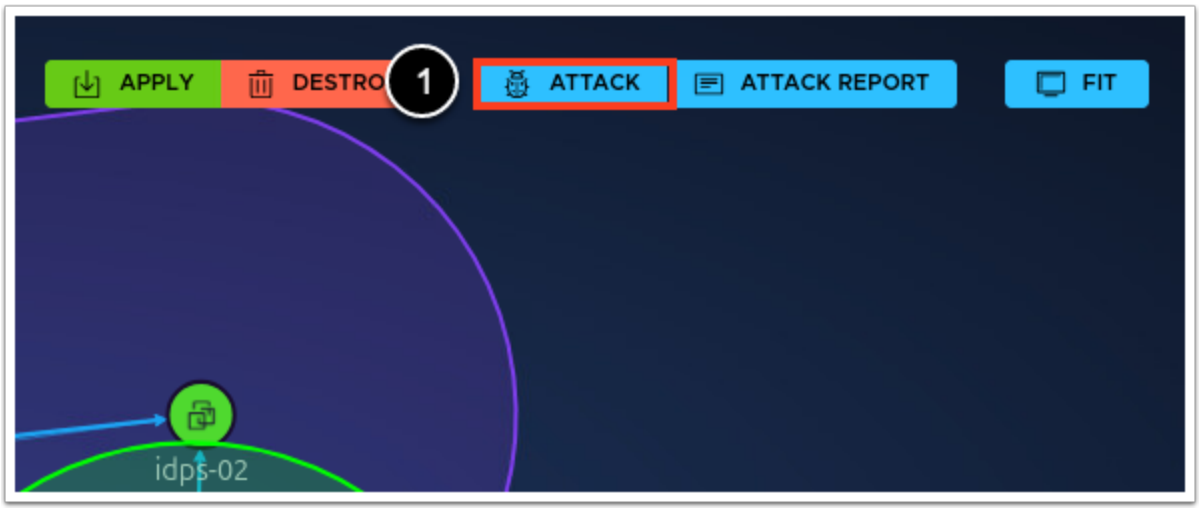
Generate Threat Events using VMFlowGen



If you were logged out of FlowGen, log in again with the user account cached in the browser

1. At the top of the browser, return to the **VMFlowGen** Tab
2. Click **Topologies**
3. Click on the **idpsdemo** tile

Open Attack sequence



1. Click on **Attack** to open the attack sequence

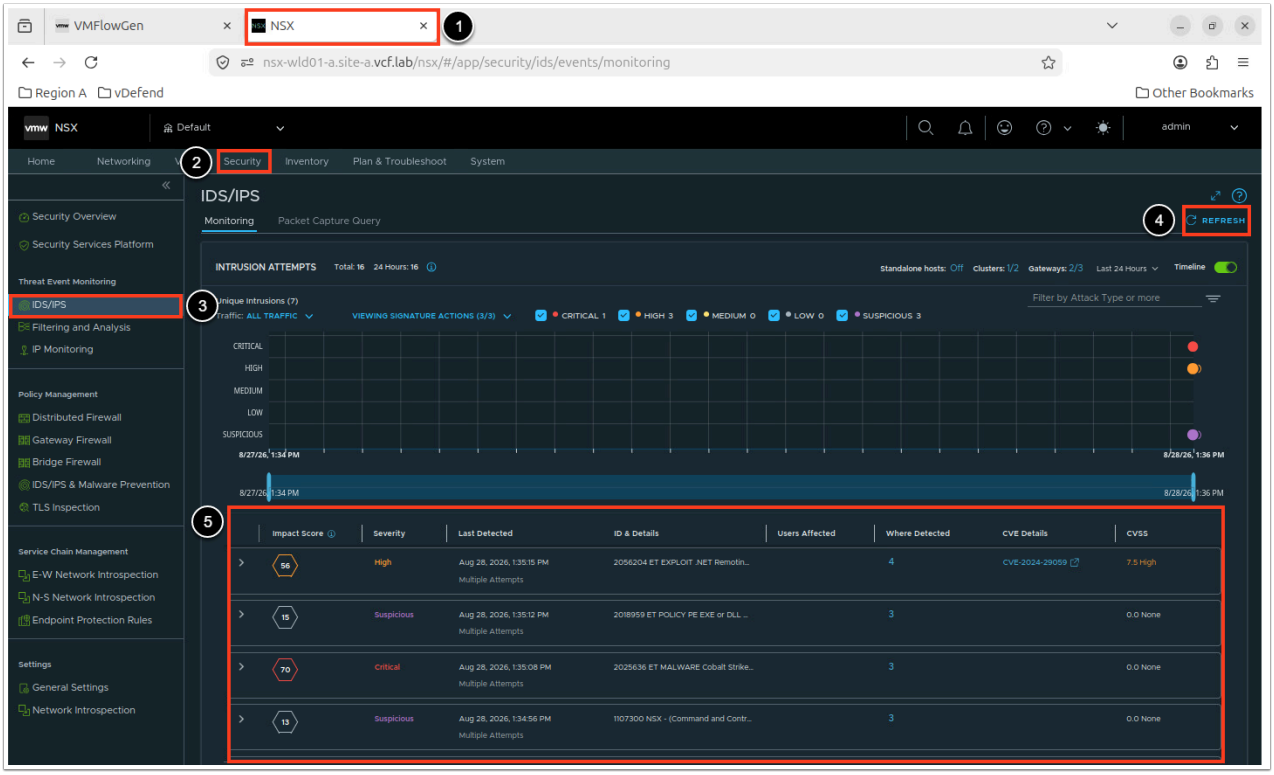
Generate events



The preconfigured attack flows can be triggered individually or all in sequence. In this exercise, we will run them all at once.

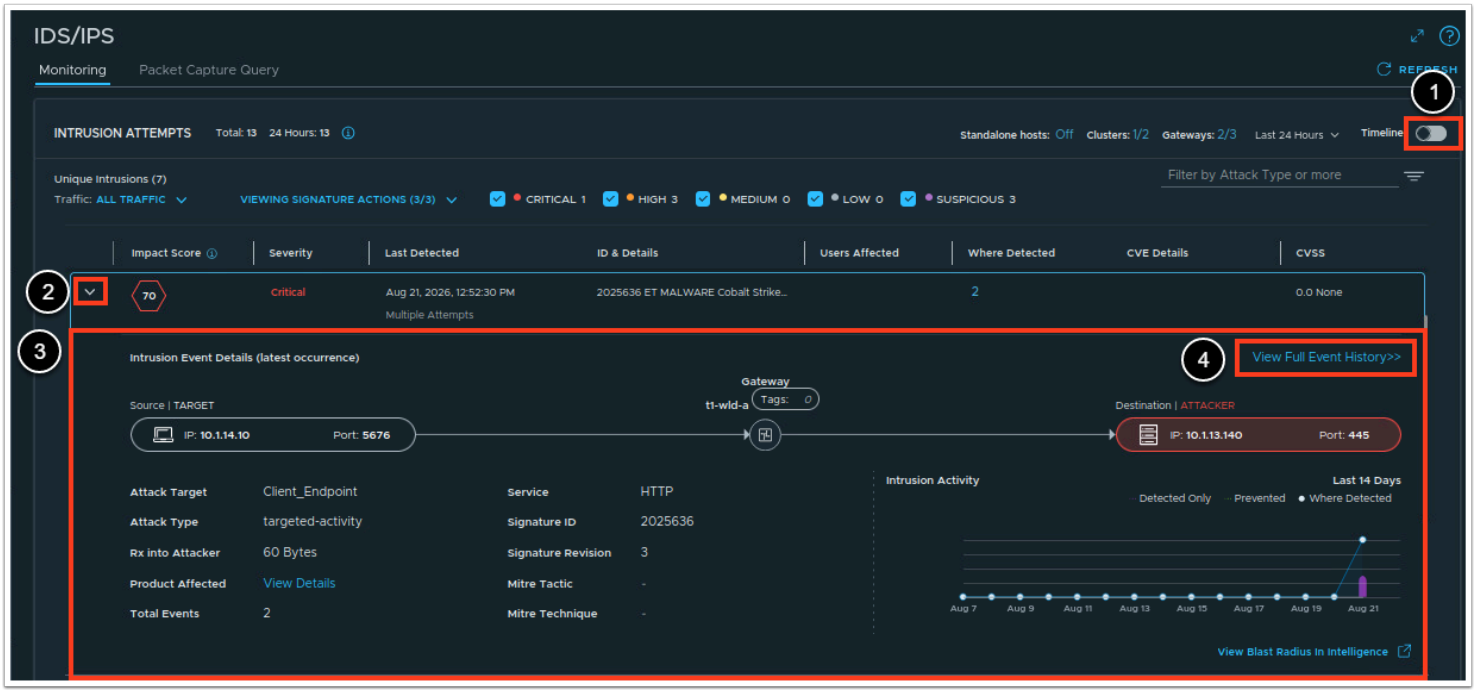
1. Click **START**, this should start each flow in sequence and takes up to 30 seconds to complete

Monitoring events from the NSX Manager



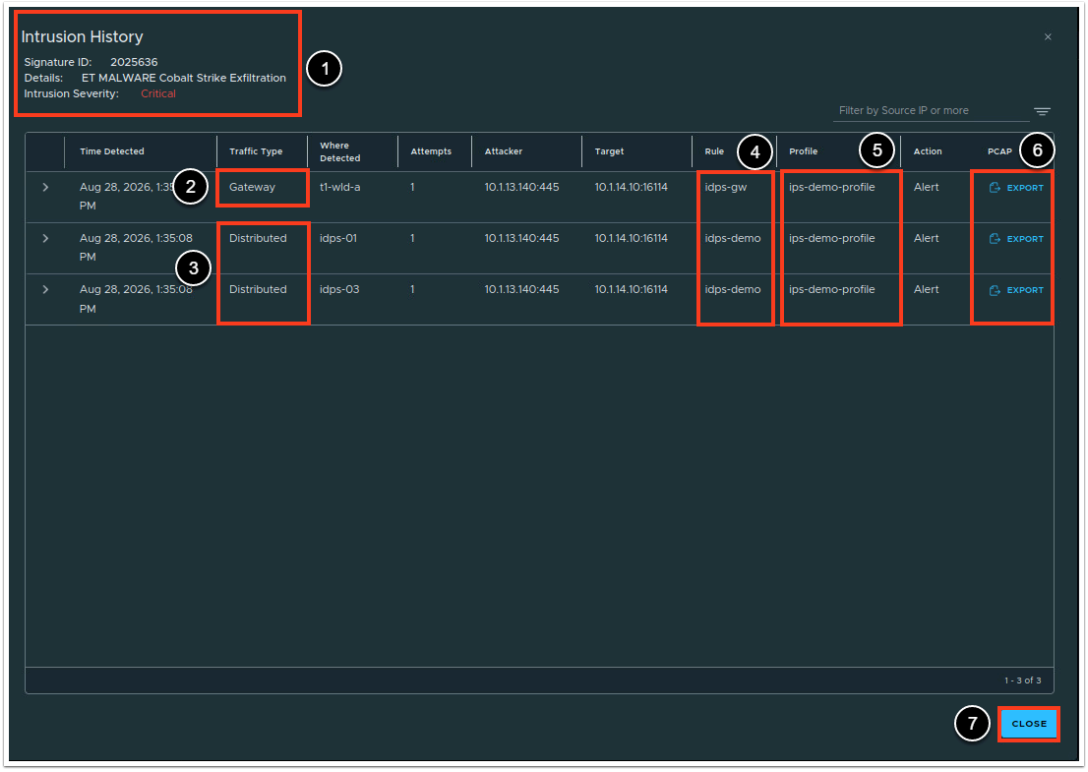
1. Return to the **NSX Manager**, Click on the **NSX** Tab on the browser
2. Click **Security**
3. Click **IDS/IPS** under Threat Monitoring
4. Click **Refresh**
5. **You should now see the events**

Expand the Critical Event



1. Click on the Timeline **toggle to close the timeline view**
2. Click on the arrow to expand the critical **event**
3. Here we see the hosts involved in this event and event-related details
4. Click on **View Full Event History**

View Full Event History

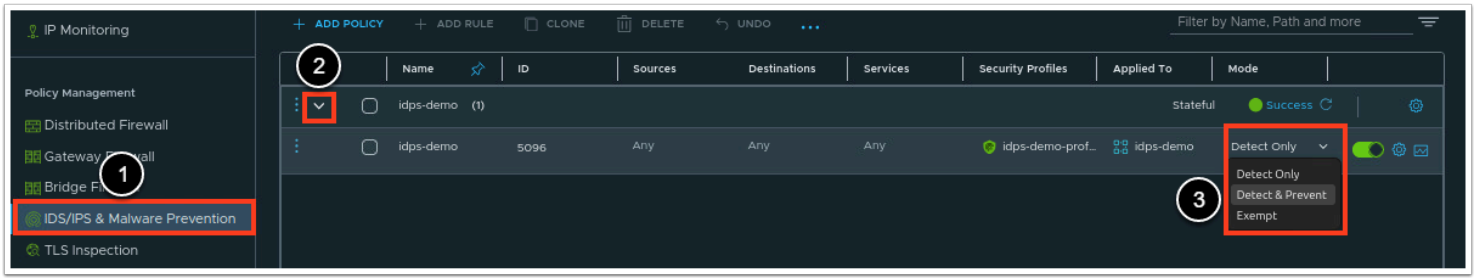


1. Here, we can see which IDS/IPS signature was triggered and the details of the attack
2. **Traffic Type and Where Detected:** This event indicates detection on the Gateway engine
3. **Traffic Type and Where Detected:** This event indicates detection Distributed IDS/IPS engine
4. **Rule Where Detected:** Specific Rule that hit for the traffic
5. **Profile:** This indicates the Signature Profile matching the event **- idps-demo-profile was the one you applied**
6. **PCAP:** Our custom profile was configured to create a PCAP capture of the event data**. We have the option to export the PCAP file**
7. Click **CLOSE**

Implement Detect & Prevent

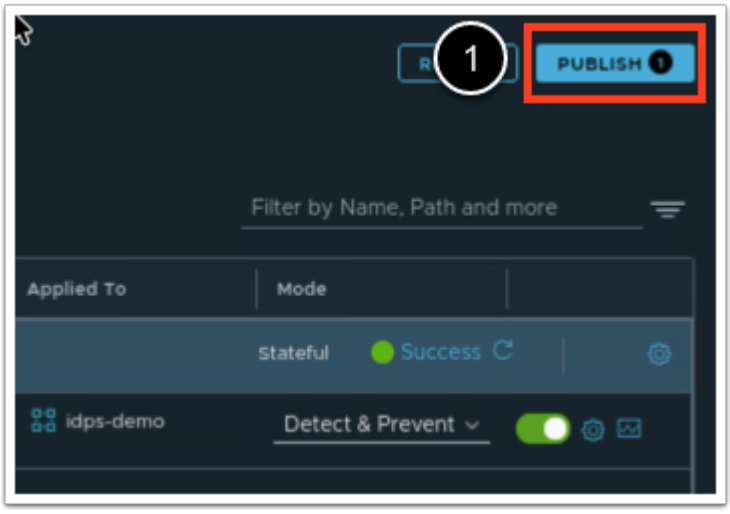
Now we will change the default action of Detect Only to Prevent to drop or reject the attacks.

Update IDS/IPS Mode



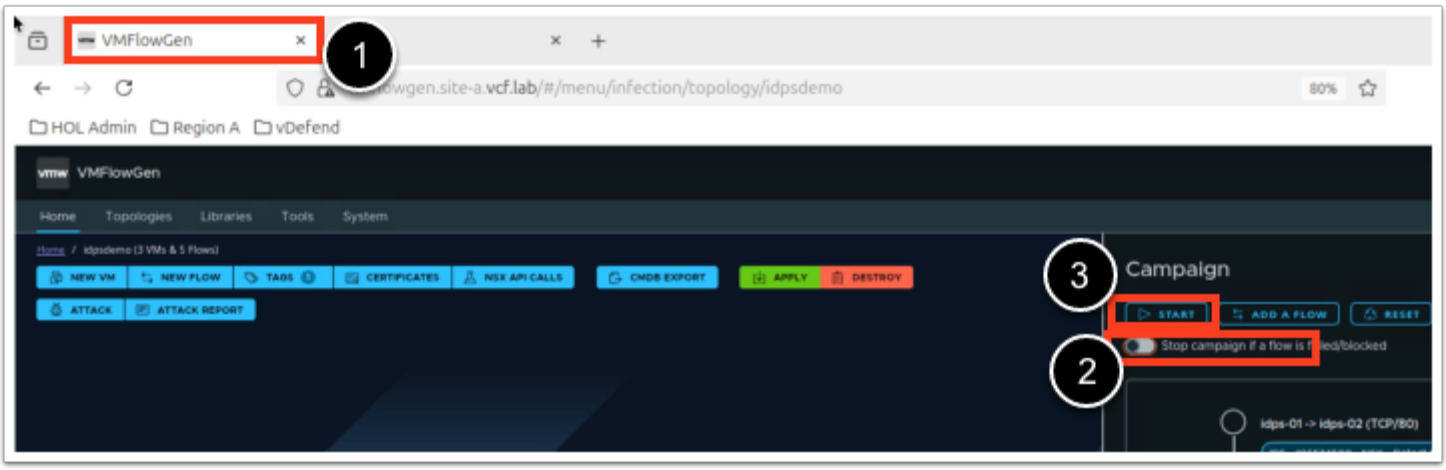
1. Click **IDS/IPS & Malware Prevention**
2. Click the **Triangle** next to the Policy Name to expose the rule below
3. Click Detect Only and Select **Detect & Prevent**

Publish the changes



1. Click **Publish**

Generate an attack from VMFlowGen

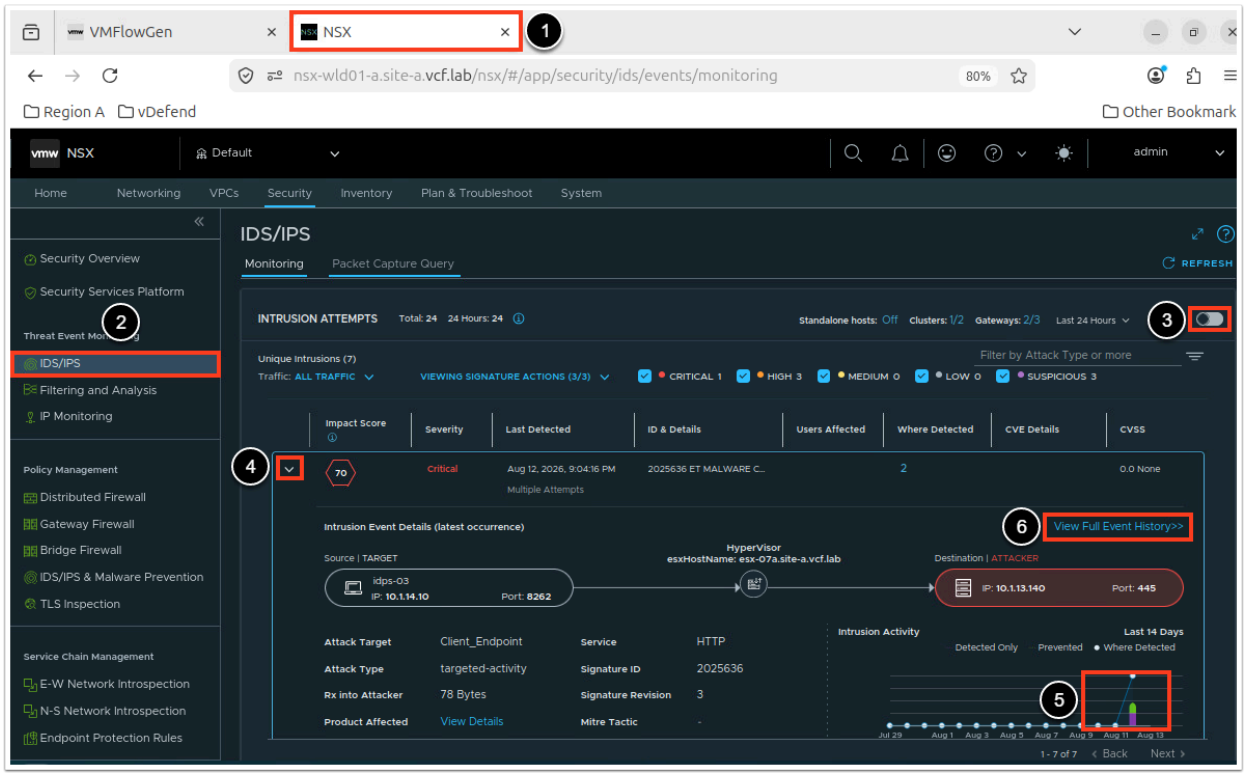


1. Return to the VMFlowGen Tab on the browser**, to the idpsdemo topology**
2. **Click** the toggle to change the default behaviour and disable "**Stop campaign if a flow is failed/blocked"**
3. **Click Start** to restart the attack sequence

Monitor updated events from the NSX Manager

We will review the events again to confirm the threats were prevented.

Access the IDS/IPS under Threat Event Monitoring



1. Return to the NSX Manager Tab on the browser
2. Click **IDS/IPS**
3. Click the toggle on Timeline **to disable/hide the timeline view**
4. Click **the Arrow on the Critical event to expose the event details**
5. You will notice a green bar here **- this indicates a block event [A blocked event could take a few seconds to appear]**
6. Click on **View Full Event History**

Viewing Full Event History



Here, we will notice that we only see an event on the distributed IDS/IPS engine and nothing on the Gateway. This is expected as we have blocked the attack at the source, and it never reaches the Gateway.

1. The Action on this event should be **Reject**
2. Comparing it to the previous event that was **Alert** only
3. Click **Close**

Virtual Patching With vDefend IDS/IPS

Virtual patching is a vulnerability-shielding tactic that protects assets by implementing a minimal layer of security policies at the network or application delivery level, front-ending that asset. These measures intercept and block exploit attempts before they can reach the vulnerable software, effectively “patching” the flaw in the communication path rather than the software itself.

Key Benefits of Virtual Patching

- **Application availability:** Ensures that applications/assets remain available while risks are mitigated.
- **No code changes:** Protects applications without requiring updated software or patches. Because software patches can introduce regressions, this approach eliminates that risk.
- **Targeted signatures:** Reduces the risk of false positives and performance impact.
- **Zero-Day Protection:** Provides a rapid response to vulnerabilities for which no official patch yet exists.
- **Legacy Support:** Shields older, unsupported systems critical to business operations where patches may no longer exist.
- **Buys Time:** Grants security teams necessary cycles to test and deploy permanent vendor patches without remaining exposed.
- **Helps with Compliance:** Meets requirements for regulations such as PCI-DSS and HIPAA that demand timely deployment of security controls.

Why Do We Need Virtual Patching

Frontier AI models have the intelligence to identify unknown (zero-day) software vulnerabilities (bugs) and find ways to exploit them faster than ever before. Attackers can leverage these exploits to infiltrate digital enterprises, propagate laterally, hopping and hunting to find high-value assets for ransom or secret theft. They can initiate widespread, volumetric, and/or targeted attacks semi-autonomously—leading to an exponential increase in the attack surface. "Security through obscurity" is no longer a viable cybersecurity strategy.

If enterprises can quickly patch software vulnerabilities, they can certainly reduce the risk of a breach or its spread. However, this is an extremely time-consuming and resource-intensive endeavor. There are thousands of software tools and apps, each with varied software versions, deployed on different types of hardware and operating systems spread across multiple data centers. In larger organizations, the "race to patch" can take weeks to months to roll out patches enterprise-wide, leaving the organization exposed to infiltration, ransom, and potential business disruption.

To help quickly protect against this tsunami of exploits unleashed against workloads and apps, and to buy down risk, enterprises need to focus on two key defenses for their private cloud workloads:

- Enable virtual patching using intrusion prevention systems and a web application firewall.

- Restrict the propagation of attacks with lateral segmentation.

How vDefend IDPS Enables Virtual Patching

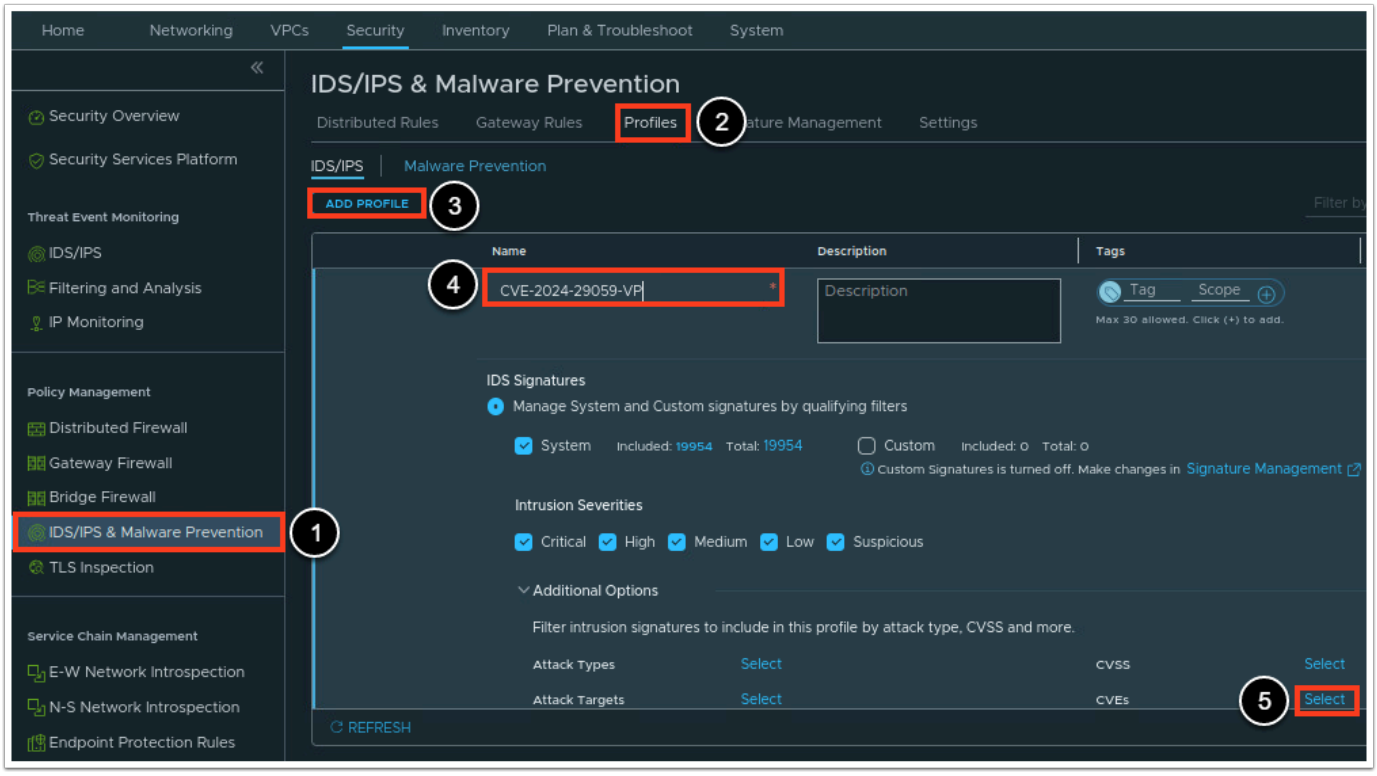
- **Hypervisor-Integrated Inspection:** vDefend's Distributed IDPS inspects network traffic at the vNIC of every VCF workload. This ensures unpatched servers cannot be exploited by attacks originating from outside or inside the network.
- **Automated Dynamic Policies:** Running a vulnerability scan identifies workloads, applies appropriate tags, and creates a virtual patching policy with a targeted set of IDPS signatures. As new vulnerable workloads are identified and tagged, policies are applied automatically so vulnerable workloads receive immediate protection.
- **East-West Security:** IDPS prevents attackers from exploiting vulnerabilities to move laterally within the environment, protecting high-value assets from compromise.

The vDefend Advantage

What makes vDefend IDPS powerful for this use case is its architectural advantage, signature strategy, and operational simplicity.

- **Speed matters in the era of AI-driven threats:** Through integration with the VCF private cloud platform, vDefend's built-in, closed-loop security architecture delivers detection and rapid mitigation, dramatically reducing attacker dwell time.
- **Frequently updated signatures:** vDefend IDPS supports signatures that are updated multiple times a day, ensuring protection against the latest global threats. Broadcom's threat intelligence team actively analyzes new exploits, creates signatures, and updates signature bundles.
- **Custom IDPS signatures:** Customers can import trusted signatures from third parties or develop signatures in-house to virtually patch their applications, providing a truly bespoke security posture.

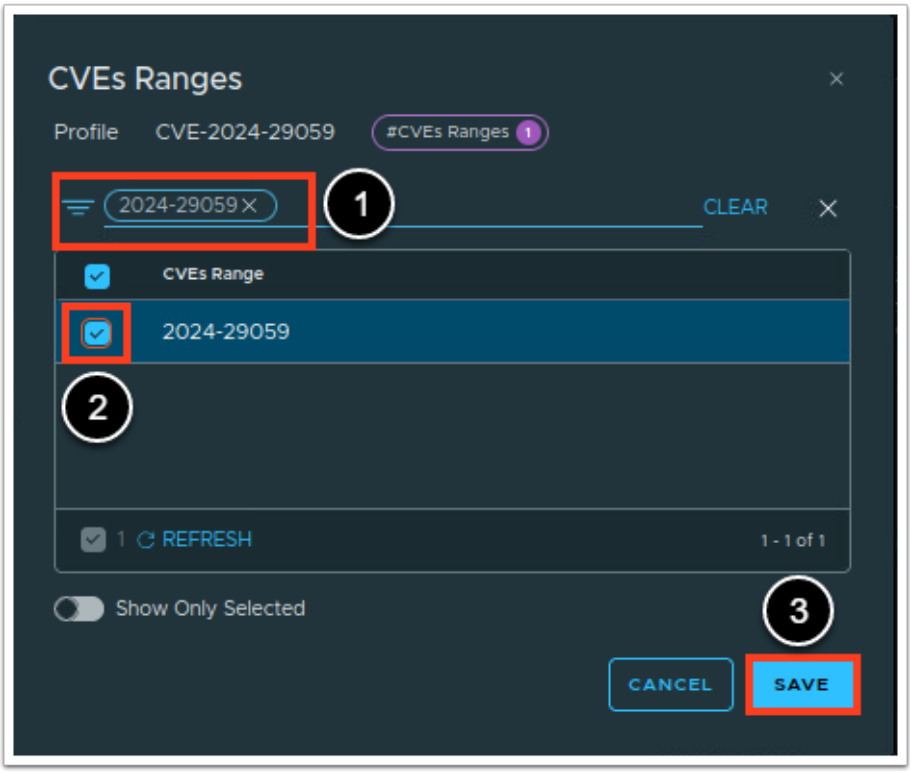
Create Custom Virtual Patching Signature Profile



As covered earlier in this module, automatic signature updates keep NSX Manager continually up to date against newly released vulnerabilities. Once a signature downloads, you can create a targeted security profile to virtually patch and shield vulnerable workloads, blocking exploits in the traffic path without needing immediate code changes until a permanent vendor patch is deployed.

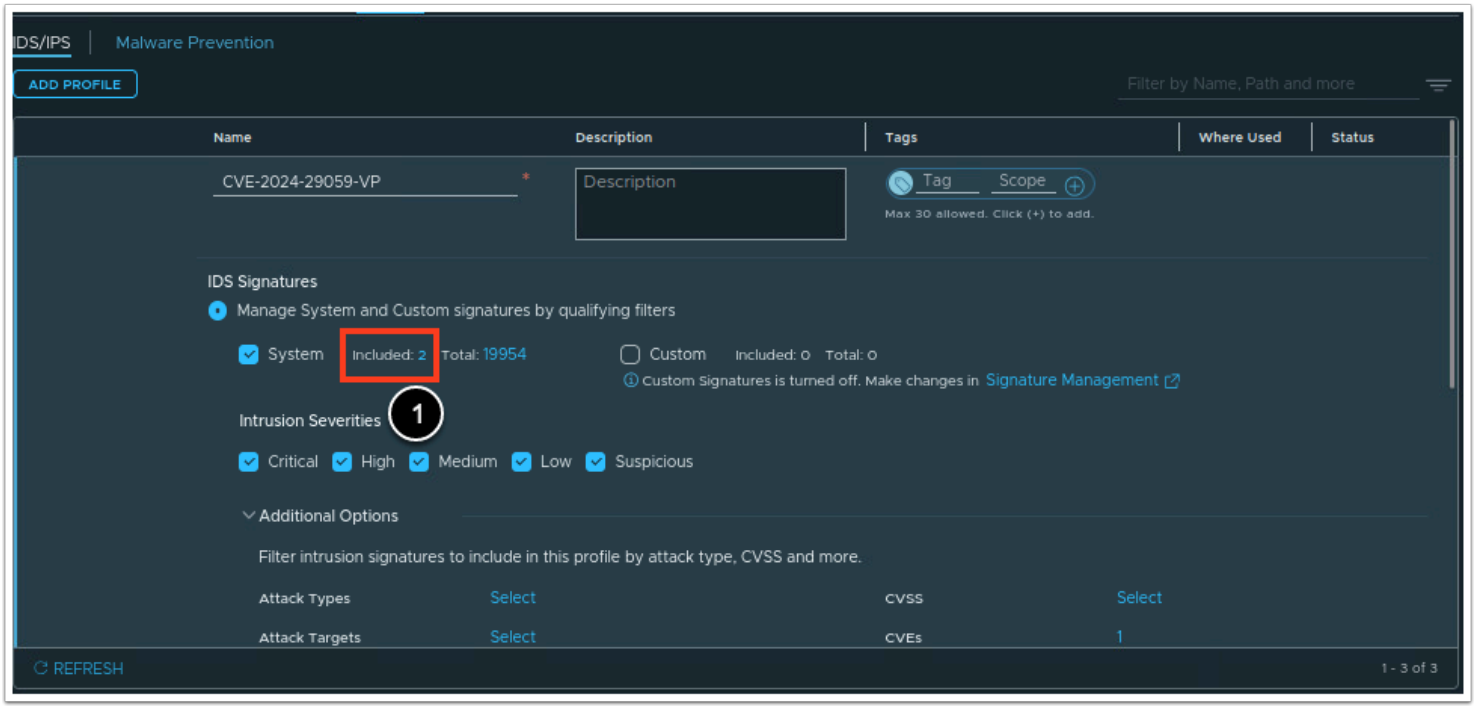
1. Click **IDPS & Malware Prevention**
2. Click **Profiles**
3. Click **ADD PROFILE**
4. Name the profile **CVE-2024-29059-VP**
5. Click **Select** next to CVEs

Enter Targeted CVE



1. In the filter field type **2024-29059** and press enter
2. Click in the box next to **2024-29059**
3. Click **SAVE**

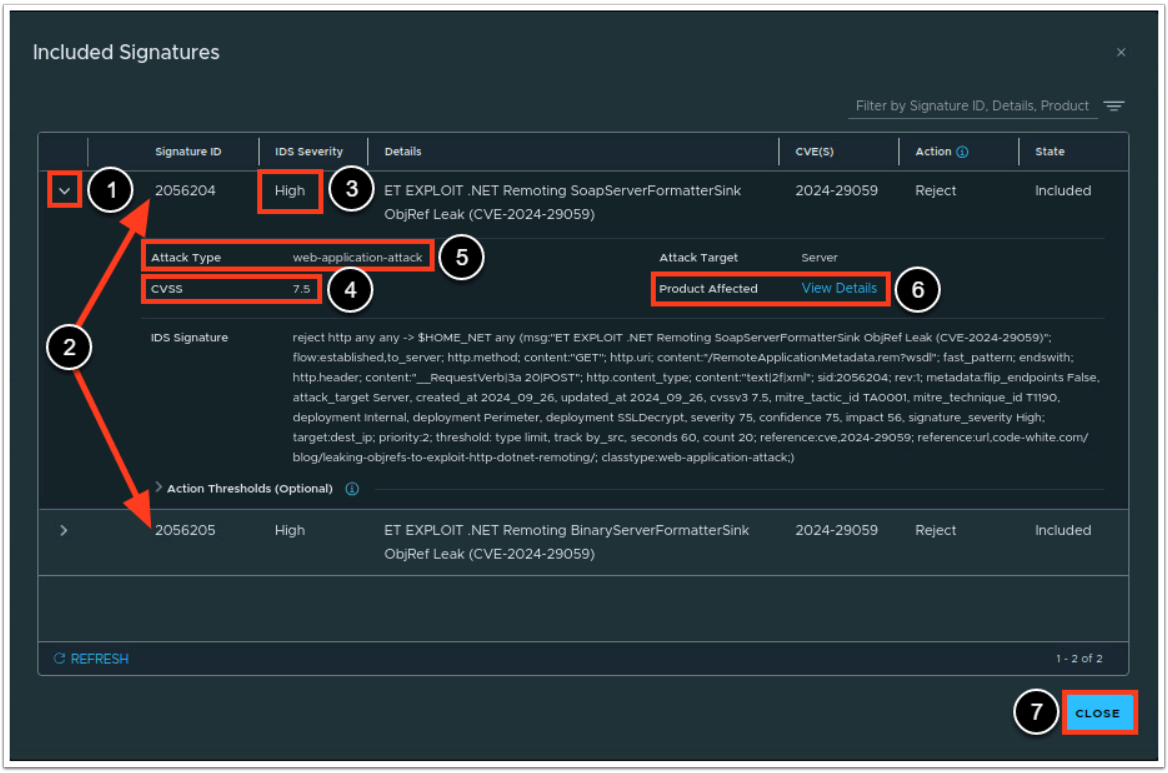
Confirm Custom Virtual Patching Signature Profile



After you have selected and saved your desired CVE it's important to confirm you have the correct vulnerability selected. While we selected a single CVE (2024-29059), this CVE has 2 different signatures registered to it. Lets take a look at these signatures.

1. Next to Included click on the **2**

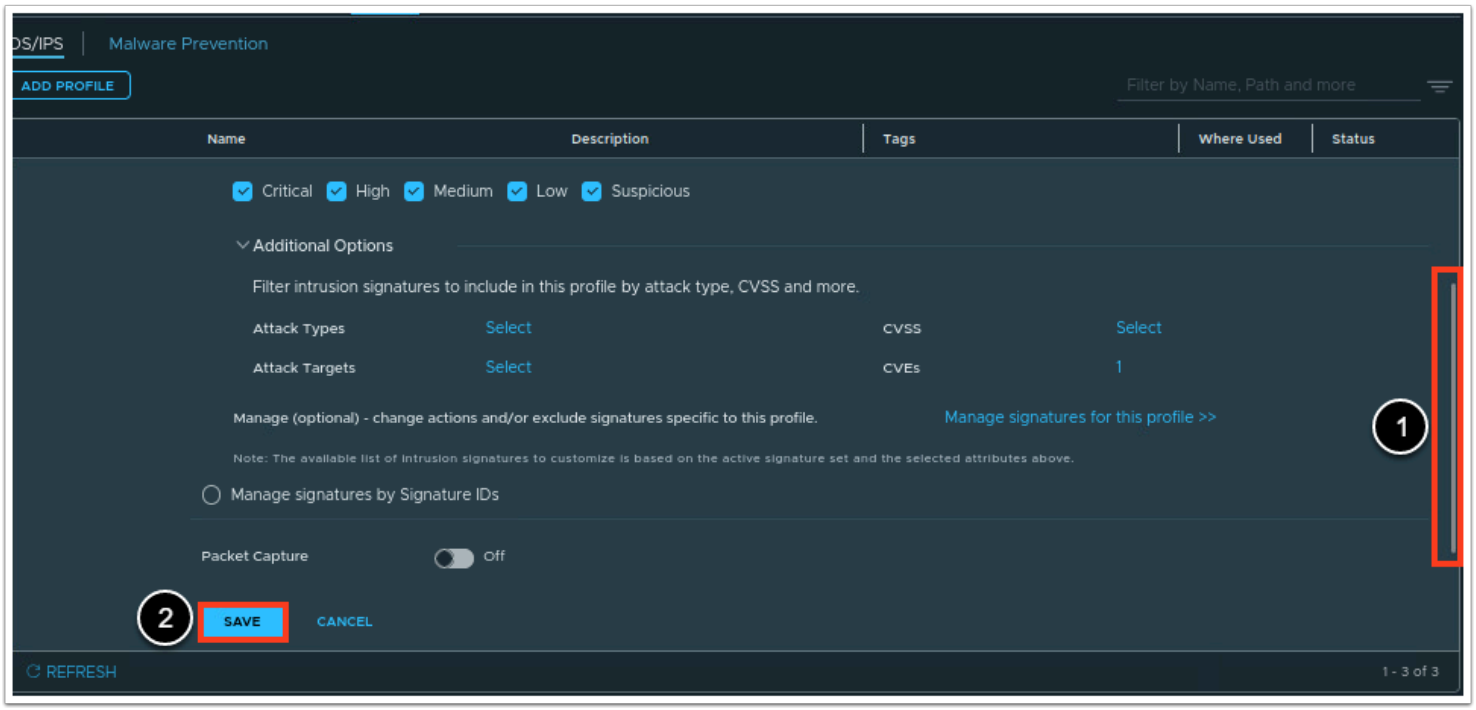
Included Signatures



We can now see that even though a single CVE was selected it has two different signatures to it. Once we expand one of the signatures we can start to gather more information about this vulnerability.

1. Click **expander**
2. Note the two different signatures
3. IDS Severity
4. CVSS - Common Vulnerability Scoring System
5. Attack Type
6. Product Affected - Provides a list of Vendors and Affected Products
7. Click **CLOSE**

Save the Custom Virtual Patching Signature Profile



1. **Click the scroll bar and page down**
2. Click **SAVE**

Create Emergency Distributed IDS/IPS Policy For Virtual Patching

Now that we have created a Emergency Virtual Patching Rule we'll need to apply that an IDS/IPS Policy.

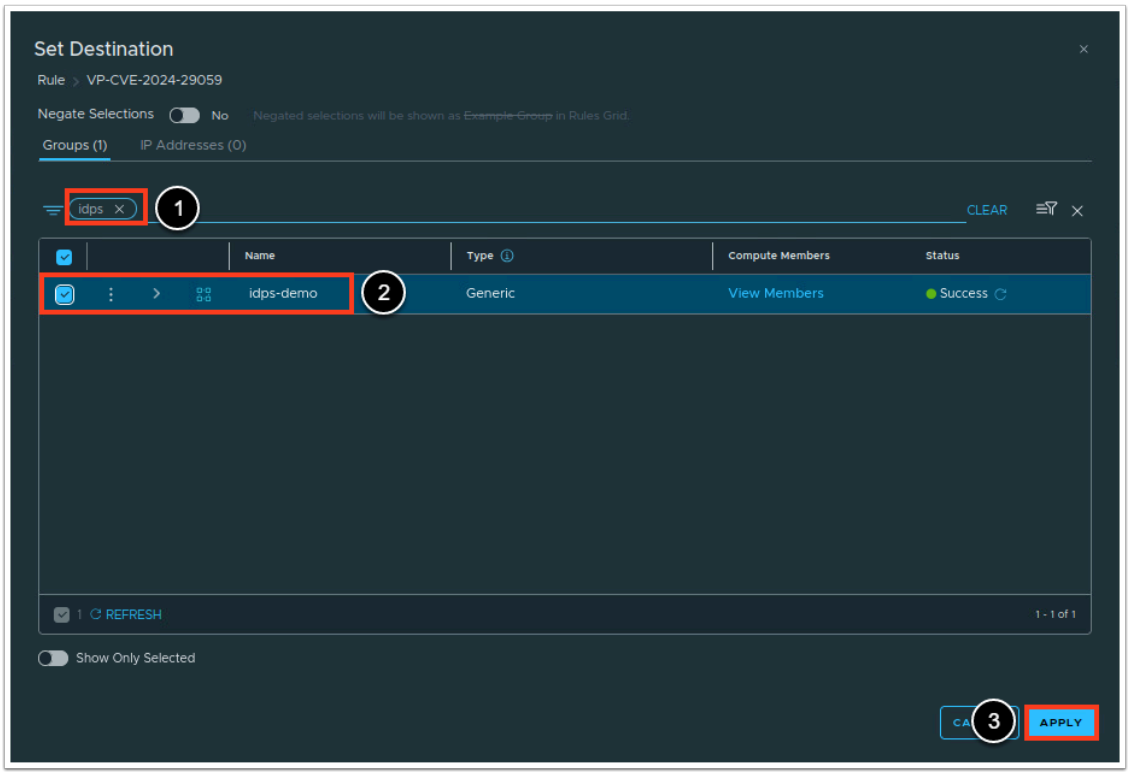
Create Emergency Distributed IDS/IPS Policy For Virtual Patching



For virtual patching our goal is to protect vulnerable workloads we're going to use the Emergency Threat Rules Section, to ensure this IDS/IPS Rule hits before other established rules.

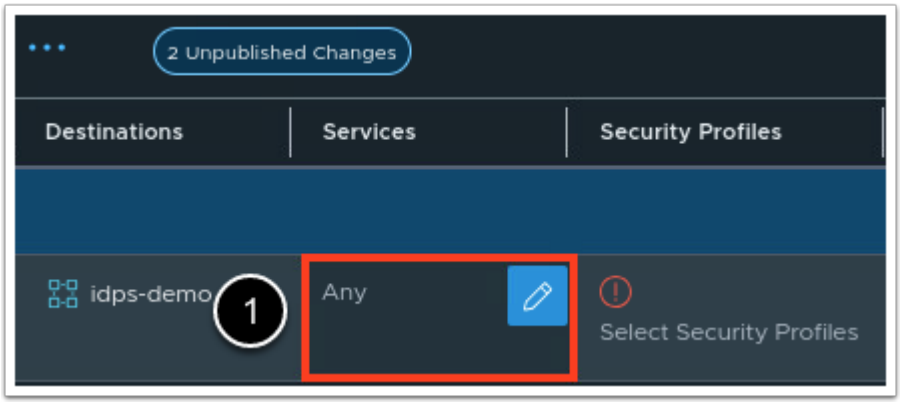
1. Click **Distributed Rules**
2. Click **EMERGENCY THREAT RULES**
3. Rename the policy to **VP-CVE-2024-29059**
4. Click **ADD RULE**
5. Rename the rule to **VP-CVE-2024-29059**
6. Click on the pencil to select **Destination**

Selecting Emergency Virtual Patching Destination Group



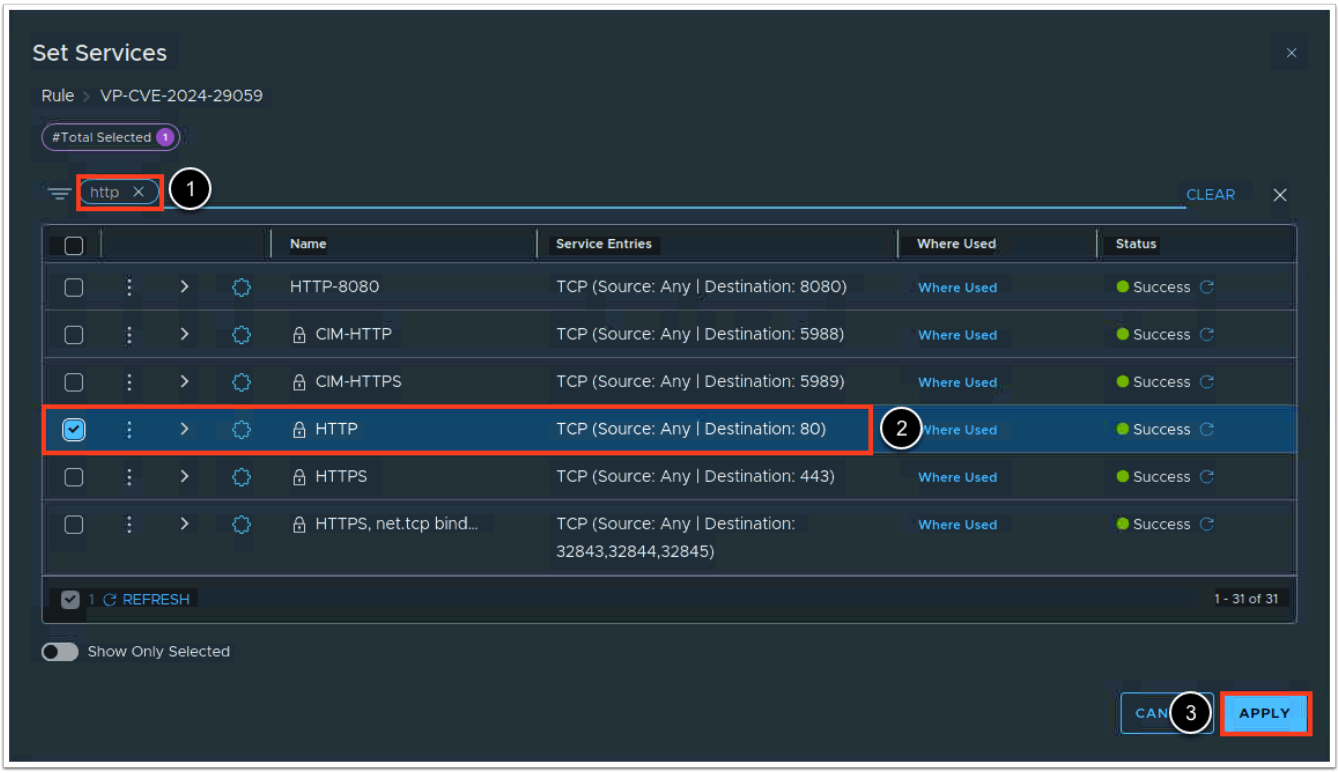
1. In the **Apply Filter, type idps** and **enter**
2. Select the group **idps-demo [This group includes the three idps demo VM's, which can be viewed by clicking on View Members]**
3. Click **Apply**

Select Emergency Virtual Patching Service



1. Click on the pencil to edit Services.

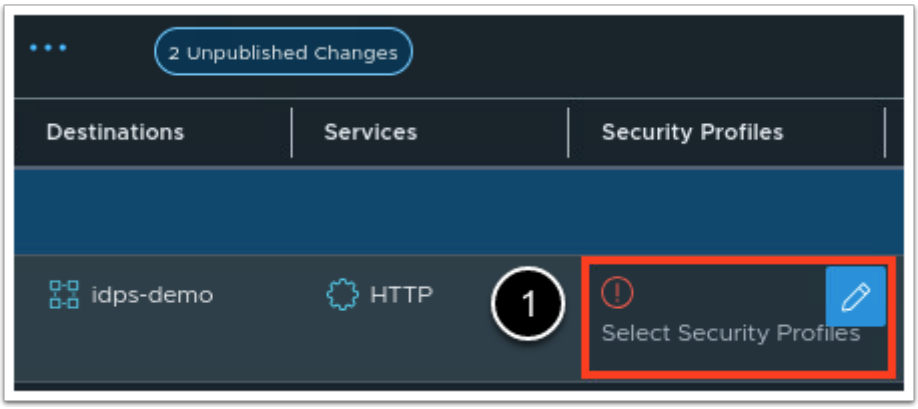
Select Emergency Virtual Patch Service



We want to limit the scope of this Virtual Patch we'll need to ensure only HTTP traffic on port 80 is inspected

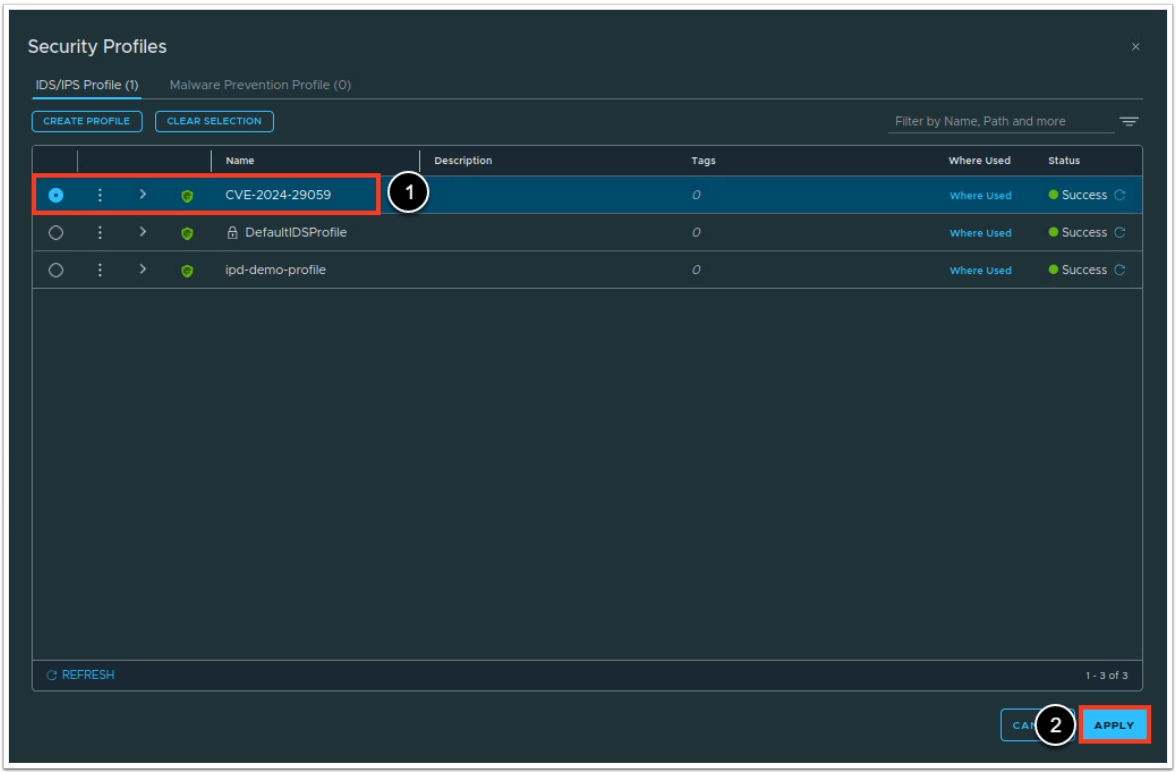
1. In the **Apply Filter**, type **http** and press **Enter**
2. Select **HTTP** with Port **80**
3. Click **Apply**

Select Virtual Patching Signature Profile



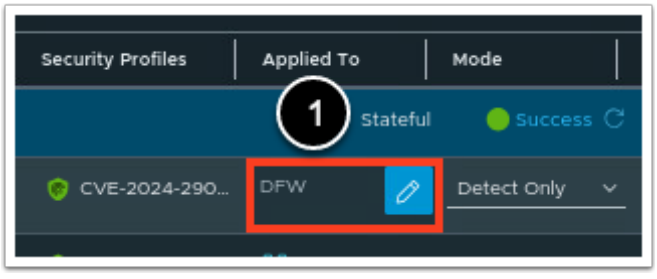
1. Click on the pencil to select Security Profiles - here you will be selecting the custom profile created from the previous step **CVE-2024-29059-VP**

Selecting Custom Virtual Patching Signature Profile



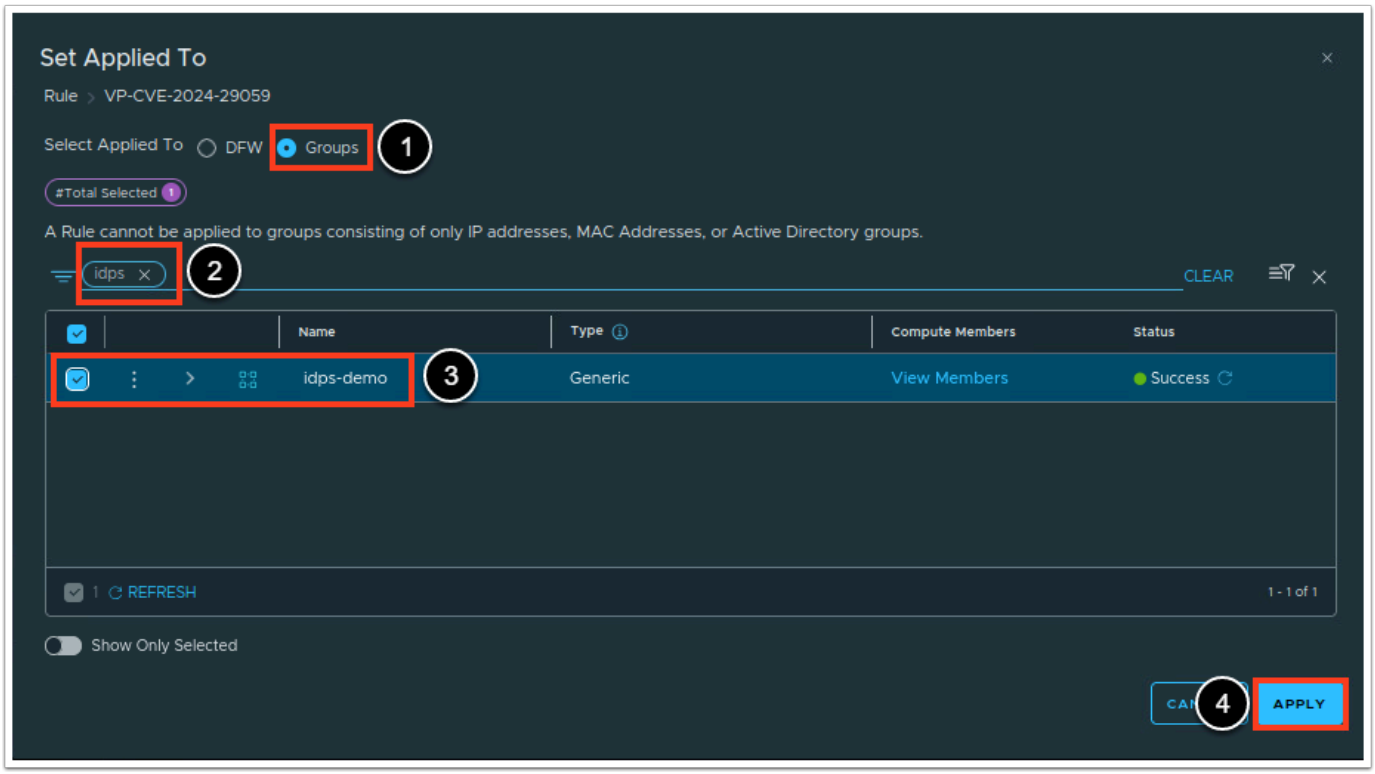
1. Click on the radio button selecting **CVE-2024-29059-VP**
2. Click **APPLY**

Apply the Emergency Distributed Policy to the Demo VM's



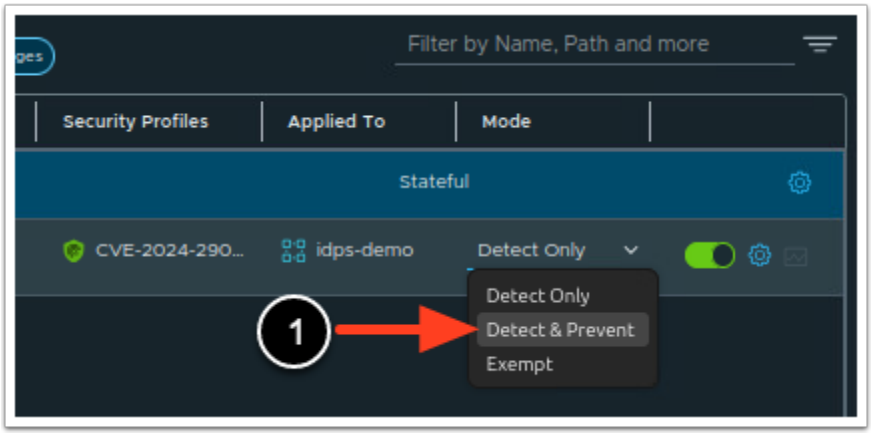
1. Click on the pencil in the Applied To field and edit the default DFW **setting**

Selecting the idps-demo group



1. Select **Groups**
2. In the **Apply Filter, type idps** and **enter**
3. Select the group **idps-demo**
4. Click **Apply**

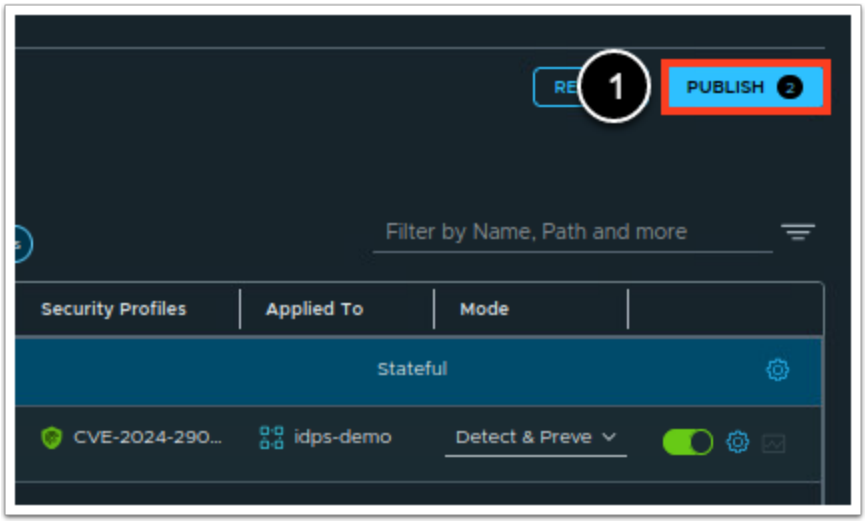
Update IDS/IPS Mode



We'll want to change from Detect Only to Detect & Prevent. This will allow us to ensure our workloads are protected until we can apply a mitigation patch.

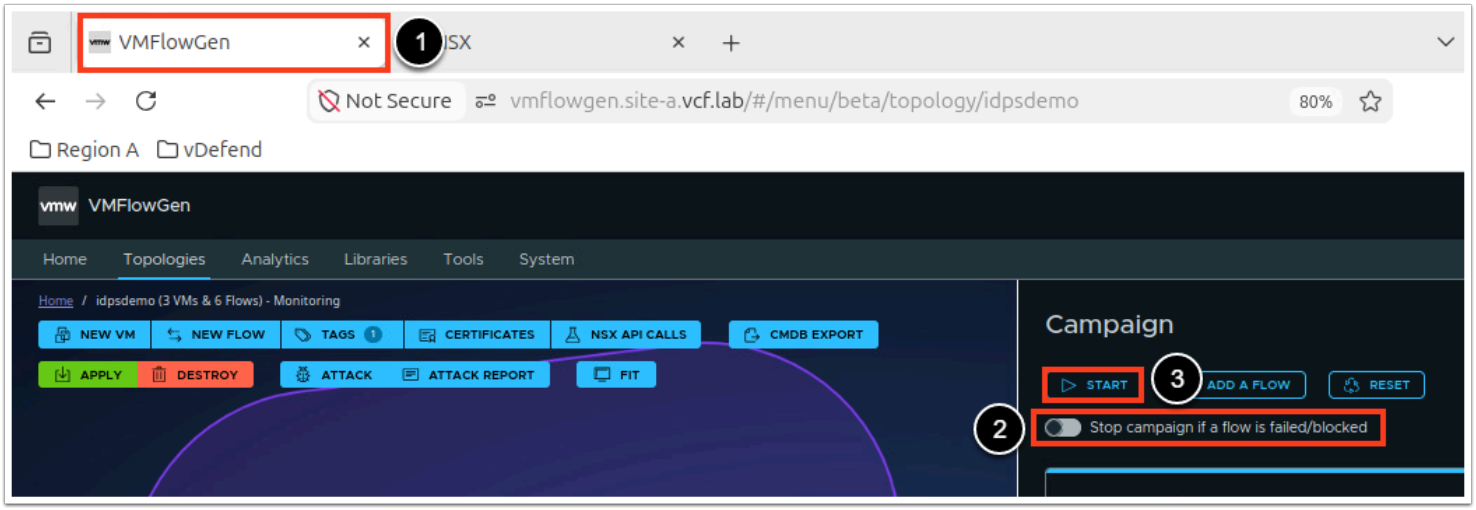
1. Click Detect Only and Select **Detect & Prevent**

Publish the Emergency Distributed IDPS Policy



1. Click **PUBLISH**

Generate an attack from VMFlowGen

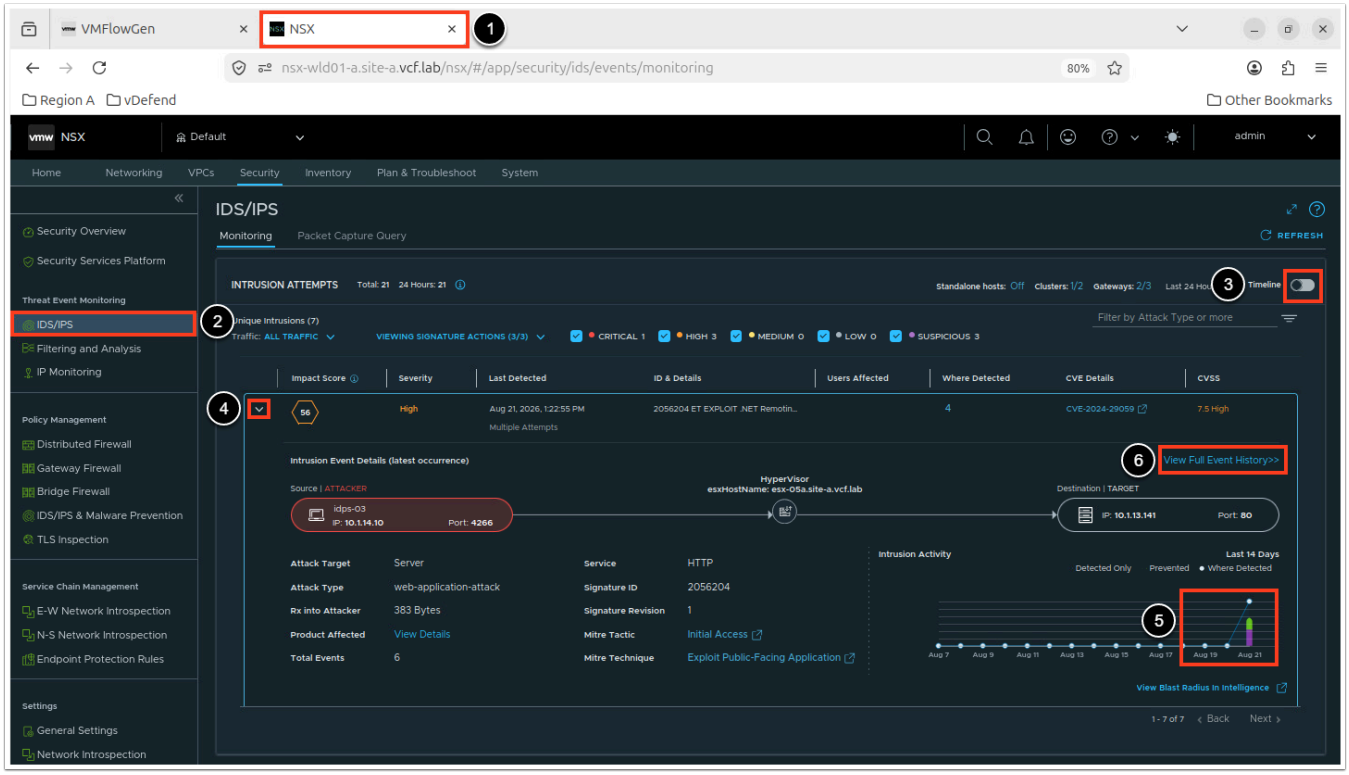


1. Return to the **VMFlowGen** Tab on the browser**,** to the **idpsdemo** topology
2. **Click** the toggle to change the default behavior and disable "**Stop campaign if a flow is failed/ blocked"**
3. **Click Start** to restart the attack sequence

Monitor Updated Virtual Patching Event from the NSX Manager

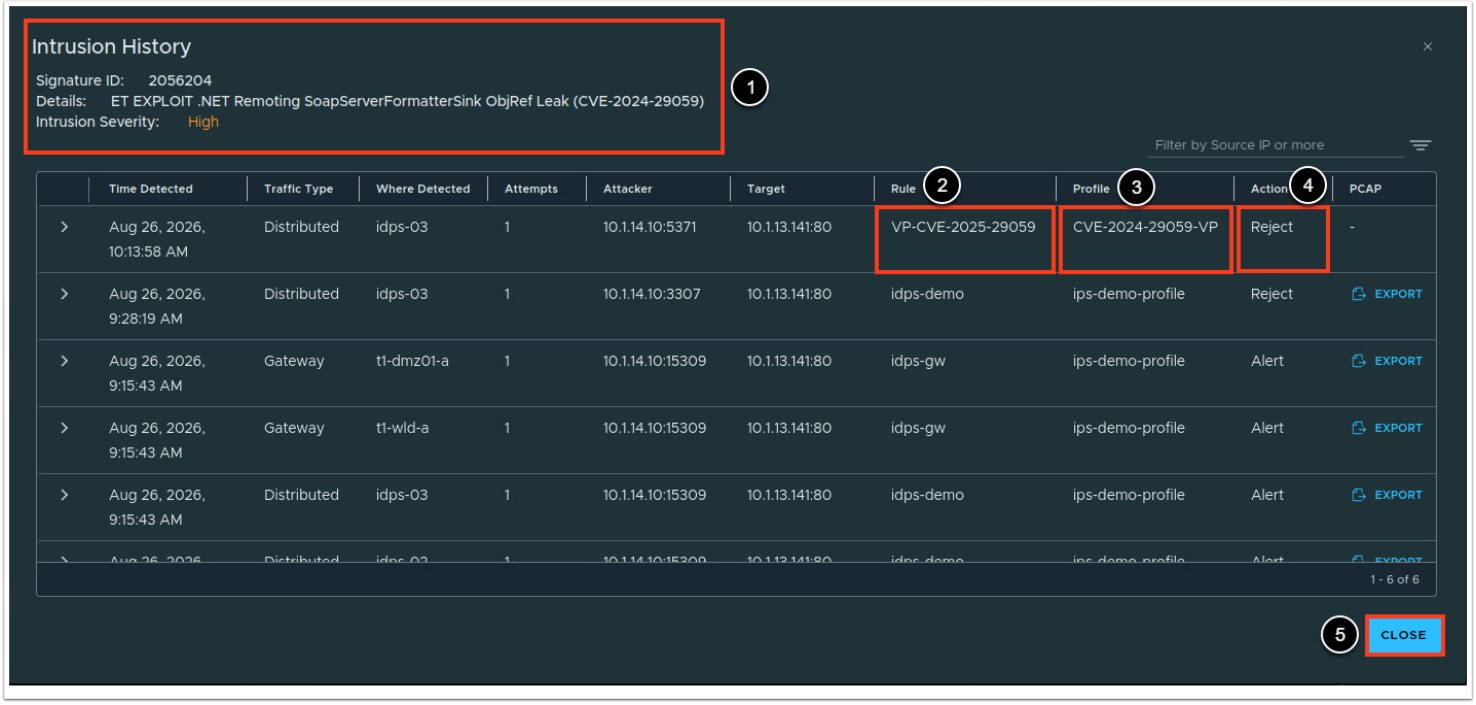
We will review the events again to confirm the threats were prevented using Virtual Patching.

Access the IDS/IPS under Threat Event Monitoring



1. Return to the **NSX Manager** Tab on the browser
2. Click **IDS/IPS**
3. Click the toggle on **Timeline** to disable/hide the timeline view
4. Click **the Arrow on the first event to expose the event details**
5. You will notice a green bar here **- this indicates a block event [A blocked event could take a few seconds to appear]**
6. Click on **View Full Event History**

Viewing Full Event History



1. Here, we can see which IDS/IPS signature was triggered and the details of the attack - **CVE-2024-29059**
2. **Rule:** This indicates the Matching Rule - **VP-CVE-2024-29059**
3. **Profile:** This indicates the Signature Profile matching the event - **CVE-2024-29059-VP**
4. **Action:** This indicates the Matching Rule Action - **Detect & Prevent = Reject**
5. Click **CLOSE**

Conclusion

In this lab, you enabled distributed IDS/IPS for East-West traffic inspection and gateway inspection for North-South flows.

You then created targeted security policies, generated attack flows between test VMs, and reviewed the resulting event logs in NSX Manager.

Finally, you implemented virtual patching by applying targeted signature profiles directly to vulnerable workloads—stopping exploit attempts and shielding them without requiring immediate application downtime or code updates.

From here you can:

- **Continue with the next lab module.**
- **Click [vlp:table-of-contents|Show Table of Contents] to jump to any module or lesson in this lab.**
- **End your lab session and return in the future.**

Module 3 - vDefend Network Traffic Analysis (10min) Beginner

Module 3 - vDefend Network Traffic Analysis (10min) Beginner

The goal of the Suspicious Traffic feature is to detect suspicious or anomalous network traffic behaviors in your NSX environment.

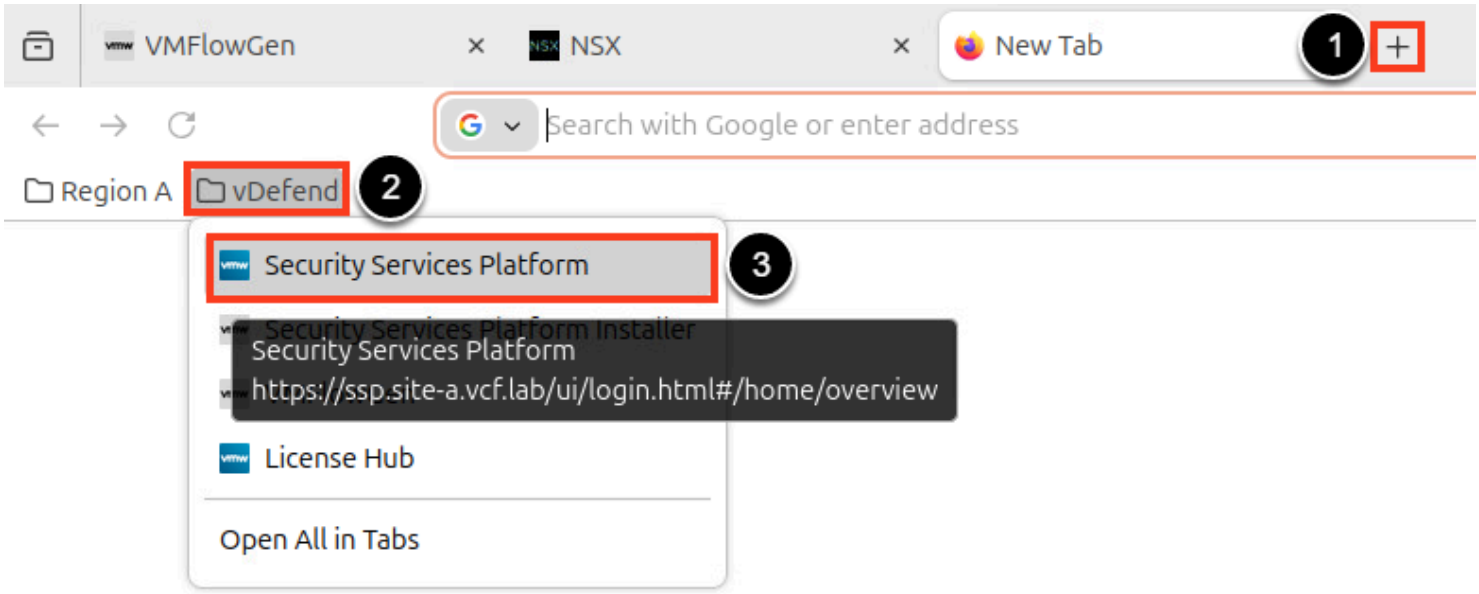
After you have satisfied the prerequisites, the Suspicious Traffic feature can start generating network threat analytics on the east-west network traffic flow data that Network Traffic Analysis has collected from your eligible NSX workloads (hosts or clusters of hosts). Network Traffic Analysis stores the collected data and persists that data for 30 days. The Suspicious Traffic engine analyzes the data and flags suspicious activities using the supported detectors.

Before any threats or suspicious network traffic data can be detected in your NSX environment, normally you'll need to manually enable Suspicious Traffic detectors that you want Network Traffic Analysis to use. Only the detectors that are activated are used for monitoring suspicious network traffic events. Within this lab we have already activated all of the detectors.

Access Security Service Platform (SSP)

Now you will open the log in page to the Security Services Platform UI

Open a New Browser Tab for SSP



1. Open a new tab in the Firefox browser.
2. Click the vDefend **Folder.**
3. Click on **Security Services Platform.**

Log in to the Security Services Platform



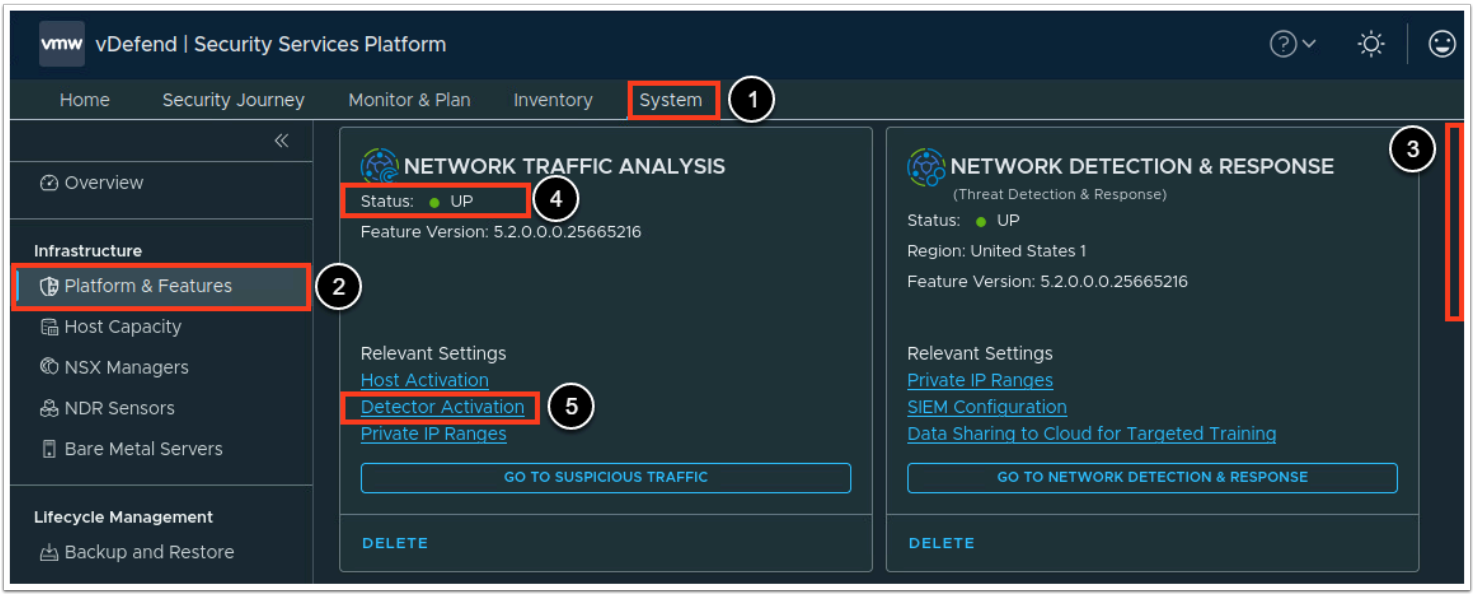
The credentials for admin **should already be cached in the browser window.**

1. Login Method: Select **Local Account**
2. **Username:** admin
3. **Password:** VMware123!VMware123!
4. Click **LOGIN**

Activate the Suspicious Traffic Detectors

The detectors have already been activated within the Security Services Platform, however lets validate they're activated.

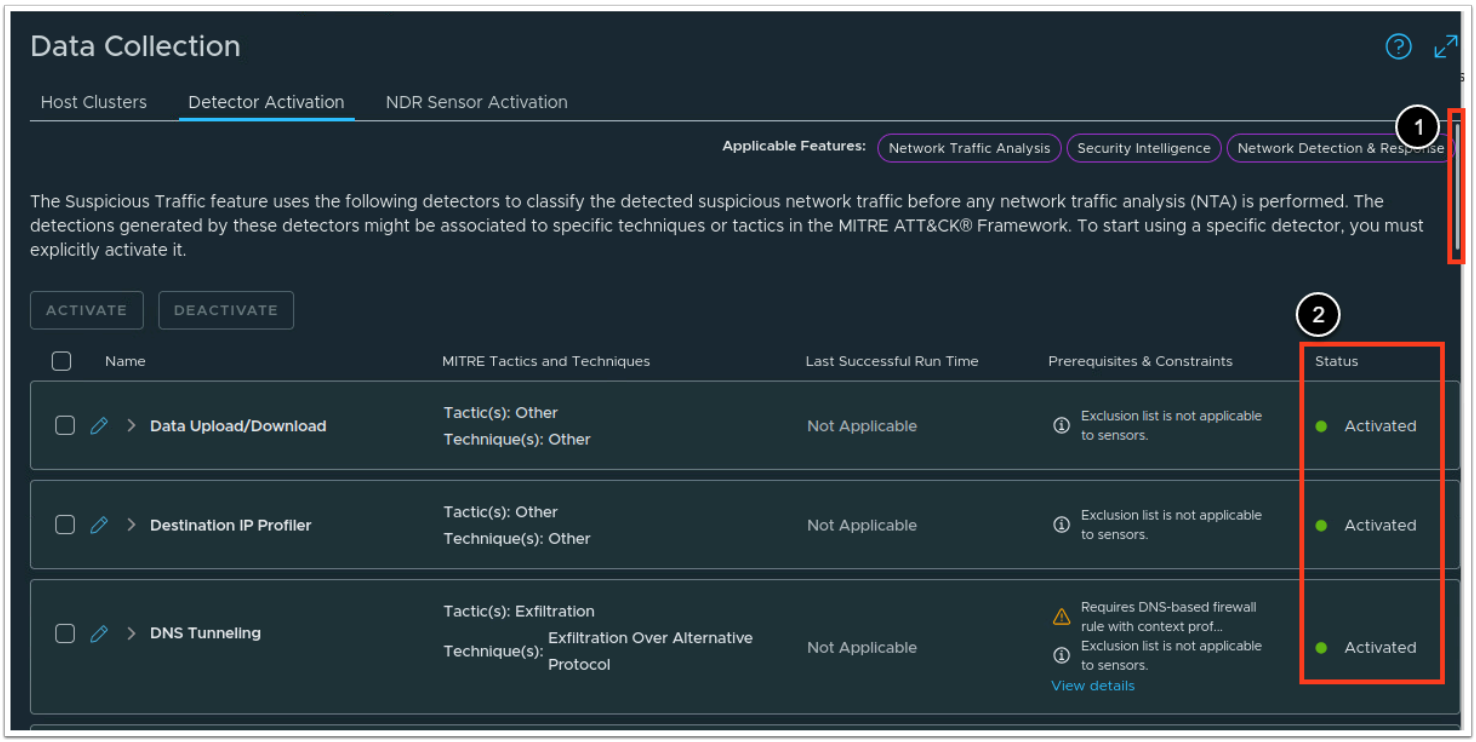
Access Security Services Platform



1. From the SSP dashboard, click **System**
2. Click **Platform & Features**
3. **Click and hold** the slide bar and page **down**
4. Validate Network Traffic Analysis is in an **UP** Status
5. Click Detector & Activation, **in the Network Traffic Analysis Tile**

> Network Traffic Analysis has already been enabled as this process can take a few minutes to enable

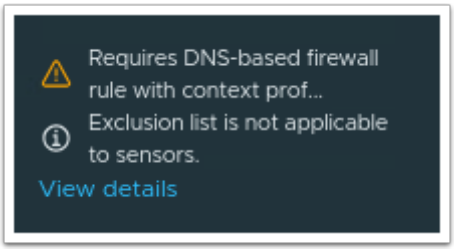
Confirm Detector Activation



In this lab ALL Detectors have been pre-enabled.

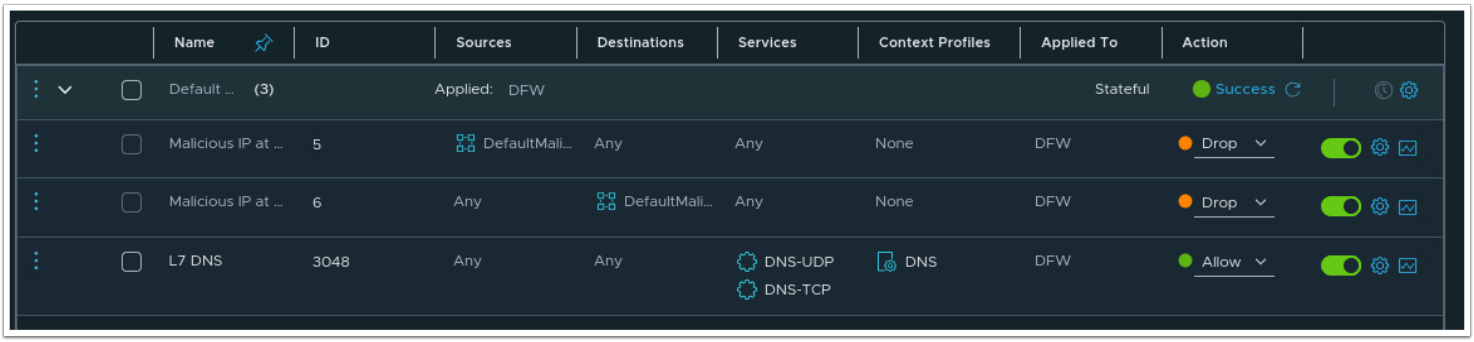
1. **Click and hold** the slide bar and page **down**
2. Validate all Detectors are in an **Activated** state

Detector Activation - DNS Detector Prerequisite



The prerequisite DNS rule is preconfigured in the lab.

Example of Prerequisite Preconfigured DNS Rule



Example of the prerequisite DNS rule that has been preconfigured within NSX for this lab.

Conclusion

There is no specific lab in this module, but it will be covered in Module 4.

From here you can:

- **Continue with the next lab module**
- **Click [vlp:table-of-contents|Show Table of Contents] to jump to any module or lesson in this lab**
- **End your lab and return in the future**

Module 4 - vDefend Network Detection & Response (25Min) (Intermediate)

Module 4 - vDefend Network Detection & Response (25Min) (Intermediate)

The main objective of the vDefend Network Detection and Response (NDR) feature is to collect key abnormal activity or malicious events from every activated event source in your NSX environment. Following the MITRE ATT&CK® model, vDefend Network Detection and Response processes the detection events that are generated from an NSX-managed network. vDefend Network Detection and Response aggregates and correlates these security-related events to present users with a visualization of the specific threats based on the tactics and techniques that are described in the MITRE ATT&CK framework.

vDefend Network Detection and Response correlates related events into campaigns. It organizes threat events in a campaign into a timeline that is available for a security analyst to view and triage threat campaigns by correlating threat signals.

In this module, you bring all the vDefend Security features together and demonstrate event correlation using Network Detection and Response. You will use the security intelligence demo topology in FlowGen to generate events after activating the NDR capabilities in the Security Services Platform.

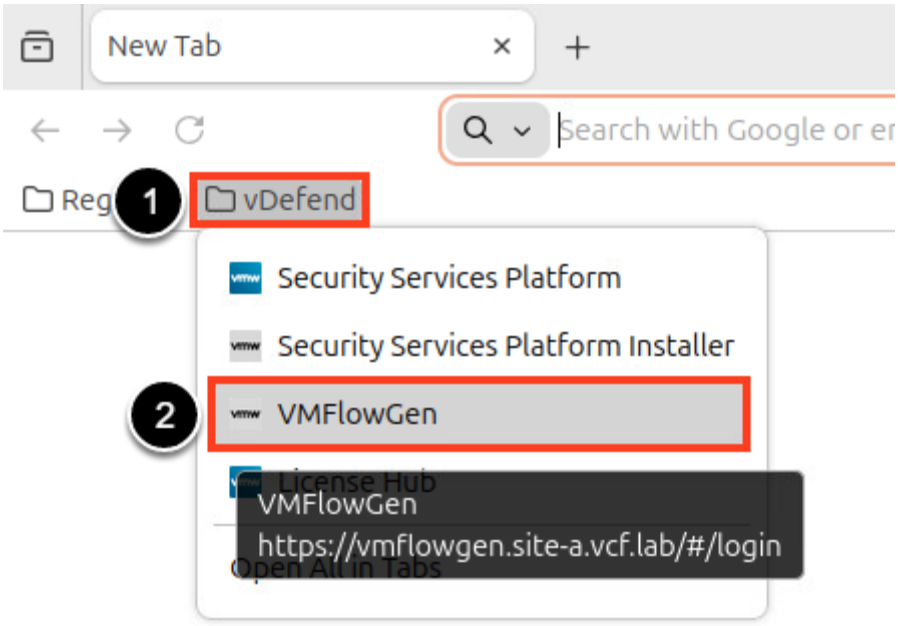


This module assumes that you have activated IDPS, and NTA from Modules 2 and 3.

Access VMFlowGen

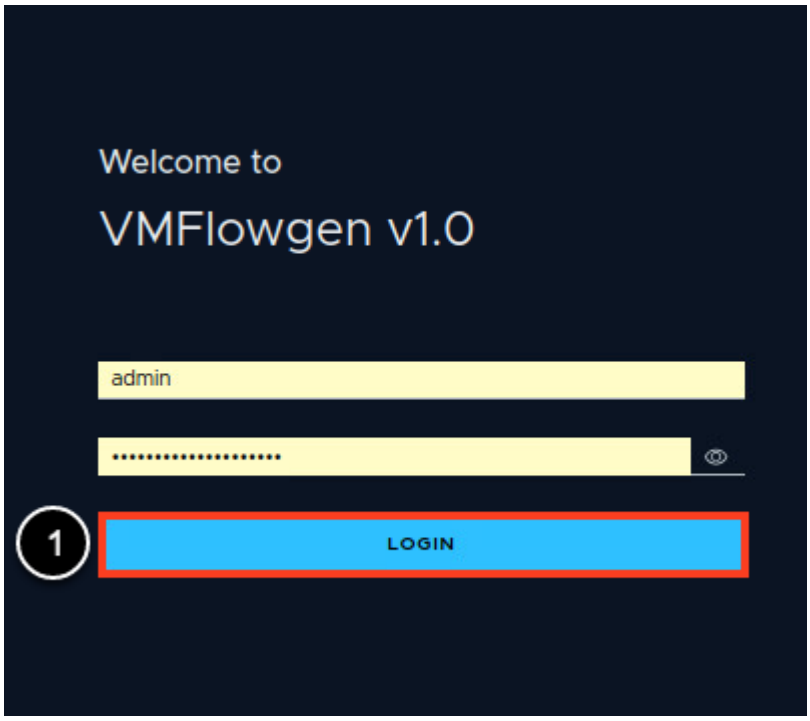
We are going to start by reviewing the application topology in VMFlowGen

Open VMFlowGen in FireFox



1. Select vDefend **from the Bookmark Bar**
2. Select VMFlowGen **from the drop-down menu**

Log into VMFlowgen



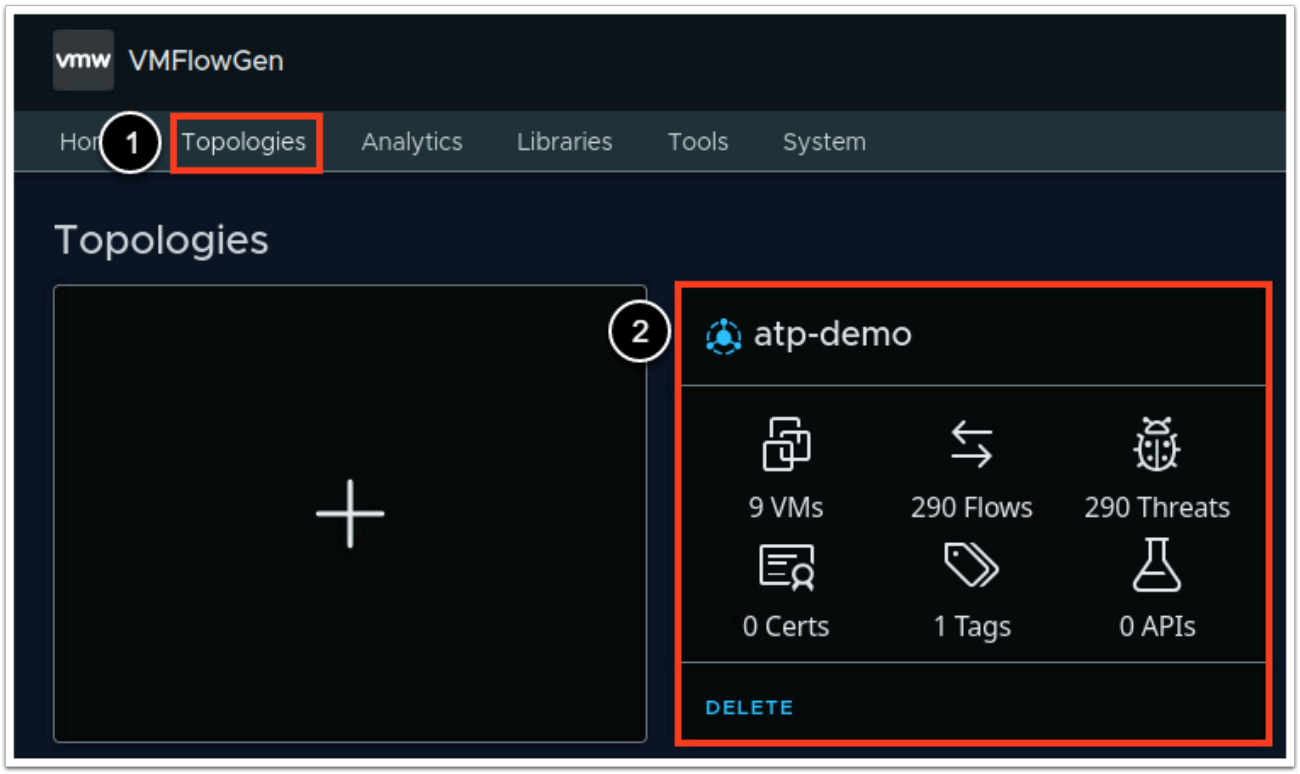
The credentials for admin **should already be cached in the browser window.**

1. Using the pre-populated user account details, select **LOGIN**

   Username: *admin*

   Password: *VMware123!VMware123!*

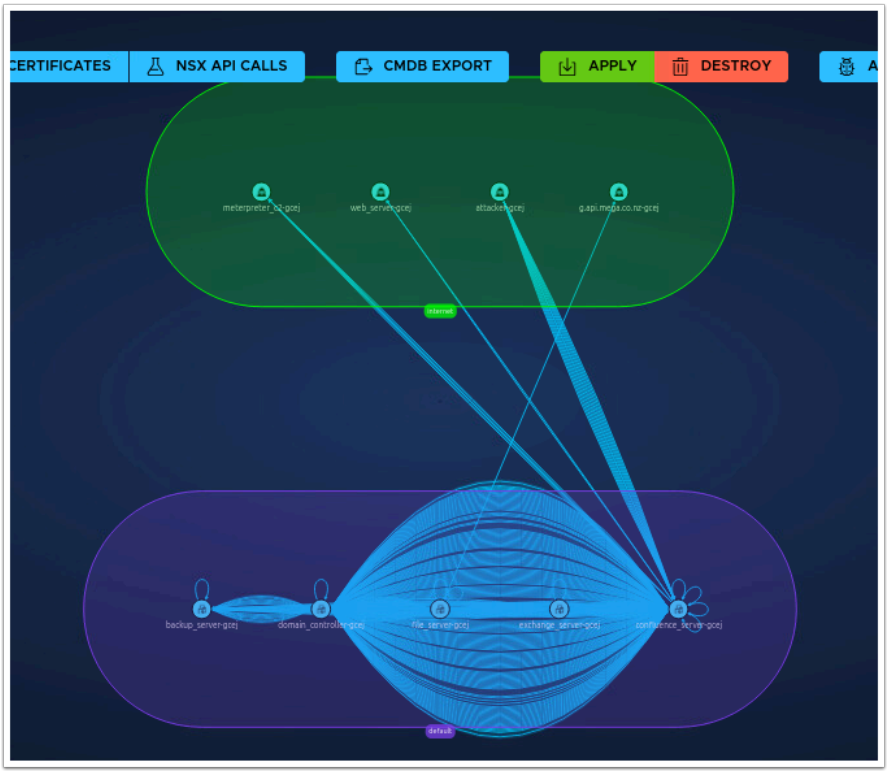
Open atp-demo Topology in VMFlowGen



1. Click **Topologies**
2. Click on the **atp-demo** tile

> The application is currently not deployed as indicated by the blue color of the symbol in the top left of the application card

ATP-Demo Topology



The demo topology used in this Module consists of 9 VM's generating multiple attacks

When the topology has not been deployed the VMs will be a blue color.

Apply Topology in VMFlowGen



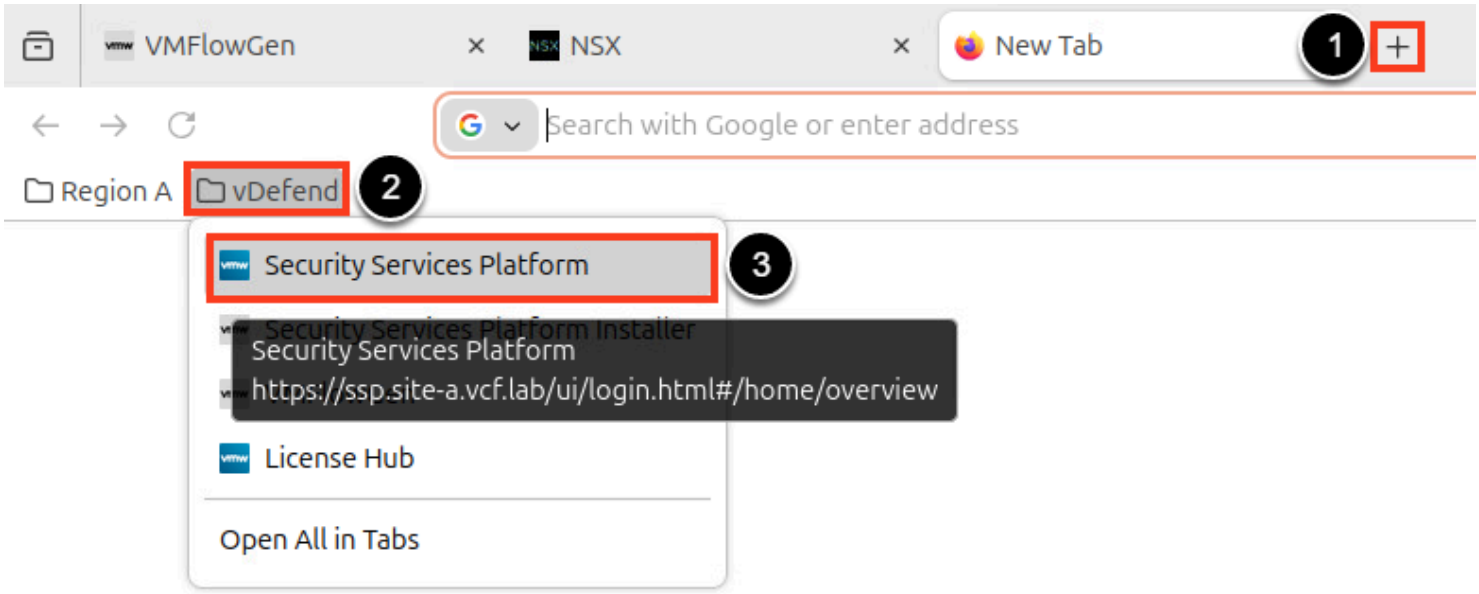
1. Click **APPLY**

> The application will take between 1 to 3 minutes to fully deploy. During this time you will notice a progress bar on the top left of the canvas. If the application fails to deploy, click APPLY again.

Access Security Service Platform

We will now access the Security Services Platform UI to activate Network Detection & Response

Open a New Tab for the Security Service Platform



1. Open a **New Tab** in the Firefox browser
2. Click the **vDefend** Folder
3. Click on **Security Services Platform** to open the login page

Log in to the Security Services Platform



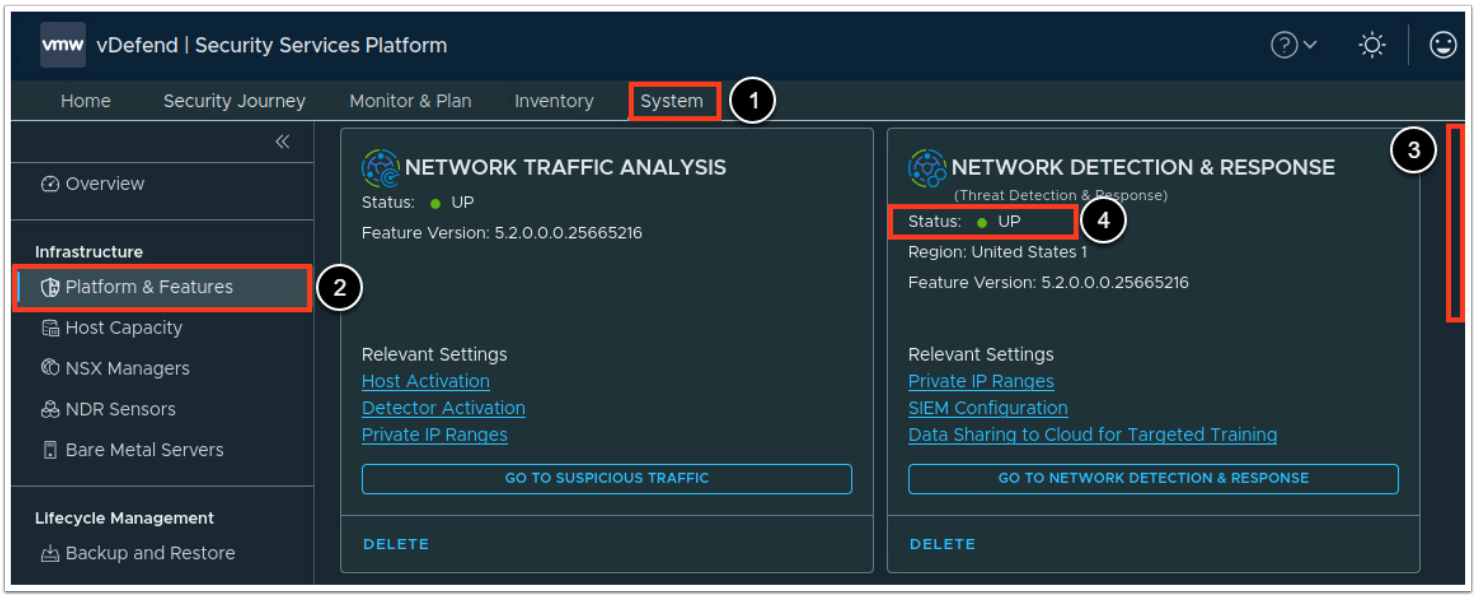
The credentials for admin **should already be cached in the browser window.**

1. **Login Method:** Select **Local Account**
2. **Username:** admin
3. **Password:** VMware123!VMware123!
4. Click **LOGIN**

Validate Features in Security Services Platform

You will now validate Network Detection & Response service is activated in the Security Services Platform. This is where all Security-detection events from IDS/IPS, malware detection, and Network Traffic Analysis will be correlated.

Validate Platform & Features - Network Detection & Response



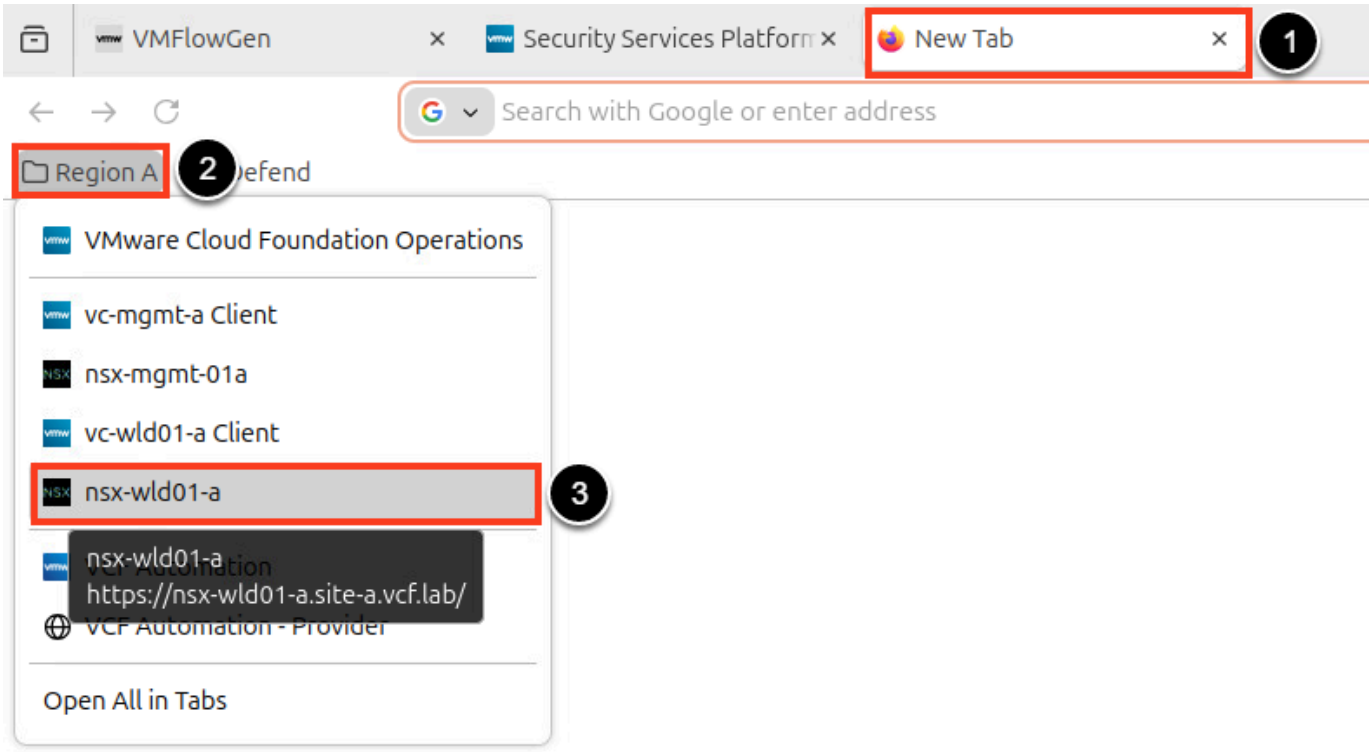
1. Click **System**
2. Click **Platform & Features**
3. **Click and hold** the slide bar and page **down**
4. Validate Network Detection & Response is in an **UP** Status

> Network Detection & Response has already been enabled as this process can take a few minutes to enable

Access the NSX Manager

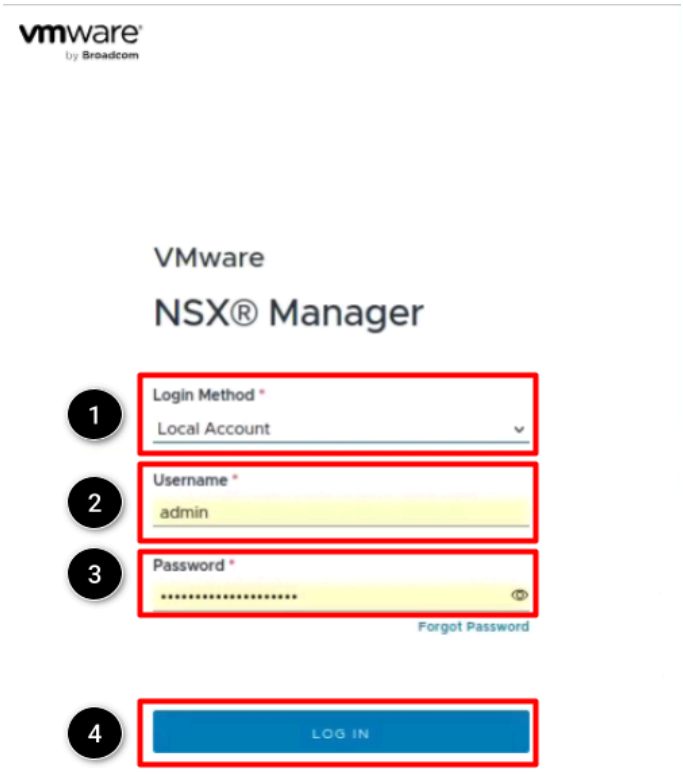
We need to add security policies in the NSX Manager and apply them to the VM's in our atp-demo topology.

Open the NSX Manager TAB



1. Open a new TAB in the browser
2. Click the **Region A** folder
3. Click **nsx-wld01-a**

Log In to NSX Manager



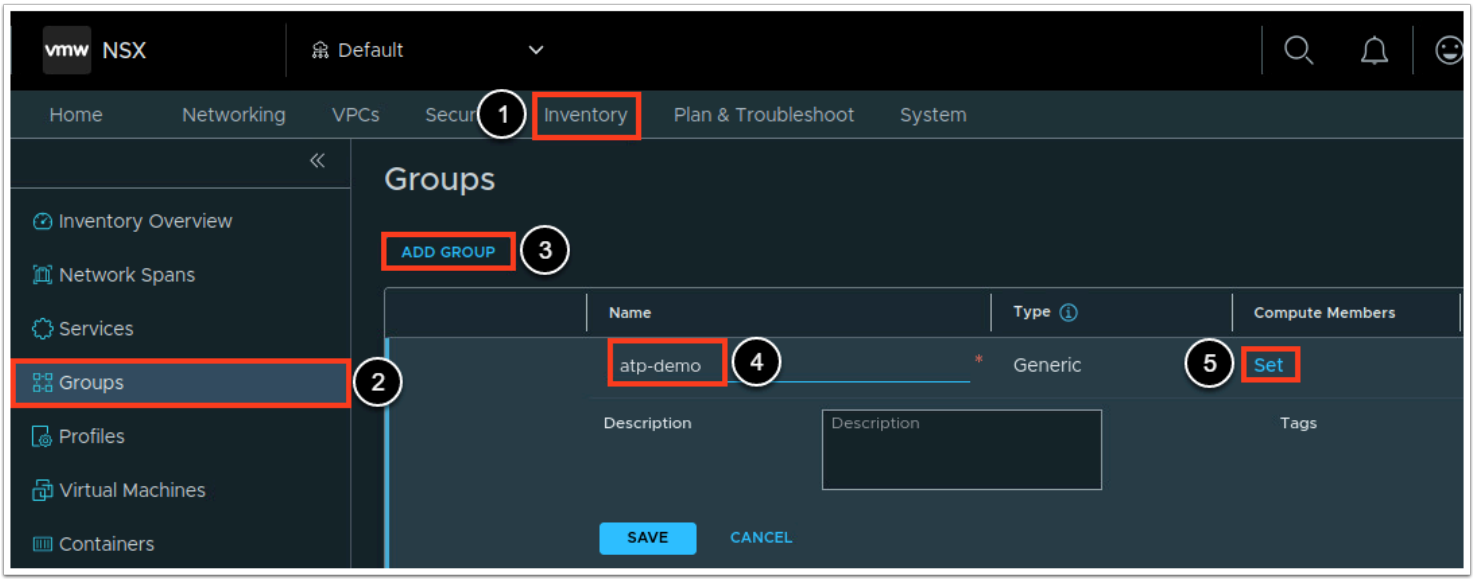
The credentials for admin **should already be cached in the browser window.**

1. Make sure the Login Method is set to **Local Account**
2. Username: admin
3. Password: **VMware123!VMware123!**
4. Select **LOG IN**

Create an NSX Group

You will be creating an NSX Group which will be used in the security policies, the group will include the three idps-demo VM's

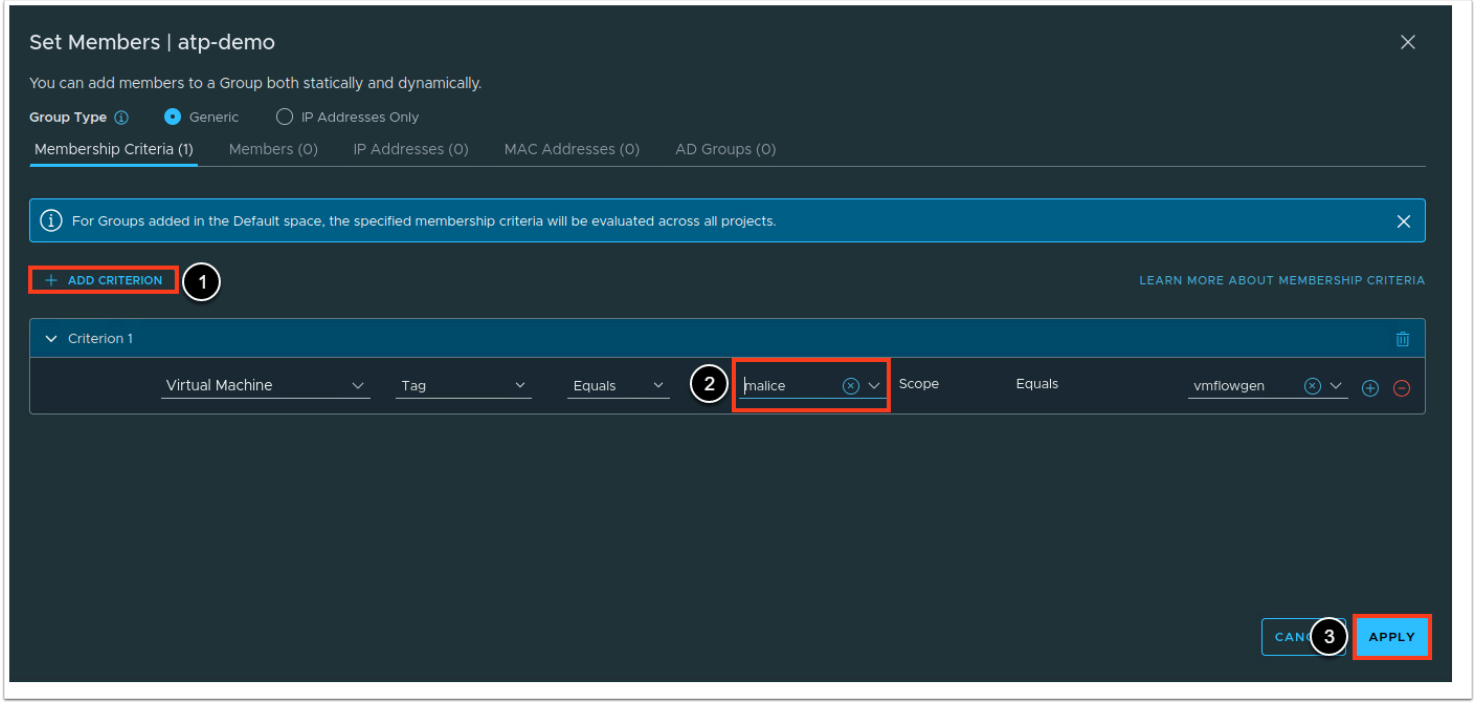
Creating an atp-demo Group



Create an NSX Group with a name atp-demo, matching a predefined Virtual Machine Tag *malice*.

1. Click **Inventory**
2. Click **Groups**
3. Click **ADD GROUP**
4. Set the group name to **atp-demo**
5. Click **Set**

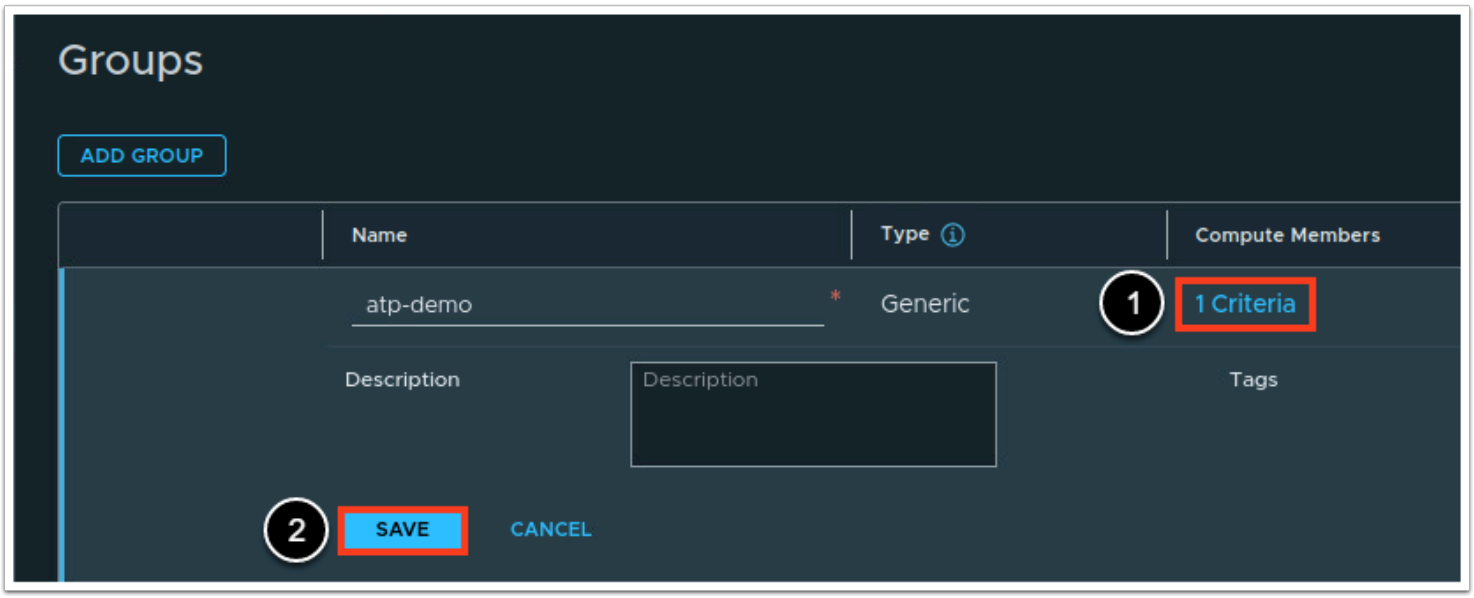
Set the group membership criteria as VM Tag



Here we need to define the match criteria in the group using the predefined tag

1. Click **ADD CRITERION**
2. Type ***malice*** and select the preconfigured tag
3. Click **APPLY**

SAVE group members

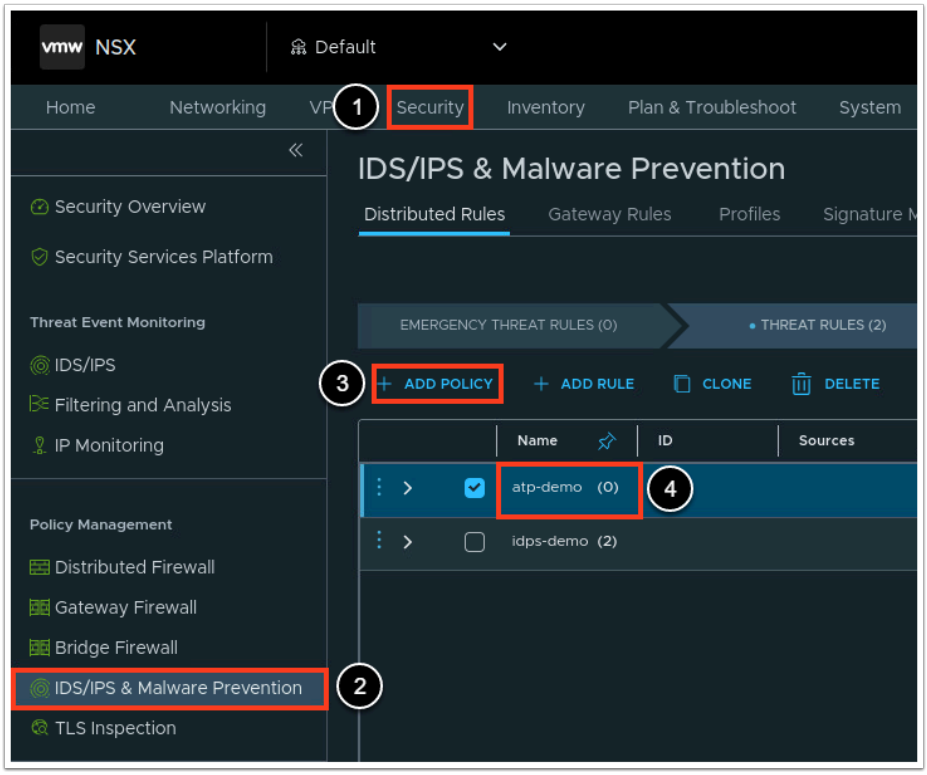


1. Confirm **1 Criteria**
2. Click **SAVE**

Adding Security Policies in the NSX Manager

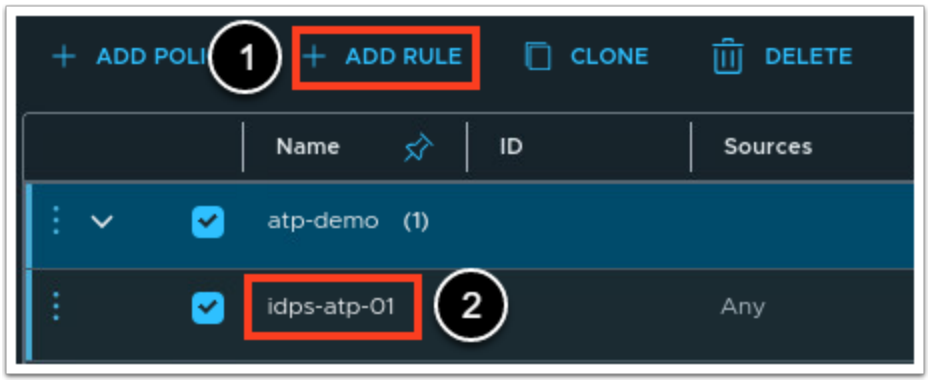
Next we'll need to create new IDS/IPS & Malware Prevention Policy.

Create Distributed IDS/IPS ATP Demo Policy



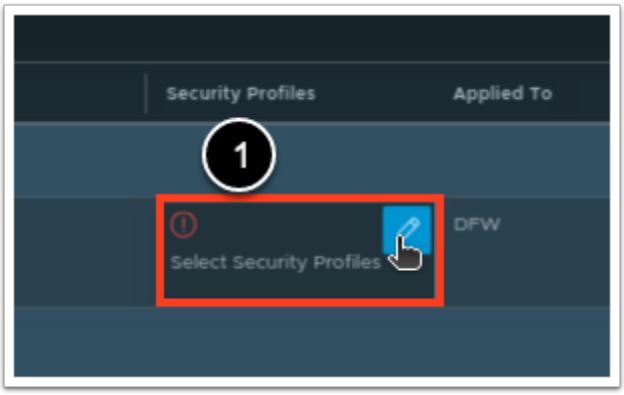
1. Click **Security**
2. Click **IDS/IPS & Malware Prevention**
3. Click **ADD POLICY**
4. Rename the new policy to **atp-demo**

Create Distributed IDS/IPS Rule



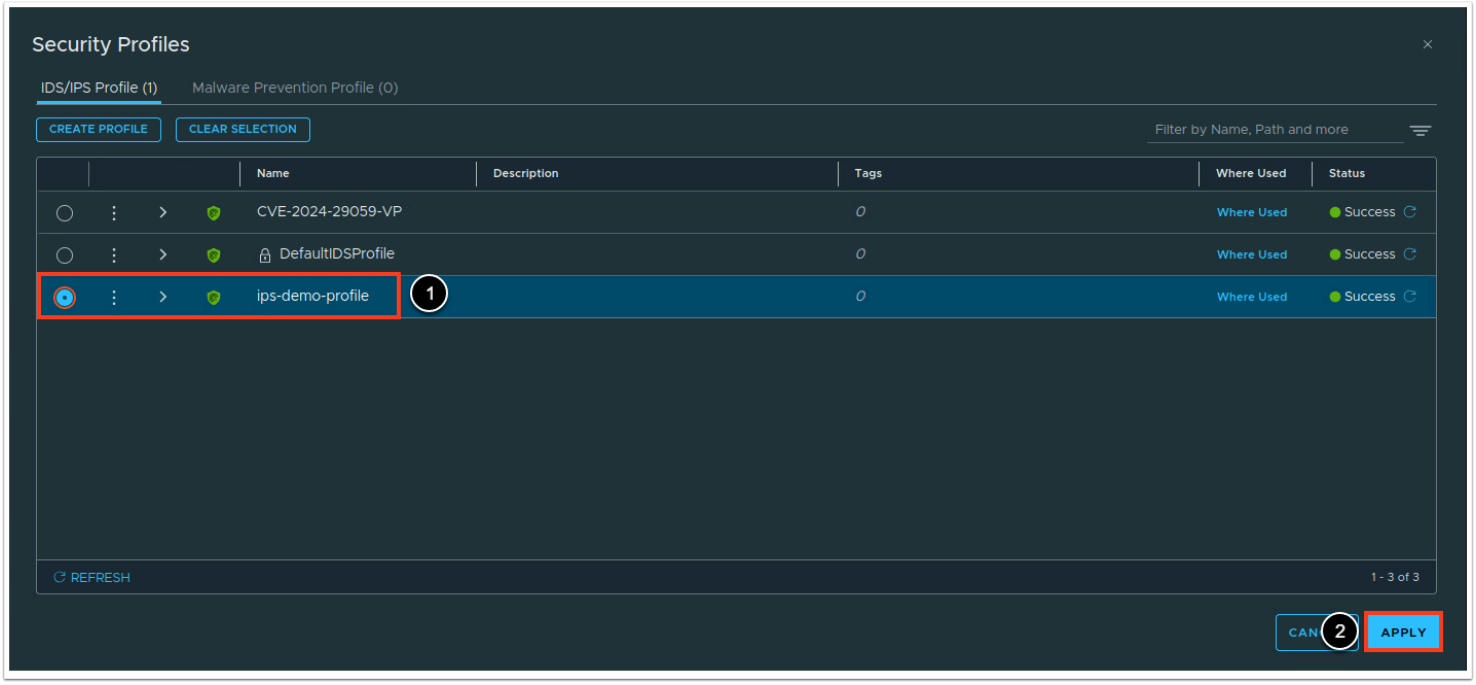
1. Click **ADD RULE**
2. Rename the new rule to **idps-atp-01**

Configure the Security Profile in the rule



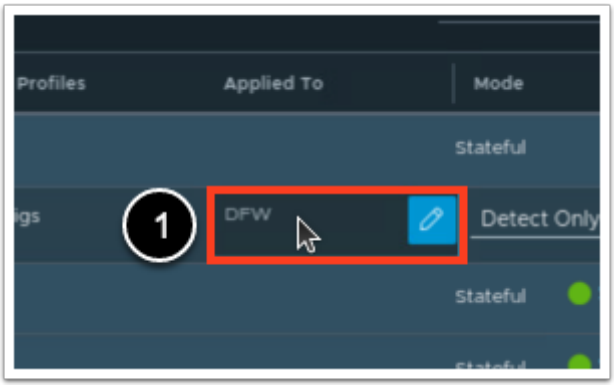
1. Click on the pencil to select a security profile

Select Distributed IDS/IPS Profile



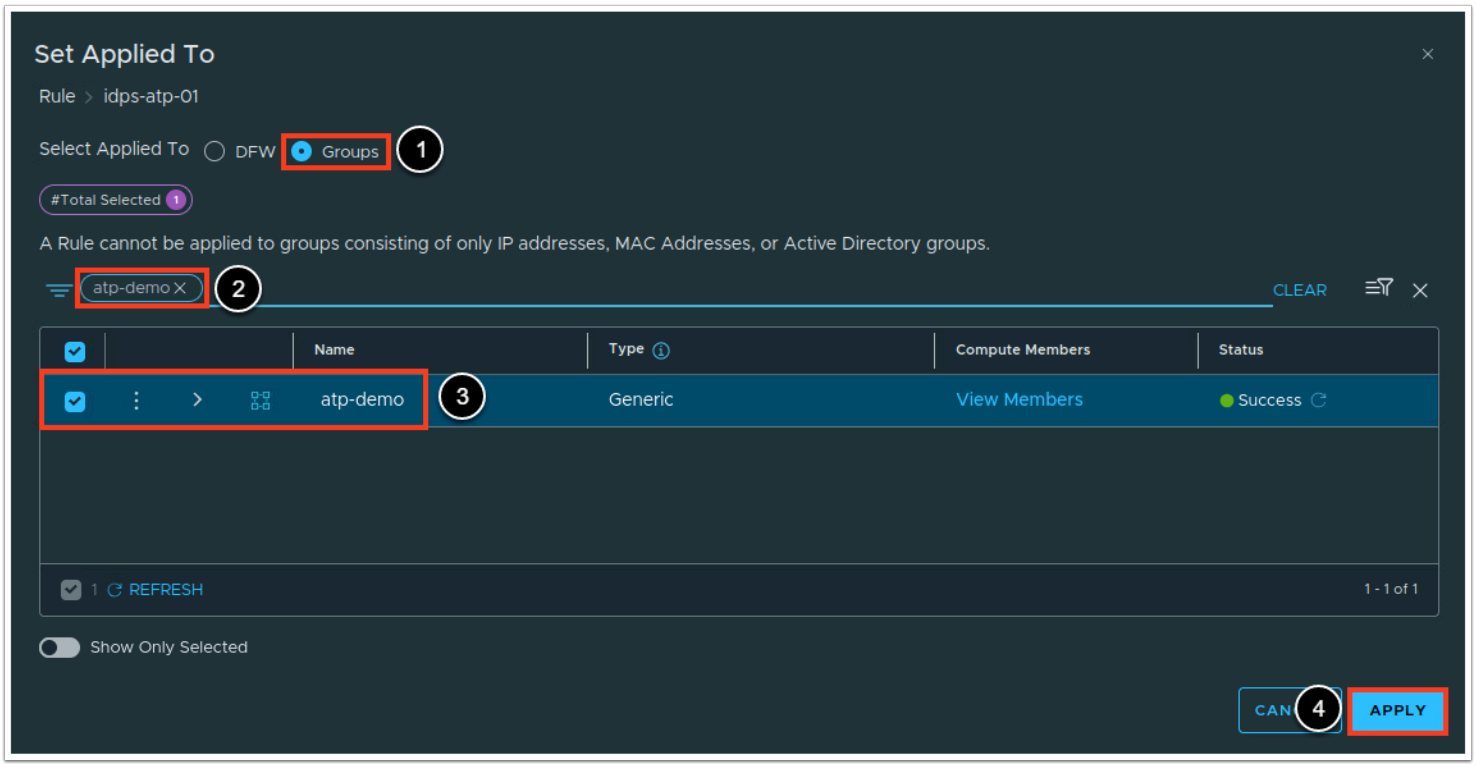
1. Select the IDS/IPS Profile: **ips-demo-profile**
2. Click **APPLY**

Edit Distributed IDS/IPS Applied To Group



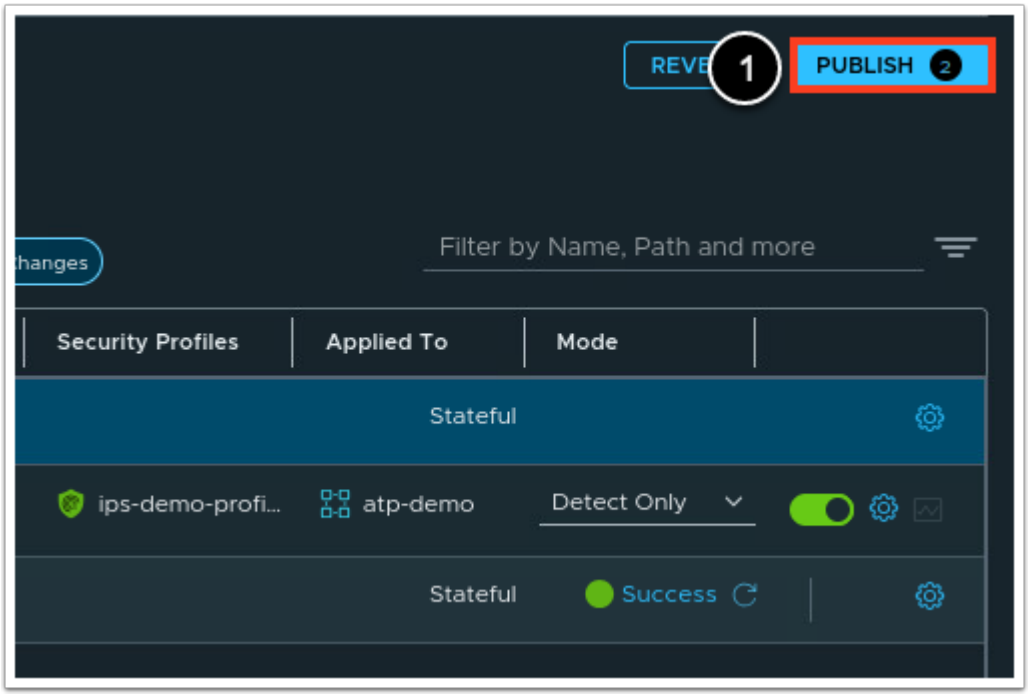
1. Click on the pencil under Applied To where it says **DFW**

Select Distributed IDS/IPS Applied To Group



1. Click **Groups**
2. Search for **atp-demo** in the search bar
3. Select the **atp-demo** group
4. Click **APPLY**

Publish the ATP-Demo Policy



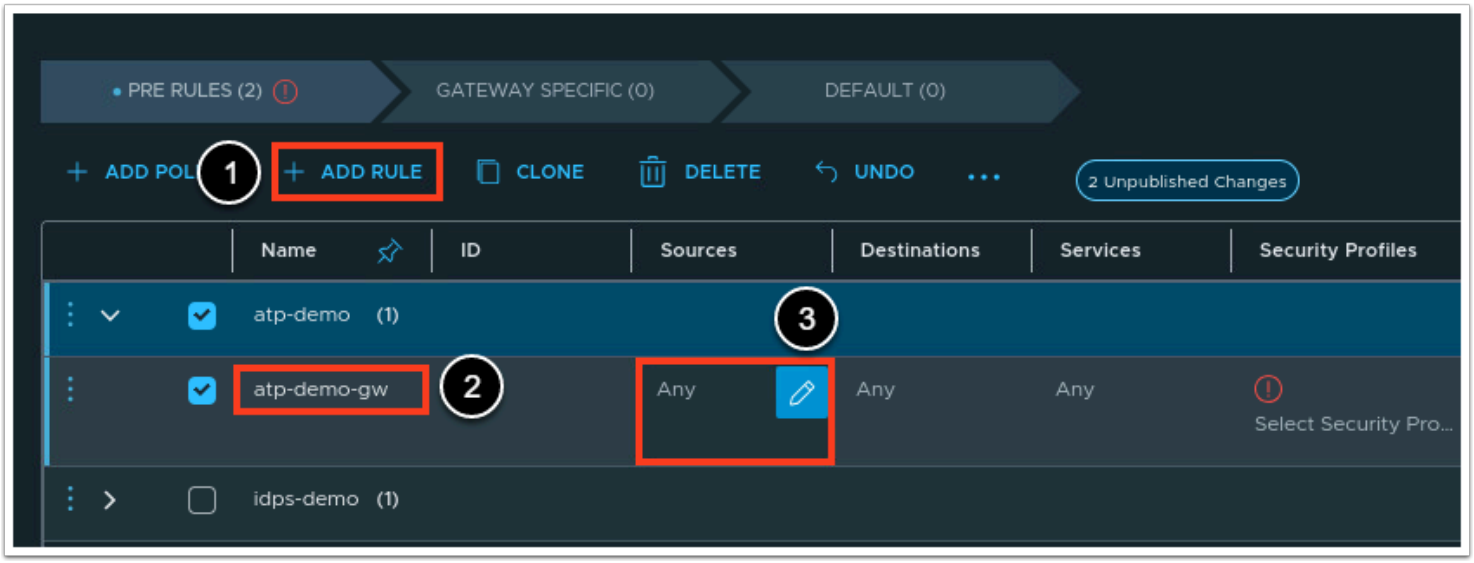
1. Click **PUBLISH**

Create ATP-Demo Gateway Policy



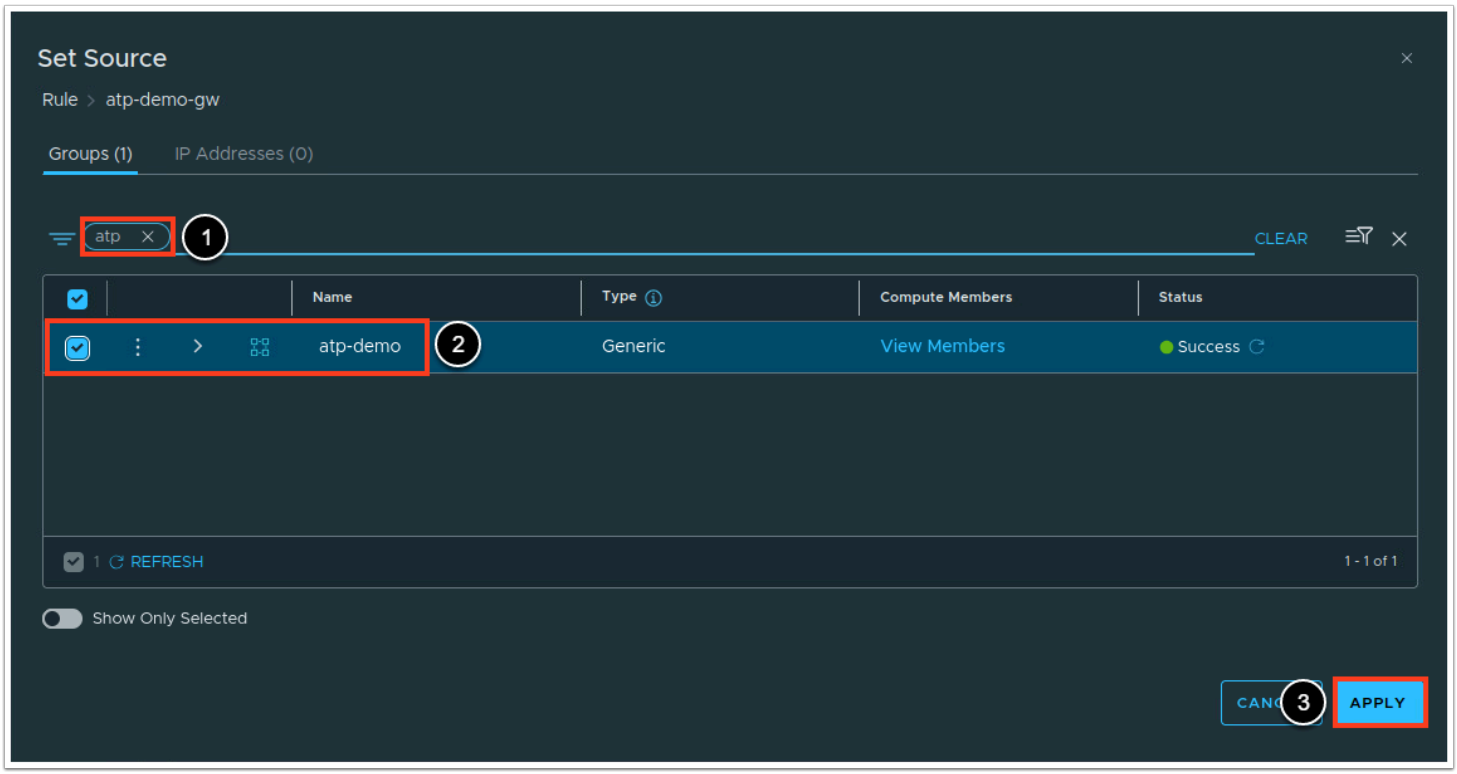
1. Click **Gateway Rules**
2. Click **All Shared Rules**
3. Click **ADD POLICY**
4. Rename the new policy: **atp-demo**

Add ATP-Demo IDS/IPS Gateway Rule



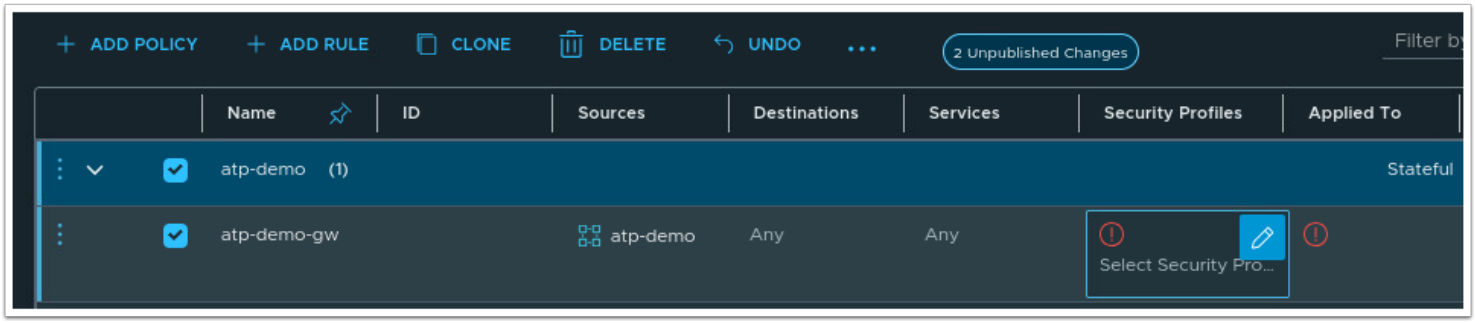
1. Click **ADD RULE**
2. Rename the new rule: **atp-demo-gw**
3. Click on the pencil in the **Sources** field

Set the Source Group



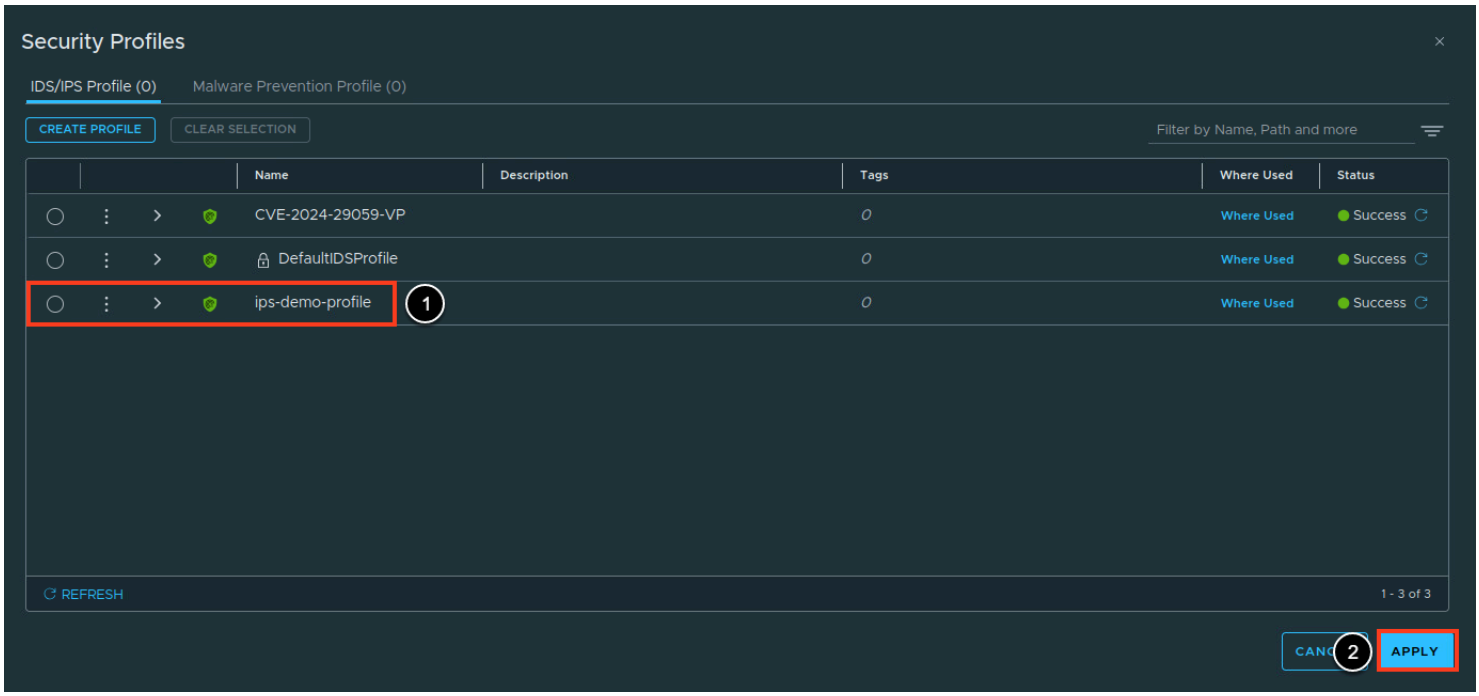
1. In the filter field, search **atp-demo**
2. Select **atp-demo** group
3. Click **APPLY**

Select IDPS/IPS Security Profile



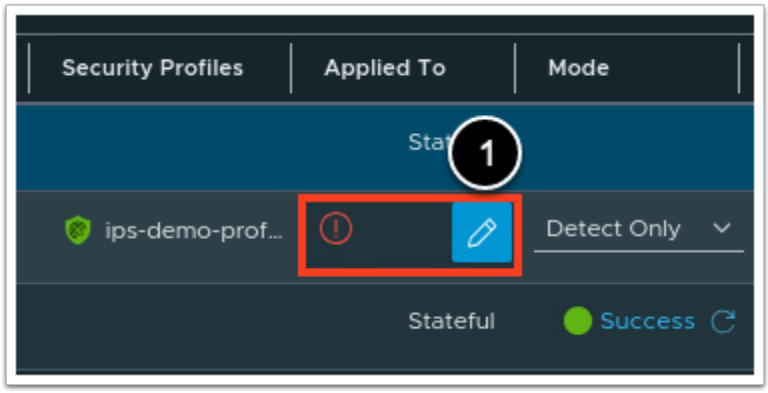
1. Click on the pencil in the **Security Profiles** field

Select IDPS/IPS Security Profile



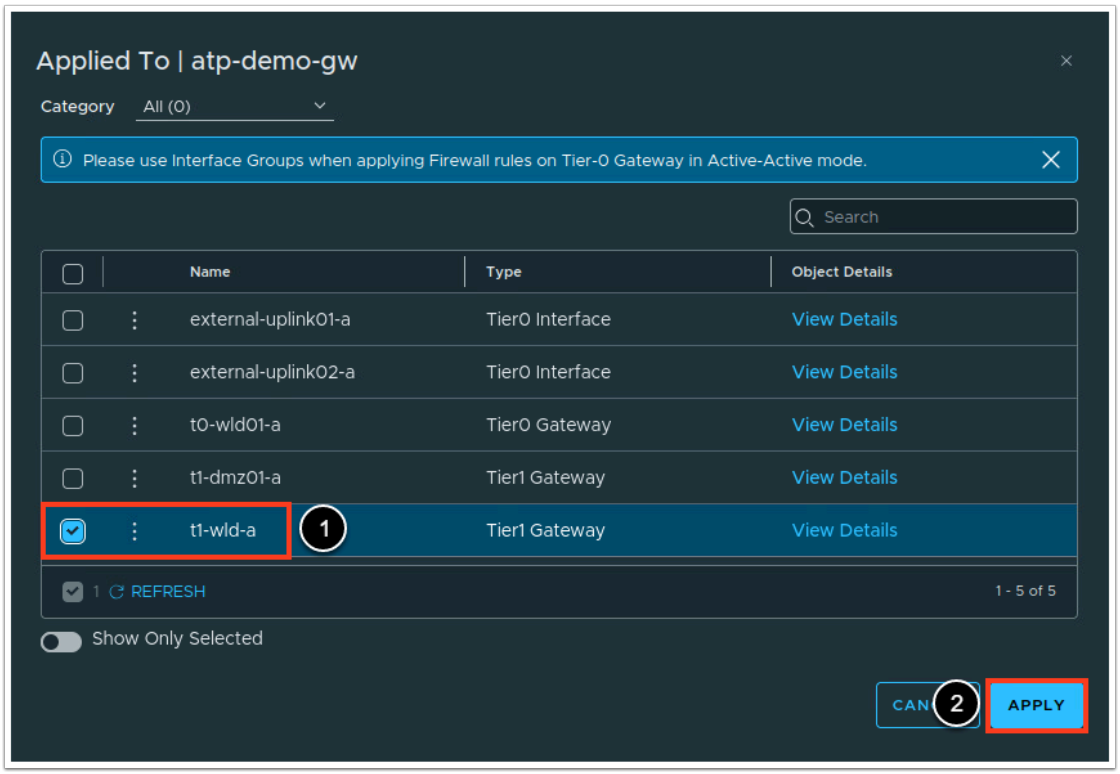
1. Select the **ips-demo-profile** Security Profile
2. Click **APPLY**

Edit the Gateway rule Applied To



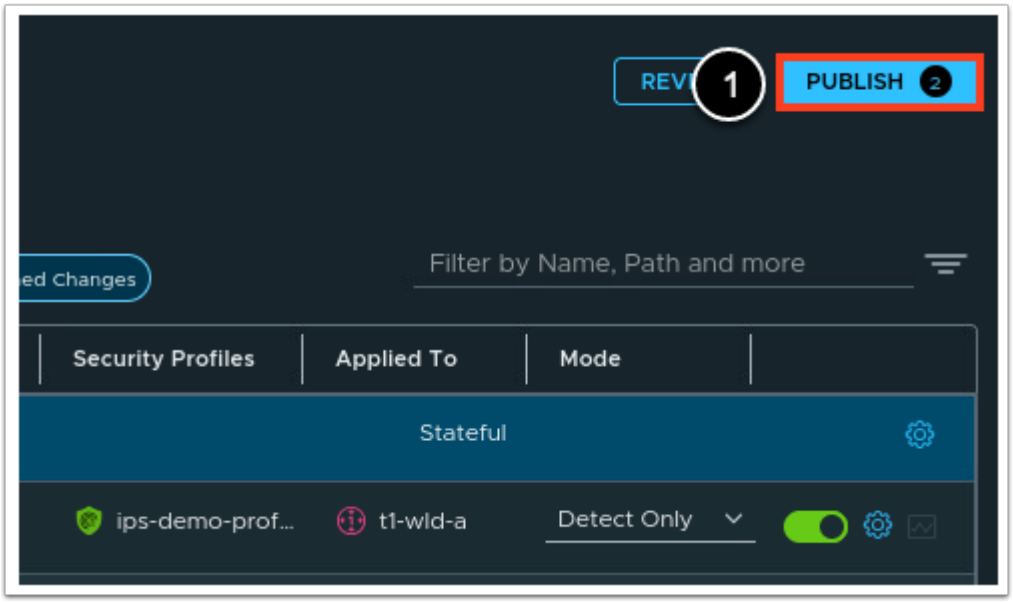
1. Click on the pencil under **Applied To** field

Apply the rule to the T1 Gateways



1. Use the scroll bar on the right and page down
2. Click on the tick box next to Name, this will select **t1-wld-a**
3. Click **APPLY**

Publish the Gateway ATP-Demo Policy and Rules

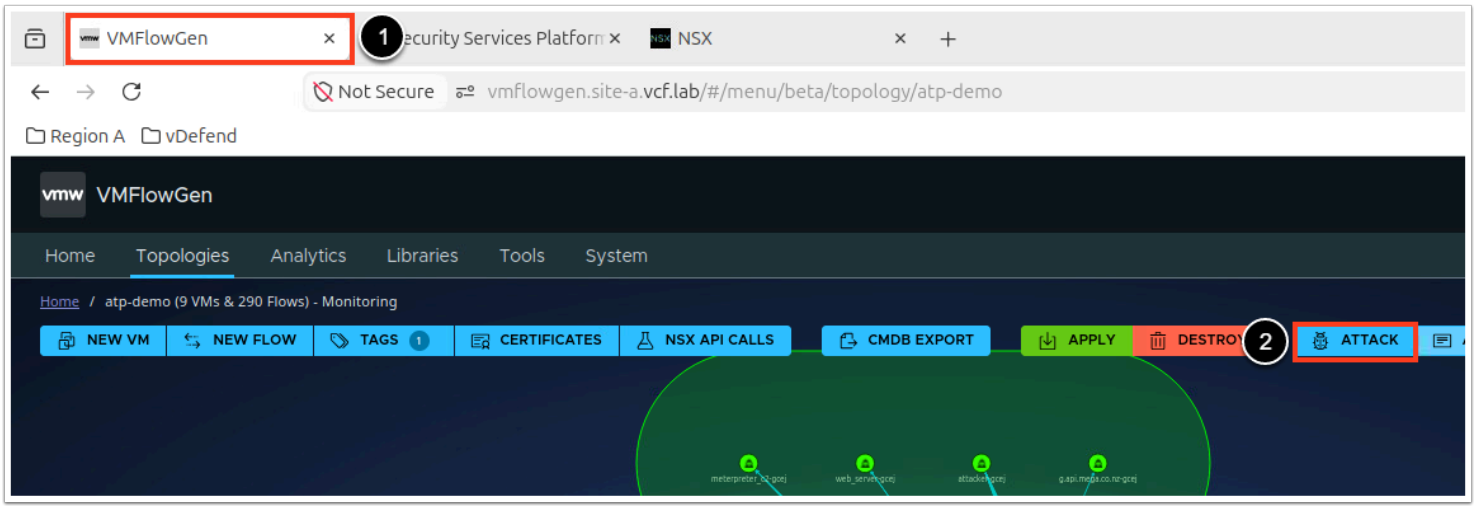


1. Click **PUBLISH**

ATP Demo

Next we'll need to run the atp-demo attack sequence.

Access VMFlowGen



If FlowGen login timed out, login in again the cached admin account and return to the ATP topology

1. Return to the **VMFlowGen Tab** in Firefox
2. Click **ATTACK** in the atp-demo topology

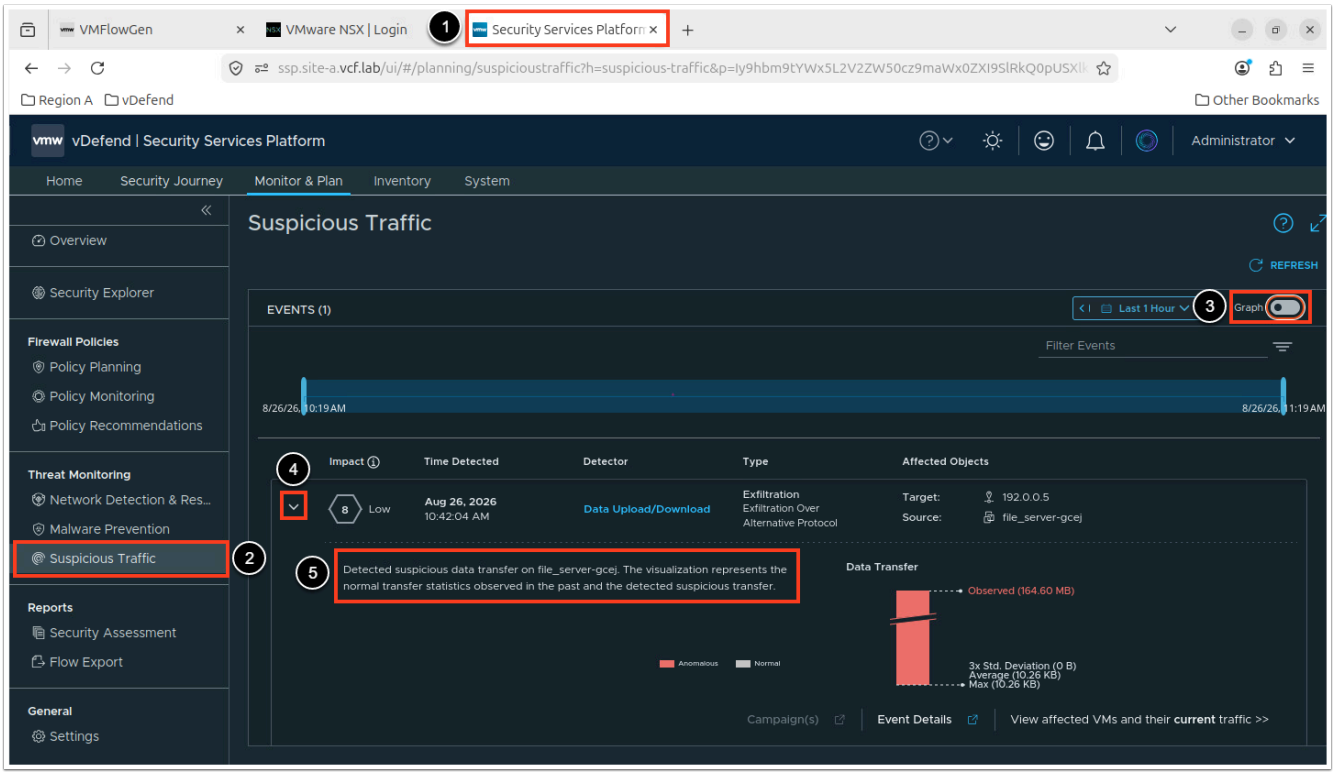
Review Flow Sequence and Start Attack



1. **Review the flow sequence**
2. Click **START** to start replaying the campaign

> The attack sequence will take about 2 minutes to complete

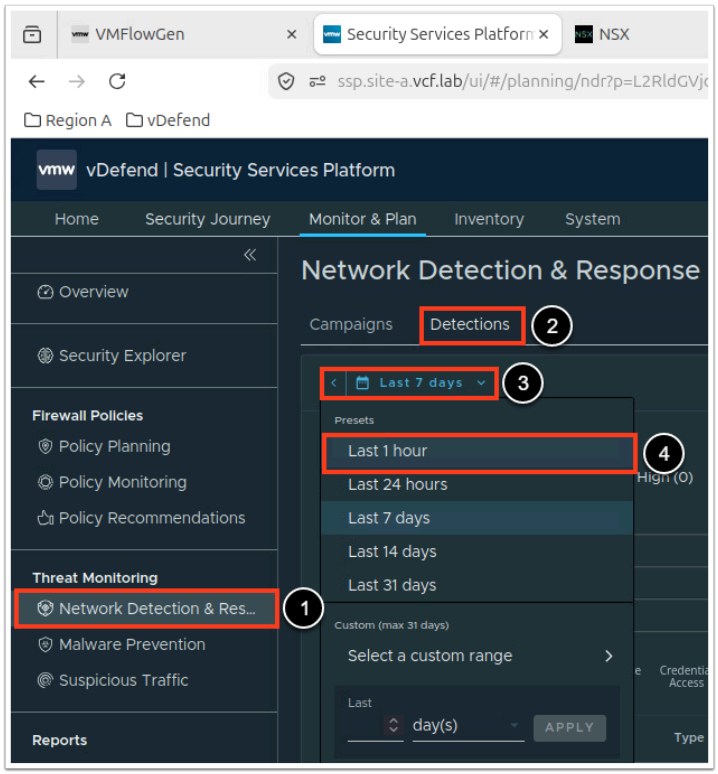
Review Suspicious Traffic



Part of the atp-demo attack sequence has a files being uploaded.

1. Click on the **Security Services Platform Tab** in Firefox
2. Click **Suspicious Traffic**
3. Click **Graph**
4. Click **Expander** for the **Data Upload / Download** Detector
5. Suspicious data transfer was detected on **file_server-gcej**

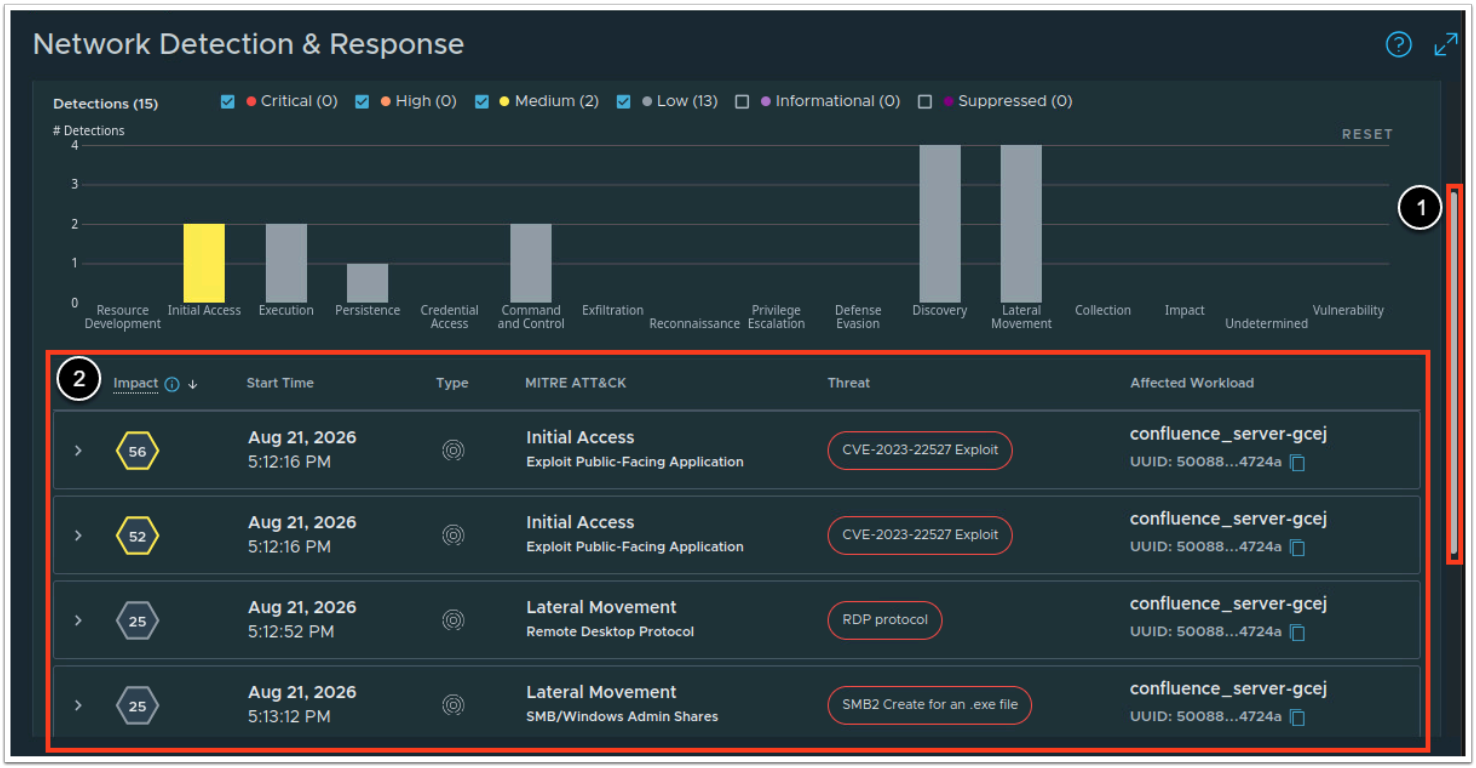
Review the events in Network Detection & Response



1. Click **Network Detection & Response**
2. Click **Detections**
3. Click on **Last 7 Days**
4. Set to **Last 1 Hour**

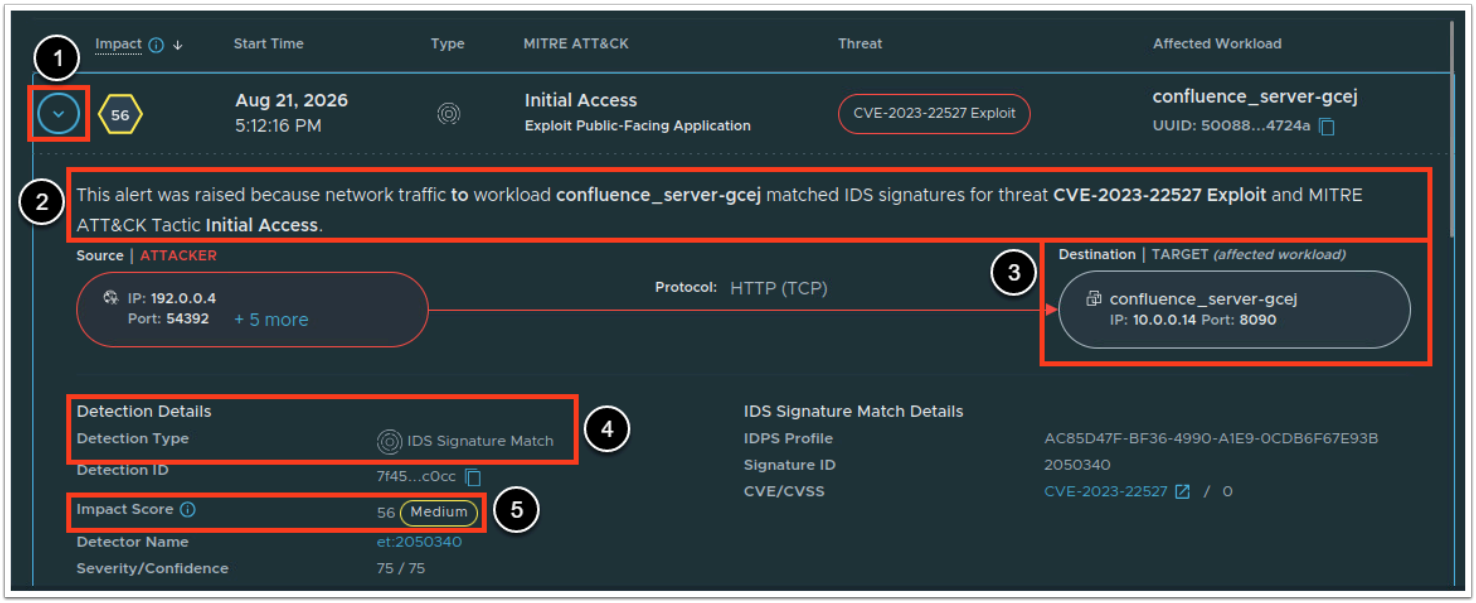
> Campaign creation and event correlation does take some time

Monitor Network & Detection Events



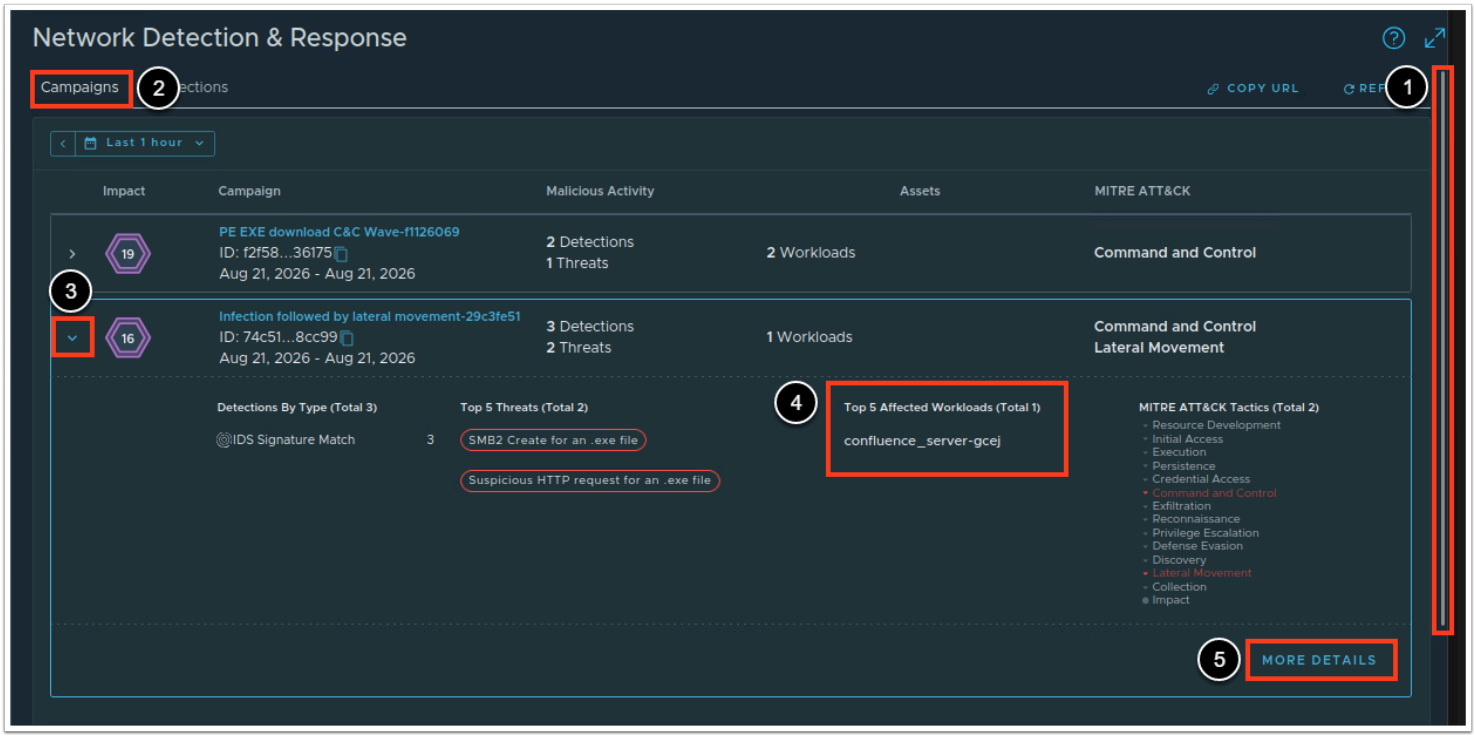
1. Use scrollbar on the right to page down see the full list of detection events
2. List of detection events

Detailed View - Network Detection & Response



1. Click the **expander** of one of the Impact **56** event for Affected Workload **confluence_server-gcej** in the list to expand the event details
2. Here we can why this alert was triggered - **IDS Signature - CVE-2023-22527**
3. Here we get to see which workload the event was triggered by - **confluence_server-gcej**
4. Here we get to see the Detection Type - **IDS Signature Match**
5. Here we can see the calculated Impact Score - **56 / Medium**

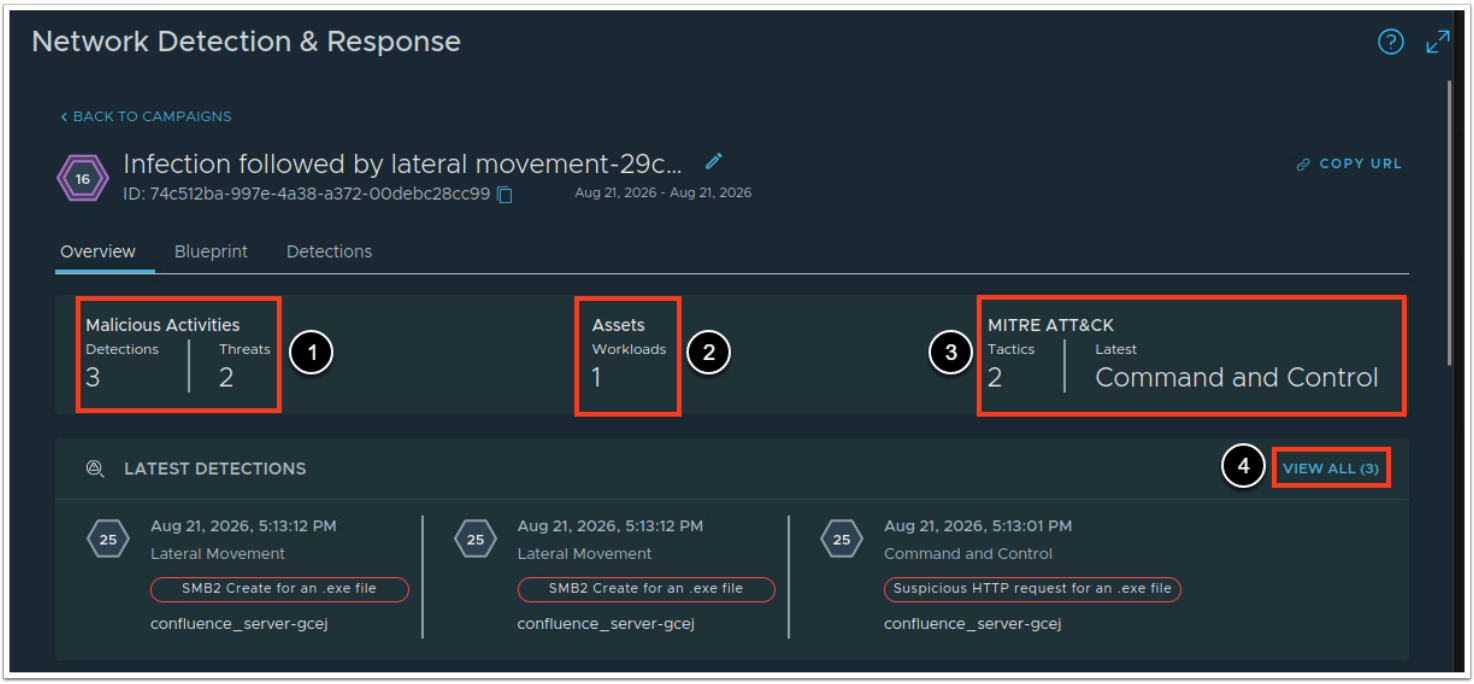
Review Campaign



1. Use scrollbar on the right to page up to the top
2. Click **Campaigns**
3. Click **expander** for the Command and Control / Lateral Movement event
4. Validate Affected Workloads is **confluence_server-gcej**
5. Click **MORE DETAILS**

> Campaign creation and event correlation does take some time

Review Campaign

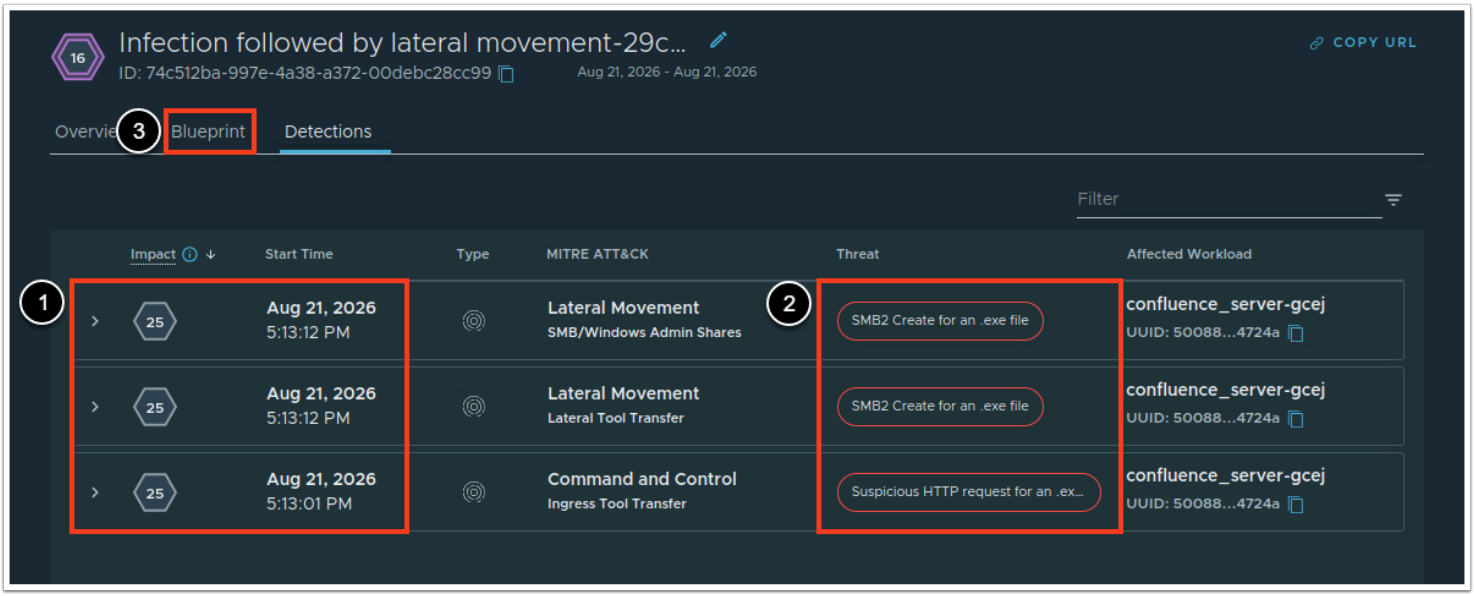


1. Here we can see that this event is part of a larger campaign of events which the NDR engine has correlated
2. We see the total numbers Assets/workloads where events have been detected and correlated into this campaign
3. Here see the number of Mitre Att&ck Tactics matched in the campaign and highlights the latest
4. Click **VIEW ALL**

Review Campaign Detections

When new detection data is received by the Network Detection and Response system, the data is compared with existing detection events to determine if the new detection can be aggregated with an existing detection event. Detection events correlate together to form a campaign. If a detection event has no correlation with any other detection, it does not get included in any campaign.

Review Campaign Detections

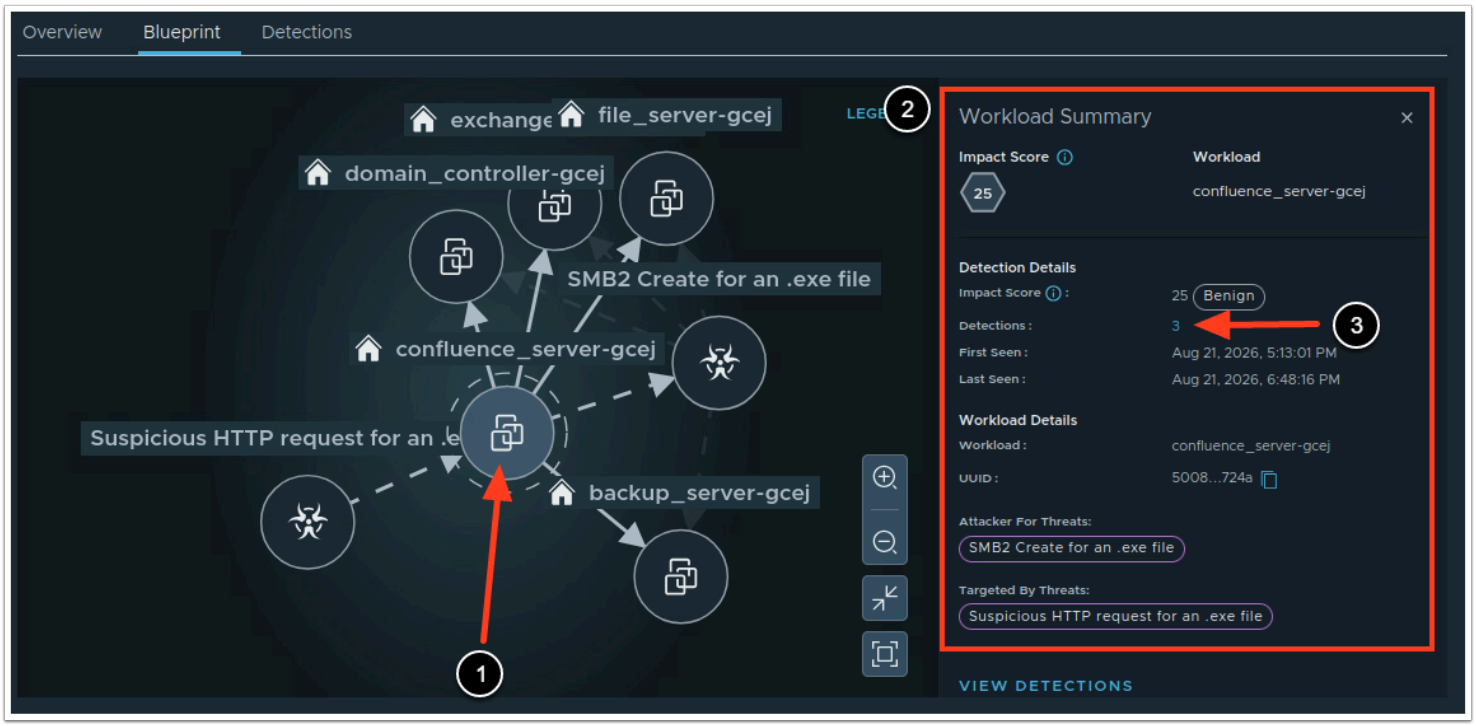


1. 3 Detections events correlated to this campaign
2. 2 Threats events correlated to this campaign
3. Click **Blueprint**

Review Campaign BluePrint

The Blueprint tab under campaign details presents an interactive graphical representation of the campaign. The Blueprint tab visualizes correlated security events, key actors, and their relationships. By simplifying triage and investigation, Blueprint helps provide clearer insight into attacks, which reduces the mean time to detect security incidents.

Review Campaign BluePrint



1. Click **confluence_server-gcej** within the Blueprint
2. Workload Summary indicating Attack Threats highlighted here as SMB2 Create for an .exe file and a list of threats targeted at this workload
3. Again we see the same 3 Detections we looked at earlier

Conclusion

In this lab we brought all the Advanced Threat Prevention detectors together. Performing a typical attack sequence, triggering IDS/IPS and Network Traffic Analysis. We got to understand how the NDR engine ingests and correlates these events in to a visual campaign.

From here you can:

- **Continue with the next lab module**
- **Click [vlp:table-of-contents|Show Table of Contents] to jump to any module or lesson in this lab**
- **End your lab and return in the future**